

Quantifying comprehensive marine vessel emissions using satellite data fusion

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Abstract

Marine vessels are central to the blue economy, but are also significant sources of greenhouse gases that drive climate change and other pollutants that negatively impact local air quality and human health. Accurately measuring these emissions is critical for mitigating them. Current approaches rely on the Automatic Identification System (AIS), a high-resolution tracking system carried by many vessels. However, not all vessels carry AIS; and even for those that do, AIS broadcast and reception quality can be inconsistent over space and time. To account for this, we developed a method for quantifying comprehensive marine vessel emissions at global scale by fusing two independent types of satellite data: 1) individual vessel tracking using AIS to estimate vessel-level emissions when possible, and 2) vessel detections using synthetic aperture radar (SAR) imagery to estimate emissions for vessels that do not use AIS or are operating in areas with poor AIS reception. We found that vessels emitted approximately 1.53 billion metric tonnes of CO₂ in 2025. In 2024, the most recent year for which global inventory estimates are available, marine vessels accounted for 16% of fossil fuel

emissions from the transportation sector, and 3.6% of all emissions globally. Current AIS-only approaches underestimate the magnitude of 2025 emissions by 22%. Meanwhile, because AIS coverage has generally been increasing over time, current AIS-only approaches overestimate the increasing emissions trend by 15% (from 2017 to 2025). By resolving the confound between changes in monitoring and actual changes in emissions, our approach provides a comprehensive estimate of the magnitude, trends, and spatial patterns of marine vessel GHG emissions and non-GHG air pollutant emissions that can enable interventions designed to mitigate them.

1 Summary

Combusting fossil fuels has important implications for both climate change and local air pollution and human health. The transportation sector contributes significantly towards these emissions through ocean-based vessels as well as through aviation, railways, and roadways. We focus on marine emissions where incomplete vessel tracking currently leaves a substantial fraction of emissions completely unaccounted for. As a result, global inventories chronically underestimate marine vessel emissions and misrepresent trends, undermining efforts to monitor, regulate, and mitigate their climate and local health impacts. Here we develop a novel approach that fuses two independent satellite-based data sources in order to monitor all large ocean-going vessels, regardless of whether they use onboard tracking systems or not. This allows us to provide, for the first time, a comprehensive and high-resolution picture of vessel emissions at-sea. It also serves as an example of how independent satellite-based data sources can be fused in order to disentangle changes in monitoring versus changes in the outcome being monitored. Our global marine vessel emissions dataset can serve as a key input to designing and evaluating interventions aimed at mitigating climate change and local health impacts.

2 Introduction

Climate change mitigation requires accurate accounting across sectors; while the transportation sector as a whole was responsible for 21% of all global CO₂ emissions in 2024 [1], fundamental gaps remain in understanding the contribution by marine vessels. These emissions come primarily from the transportation of goods via shipping vessels, fuel and chemicals via tanker vessels, carriage of passengers, fishing activity, and industrial activities such as dredging and offshore infrastructure maintenance. As such, marine vessels play a central role in the blue economy [2]. Because these diverse vessel types vary dramatically in the extent to which vessel behavior is tracked and vessel characteristics are understood, we lack a comprehensive estimate of their emissions. This knowledge gap limits both our basic understanding of marine vessel emissions, and the data granularity and temporal frequency that are needed to underpin policy or private sector interventions aimed at reducing emissions.

The currently accepted state-of-the-art method for estimating marine vessel emissions relies on an AIS-based model that uses tracking data for individual vessels from the Automatic Identification System (AIS). This approach combines information on vessel movement (e.g., speed, time traveled), vessel characteristics (e.g, main and auxiliary engine powers, boiler power), and emission factors that specify the kg of emissions per kilowatt-hour of energy output. These emissions factors vary for each GHG and non-GHG pollutant and by vessel type, vessel size, engine type, and fuel type [3, 4]. The International Maritime Organization (IMO) has validated this AIS-based approach against multiple independent data sources including: high-frequency continuous monitoring sensor data installed directly on vessels; vessel-level emissions data from the “EU Monitoring, Reporting, and Verification emissions database”; and other global emissions inventories [4]. The IMO has also shown this AIS-based approach to be more reliable and substantially more granular than approaches based on aggregate fuel consumption data [4]. Consequently, the AIS-based approach has been widely adopted by the scientific community (a Google Scholar search in January 2026 for articles using the search term “Automatic Identification System” “emissions” yielded 6,880 results).

However, this AIS-based method faces multiple challenges. AIS coverage is incomplete, variable, and not fully representative of global maritime vessel activity. Many vessels are not equipped with AIS systems or are not required to broadcast AIS [5]. Satellite imagery analysis has shown that about 75% of industrial fishing vessels and over 25% of vessels associated with transport (e.g., cargo, passenger) and offshore energy infrastructure (e.g. tankers, support vessels) are not tracked with AIS [6]. Even for vessels that do carry AIS, there are often gaps in data transmission due to challenges with the reception quality of satellites and terrestrial receivers, or due to intentional AIS disabling that may be associated with illegal activities or with legal activities such as captains hiding their vessel location in areas prone to piracy [7]. Changes in AIS adoption and the evolution of AIS technology over time also make it challenging to assess the trends of vessel activities using AIS data alone. For example, the introduction of a new technology in 2022 whereby AIS receivers, which are typically on land or via satellite, are installed directly on shipping vessels (known as Dynamic-AIS) has increased the AIS reception quality in some high traffic regions [8], resulting in significant increases in the vessel activity captured by AIS that do not necessarily reflect the trend in the actual amount of activity. The landscape of AIS data providers (Orbcomm, Spire Global, Exact Earth, Marine Traffic, Kpler) and the type of AIS data each provides (satellite-based, terrestrial, dynamic) has also changed over time, further confounding the ability to study trends based on AIS data alone. Finally, key technical vessel characteristics necessary for AIS-based emissions modeling are often unknown, particularly for low-information vessels that are not listed in publicly available registries [9]

Here we develop a method to estimate high-resolution marine vessel emissions globally from 2017 through 2025 by fusing AIS data from broadcasting vessels with Sentinel-1 (S1) Synthetic Aperture Radar (SAR) detections of vessels that are not broadcasting AIS. The 20m-resolution S1 SAR imagery allows us to detect and estimate the lengths of most vessels larger than 15 m regardless of whether they broadcast

139 AIS, cloud cover, weather conditions, or solar illumination [6]. The two S1 satel-
 140 lites active during our study period (full period for S1A, 2017-2021 for S1B) operate
 141 in a polar, sun-synchronous orbit, providing a global revisit frequency of 6 to 12
 142 days. For over 775,000 AIS-broadcasting vessels, we quantify emissions for each vessel
 143 using precise latitude/longitude/timestamp information for over 119 billion individ-
 144 ual AIS messages. We do so using previously published and validated methods for
 145 the AIS-based emissions model itself [3, 4], while enriching key vessel characteris-
 146 tics for low-information vessels using machine learning models to infer main engine
 147 power, design speed, and vessel type. To account for the vast missing emissions from
 148 non-broadcasting vessels (also sometimes known as ‘dark vessels’ [10]), we develop a
 149 machine learning framework that leverages over 31.9 million vessel detections from
 150 across 924,000 S1 scenes to estimate non-broadcasting emissions at a monthly global
 151 1x1-degree gridded resolution, from 2017 through 2025, and disaggregated by ves-
 152 sel type (fishing or non-fishing). Our method leverages computing infrastructure and
 153 technical expertise developed by Global Fishing Watch (GFW) [11] to process AIS
 154 data [12] and S1 vessel detection data [6], and to estimate key vessel characteristics
 155 for vessels lacking readily available public information [13]. Our method, therefore,
 156 effectively addresses the two major challenges currently present in remotely measuring
 157 marine vessel emissions.

158 Our novel approach to estimating emissions from non-broadcasting vessels actu-
 159 ally makes use of information from broadcasting vessels by combining data fusion and
 160 machine learning. Because we know the exact time and location of each S1 scene, we
 161 can classify every vessel detected by S1 as either being matched to an AIS-broadcasting
 162 vessel or not [6]. Since we estimate vessel level emissions for all AIS-broadcasting ves-
 163 sels using an AIS-based model, we are able to exploit the way S1 data are differentiated
 164 by: 1) building a machine learning model that predicts AIS-broadcasting emissions
 165 from the density of matched S1 vessel detections over space and time; and 2) using
 166 that model to predict non-broadcasting emissions from unmatched S1 vessel detec-
 167 tion density over space and time (i.e., for 1x1 degree pixels each month). To estimate
 168 unmatched vessel detection density over space and time, we must first account for the
 169 fact that S1 coverage is spatiotemporally heterogeneous. The S1 footprint only ever
 170 covers certain regions of the globe, so there are some areas that are never imaged at
 171 all (here we define the ‘S1 footprint’ as the spatial union of all S1 scenes during our
 172 2017 – 2025 study period). And even for those areas within the S1 footprint, different
 173 areas are imaged by S1 scenes at different frequencies over time, which means that
 174 not all areas within the S1 footprint are even imaged every month. To account for
 175 this heterogeneity, we build an additional set of machine learning models to predict
 176 unmatched vessel density for areas that are not imaged by S1 in a given month (either
 177 because this area is within the S1 footprint but did not receive S1 coverage in this par-
 178 ticular month; or because this area falls completely outside the S1 footprint). Using
 179 this predicted unmatched vessel density, we can predict non-broadcasting emissions
 180 for the entire ocean every month, even in areas and times that are not imaged by S1.

181 Our approach yields monthly, spatially explicit estimates of marine vessel emissions
 182 for three GHGs that drive climate change (carbon dioxide: CO₂, methane: CH₄, and
 183 nitrous oxide: N₂O) and six other non-GHG air pollutants that impact ambient air
 184

quality and health (carbon monoxide: CO, nitrogen oxides: NO_x, sulfur oxides: SO_x, particulate matter 2.5: PM_{2.5}, particulate matter 10: PM₁₀, and volatile organic compounds: VOCs). Our vessel-level AIS-broadcasting data further allows for quantifying emissions across individual port stays, domestic trips, and international trips. Focusing our main results on CO₂ as the primary greenhouse gas (with other GHGs and non-GHG air pollutants described in the supplement), we report trends in emissions from 2017 through 2025 (the last full year of available data), and compare trajectories of emission estimates based on either the AIS data alone or the fused AIS-S1 data. We compare the trend in marine vessel CO₂ emissions to those from other transportation sectors over the same time period. Our data allows for detailed mapping of emissions from vessel activities at a global scale with a breakdown across vessel types. By comparing emission trends from AIS-only models and our fused AIS-SAR approach, we reveal the true magnitude and evolving dynamics of the marine sector’s contribution to global and regional emissions.

3 Results

To correctly estimate both the magnitude and trends of marine emissions, it is essential to fuse data from sources that monitor both broadcasting and non-broadcasting vessels. Using this data fusion approach, we find that total global CO₂ emissions at sea in 2025 was 1.53 billion MT (Fig. 1A), with an increase of 38% from 2017 to 2025 (Fig. 1B). If only considering emissions from vessels that broadcast AIS, in line with currently accepted state-of-the-art models, global CO₂ emissions in 2025 would be only 1.19 billion MT (Fig. 1A), with a more pronounced increase from 2017 to 2025 of 44% (Fig. 1B). Thus using only AIS data from broadcasting vessels underestimates the magnitude of 2025 global CO₂ emissions by 22%, and overestimates the increasing trend for 2017–2025 by 15% (see Tables S1 and S2 for summaries of all other GHGs and pollutants). Fusing AIS and S1 data also provides a comprehensive picture of spatial marine vessel emissions patterns and how those patterns change over time (Fig. 1C and Fig. S2).

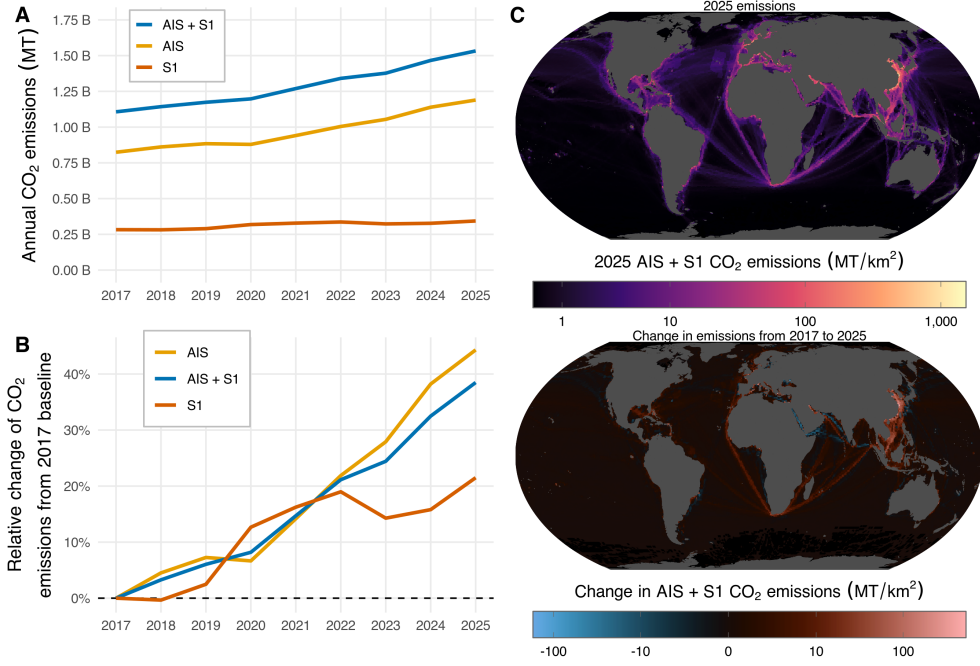


Fig. 1 (A) Annual marine vessel CO₂ emissions, with each line colored based on the data source. (B) Relative change in annual marine vessel CO₂ emissions from a 2017 baseline, with each line colored based on the data source. (C) Spatial map of CO₂ emissions in 2025 (top) and change in CO₂ emissions between 2017 and 2025 (bottom), aggregated across AIS-broadcasting vessels and non-broadcasting vessels, and shown at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale).

Estimating emissions from non-broadcasting vessels offers new insights into the temporal trends of and spatial patterns of global emissions (Fig. 2). Although total emissions has generally been increasing (Fig. 2A), the share of emissions from non-broadcasting vessels has generally been declining since 2021 (Fig. 2B), consistent with a steady increase in AIS monitoring over the period. The time series data reveal strong seasonal trends and event-driven variations in marine GHG emissions. We see consistent dips during major holidays including Christmas, New Year's Day, and Chinese New Year, as well as during the COVID-19 pandemic (Fig. 2A and Fig. S3). Being able to spatially quantify emissions from non-broadcasting vessels fills important gaps in our current understanding of where marine vessel emissions take place, particularly in southeast Asia, China, the Caribbean, and the Mediterranean (Fig. 2C), crucial for targeting policy or private sector interventions. Our dataset also provides this sort of nuanced spatiotemporal information disaggregated by fishing and non-fishing vessels (Fig. S1). In 2025, 36% of CO₂ emissions from fishing vessels were from non-broadcasting vessels, and 21% of CO₂ emissions from non-fishing vessels were from non-broadcasting vessels.

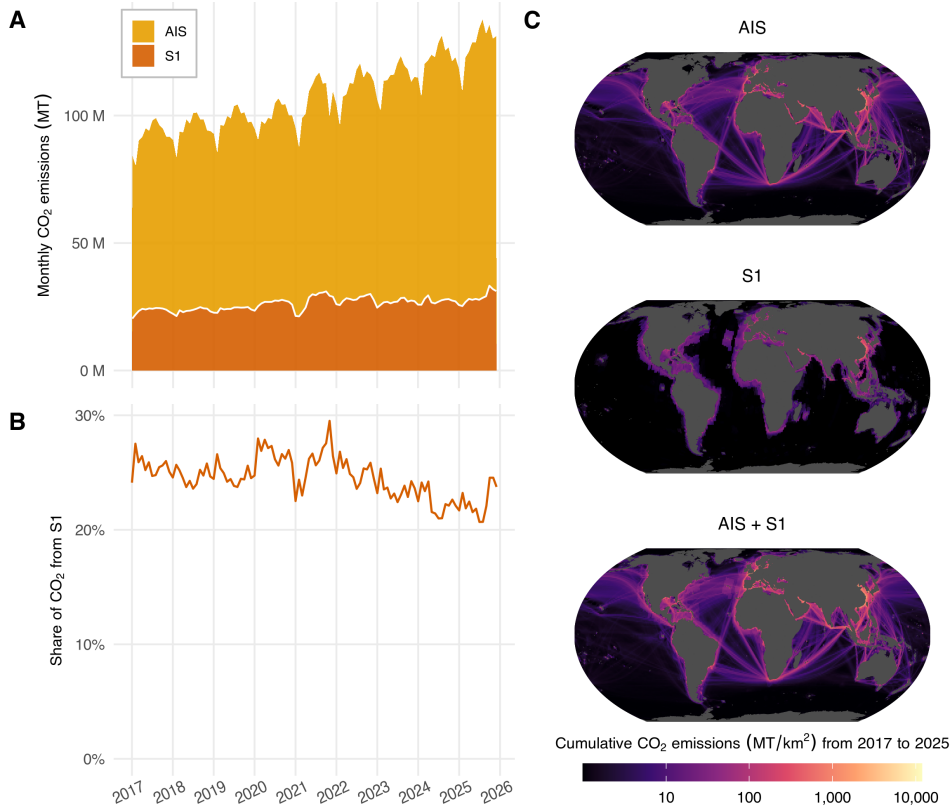


Fig. 2 (A) Time series of monthly global CO₂ emissions, shown for both AIS-broadcasting and S1-detected non-broadcasting vessels. (B) Time series of the monthly share of global CO₂ emissions attributable to S1-detected non-broadcasting vessels. (C) Spatial distribution of cumulative 2017-2025 CO₂ emissions from AIS-broadcasting vessels (top), S1-detected non-broadcasting vessels (middle), and both fleets combined (bottom), shown at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale).

Across the entire 2017 to 2025 time series, almost all non-broadcasting emissions (96.4%) occurred inside the S1 footprint and in pixel-months that were imaged by S1 (Fig. 17). This makes sense because part of the Copernicus system’s observation strategy for S1 is specifically focused on imaging coastal and shipping areas, where ship density is highest [14]. Our machine learning model for predicting these emissions directly from observed S1 detection density achieves a high out-of-sample predictive performance (traditional $R^2 = 0.94$ and total percent error of -8.6% for CO₂, Table 3). Only 3.5% of non-broadcasting emissions occurred inside the S1 footprint but in pixel-months that weren’t imaged by S1, requiring us to use a two-stage hurdle framework using both a classification model (F1 Score = 0.72) and a regression model (traditional $R^2 = 0.73$) to predict S1 detection density before predicting emissions (traditional $R^2 = 0.94$). Only 0.08% of non-broadcasting emissions occurred outside the S1 footprint entirely, since most vessels operating in these open ocean areas use AIS.

Our vessel-level AIS-broadcasting emissions data provide a detailed understanding of marine vessel emissions across vessel classes and activity types (Fig. 3). We are able to distinguish activity between 37 classes of vessels, including 14 classes of fishing vessels; this makes our inventory the first to our knowledge that provides this level of emissions detail for fishing vessels. Container and bulk carriers, which transport much of the world’s goods by sea, are the highest-emitting cargo classes (Fig. 3A-B). Oil tankers, which themselves transport fossil fuels, are the highest-emitting tanker type. These vessel classes are ranked similarly in other inventories [15–18], with the notable exception of the passenger category. Passenger vessels have a highly aggregated definition in our model, composed of several diverse vessel subclasses including ferries, cruise, Ro-Pax and yachts, which makes it the most numerous class in our fleet and consequently the largest emitting group in 2025 (Fig. S18). This and any further departure from other studies reflects our broader vessel coverage, which includes classes and sizes that are absent or underrepresented elsewhere—vessels below 25 m account for most passenger emissions in our inventory (Fig. S16, panel B), and fishing vessels are our second-largest class by vessel count (Fig. S18). Trawlers, which use an energy-intensive fishing technique in which nets are dragged through the water [19], are the highest-emitting type of fishing vessel. Emissions from AIS-broadcasting vessels can also be assigned to port stays in particular countries and port-to-port voyages within- and between-countries by analyzing the movement tracks of each individual vessel (Fig. 3C and Fig. S4). Most emissions for cargo and tanker vessels occur during international trips, while most service and passenger vessel emissions occur within ports (Fig. 3C). China, Japan, and United States had the most CO₂ emissions during within-country domestic trips in 2025. China, Malaysia, and United States had the most CO₂ emissions during international trips that started or ended in these countries in 2025 (where the emissions from each trip are partitioned equally to its departure and arrival country). China, United States, and Indonesia had the most CO₂ emissions during port stays. Country totals for every activity type are summarized in Fig. S4 and listed in Table S6. We note that not all AIS messages are broadcast by vessels; we are therefore careful to exclude all non-vessel AIS messages from our emissions estimates. Roughly 47% of AIS-enabled signals are attributable to buoys, FADs, navigation aids, offshore structures, fishing gear, and other objects. These are classified as a separate ‘gear’ class by our vessel classification model, and are not included in this analysis.

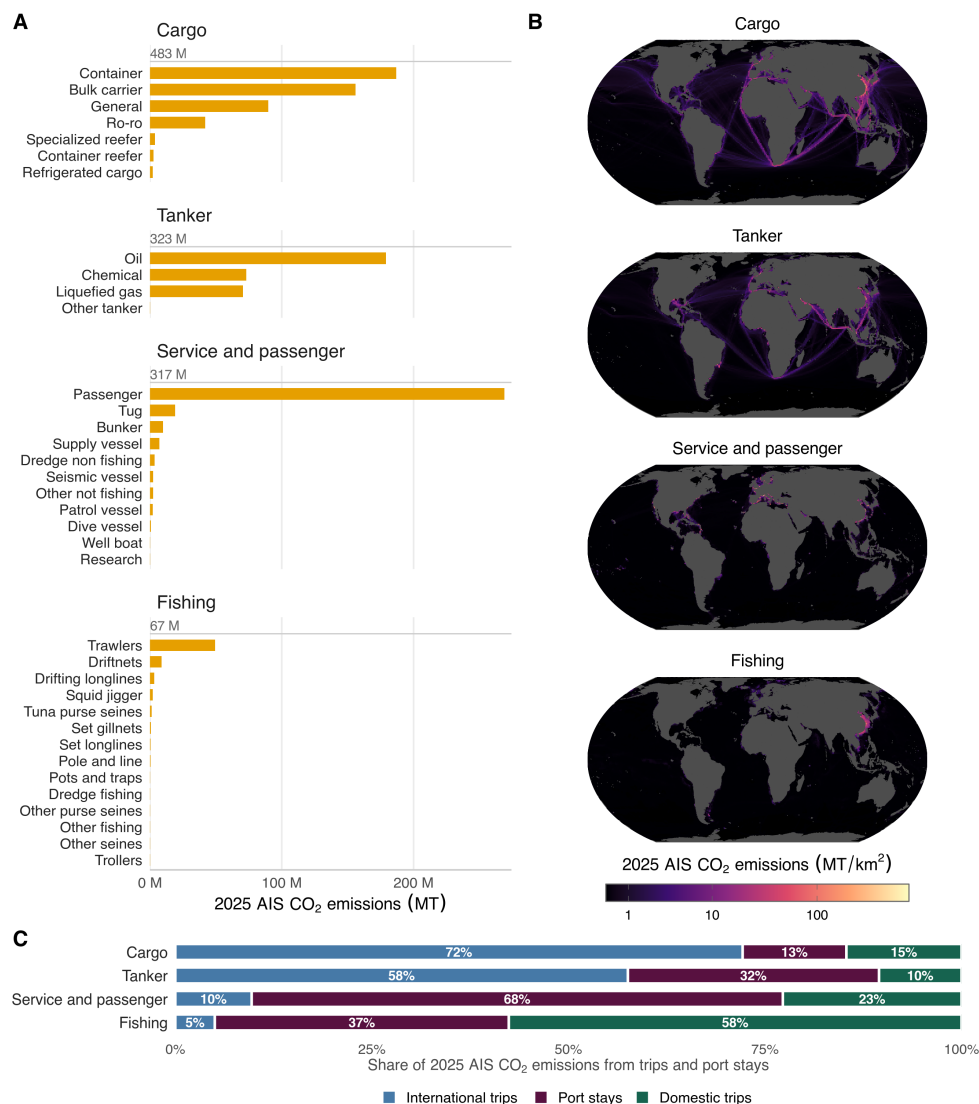


Fig. 3 Richness of AIS-broadcasting vessel emissions data, showing: (A) Total 2025 CO₂ emissions by vessel class for AIS-broadcasting vessels. Vessel classes are grouped into four families, with the total for each family shown above its bars; every class in a family is shown as its own bar, so panel heights differ with the number of classes. These totals are aggregated from the same gridded emissions shown in panel B, so the two panels describe exactly the same emissions and the bars sum to the mapped total. (B) Spatial distribution of 2025 CO₂ emissions from AIS-broadcasting vessels for each of those same vessel class families, shown at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale). Families appear in the same order in both panels. (C) For each family, the share of CO₂ emissions attributable to international trips, port stays, and domestic trips (note that Panel C is not a decomposition of panel A: it covers only emissions that could be assigned to a trip or port visit that ended in 2025; each family sums to 100% of its own assigned emissions).

415 By developing new methods for inferring key missing vessel characteristics that
 416 are necessary for AIS-based emissions modeling, we can accurately estimate vessel-
 417 level emissions for a huge swath of the world’s fleet that does not otherwise have the
 418 necessary vessel characteristic information in public registries. Of the roughly 775,000
 419 AIS-broadcasting vessels in our analysis, we matched 139,000 (18%) to public reg-
 420 istries or proprietary databases, the majority of which (93,000) were IMO-registered.
 421 This allowed us to leverage vessel characteristics provided for these vessels from these
 422 databases. However, the vast majority of AIS-broadcasting vessels were not present in
 423 these registries, meaning that we needed to use model-derived vessel characteristics for
 424 between 83% and 92% of vessels, depending on the characteristic (S8). Because many
 425 large vessels are already listed with the IMO, the majority of our total CO₂ emissions
 426 come from these vessels (39%); 19% come from vessels listed on other registries; and
 427 42% come from vessels with no public registry information and which rely entirely on
 428 modeled vessel characteristics.

429 Using the vessel characteristic values that are derived from our machine learning
 430 models, we achieve an out-of-sample model performance of 0.85 traditional R² when
 431 comparing our daily vessel-specific CO₂ emissions estimates against those derived
 432 from a proprietary vessel registry dataset of over 15,000 vessel-specific fuel consump-
 433 tion values that were measured during at-sea trials (Fig. 6). If we were to instead
 434 estimate emissions with our model using only currently available public vessel charac-
 435 teristic data (i.e., the IMO provides lookup tables for average main engine power and
 436 design speed for different vessel types and sizes [4]), we would only achieve a predic-
 437 tive performance value of 0.73 traditional R². We also measured the performance of
 438 our model’s ability to estimate annual vessel-level CO₂ emissions estimates by com-
 439 paring our estimates to those available in the publicly available European Maritime
 440 Safety Agency’s Monitoring, Reporting, and Verification (MRV) database [20]. Using
 441 a set of over 27,000 matched vessel-specific annual emissions estimates, we found a
 442 performance value of 0.84 traditional R² (Table 1).

443 The dataset also allows us to examine large-scale trends in marine vessel emis-
 444 sions for different parts of the ocean, and for different sectors of the global economy.
 445 Although vessel activity in the Southern Ocean represents a very small fractions of
 446 the world’s marine vessel emissions (0.04%), it represents the part of the world experi-
 447 encing the highest growth rate of marine vessel emissions (267%) (Fig. 4A, Table S3).
 448 These polar estimates are primarily driven by AIS-broadcasting vessels, with only a
 449 small fraction coming from non-broadcasting vessels (1.02% in the Southern Ocean;
 450 Table S3, Figure S5). Polar regions have extremely limited S1 coverage (only 0.06% of
 451 pixel-months in the Southern Ocean were imaged by S1), meaning that trends there
 452 may be more susceptible to AIS adoption biases. This reflects the growing importance
 453 of monitoring marine vessel activity in the world’s extreme latitudes.

454 Using the latest data available from the EU’s Emissions Database for Global Atmo-
 455 spheric Research (EDGAR), we estimate that in 2024 marine vessels accounted for
 456 16% of all transportation-related CO₂ emissions, and 3.6% of all emissions globally [1].
 457 This is consistent with a report from the European Maritime Safety Agency that found
 458 that 14.2% of all transportation-related CO₂ emissions in the EU were from the mar-
 459 itime sector in 2022 [21]. Marine vessel CO₂ emissions increased by 32% between 2017
 460

and 2024, five times faster than the growth rate of global emissions as a whole, and ten times faster than the growth rate of emissions from all other transportation-related sectors including railways, road transportation, aviation, and pipelines (Fig. 4B, Table S4). This provides new evidence supporting the concept of the ‘blue acceleration’ and the expansion of human activity in the oceans [22].

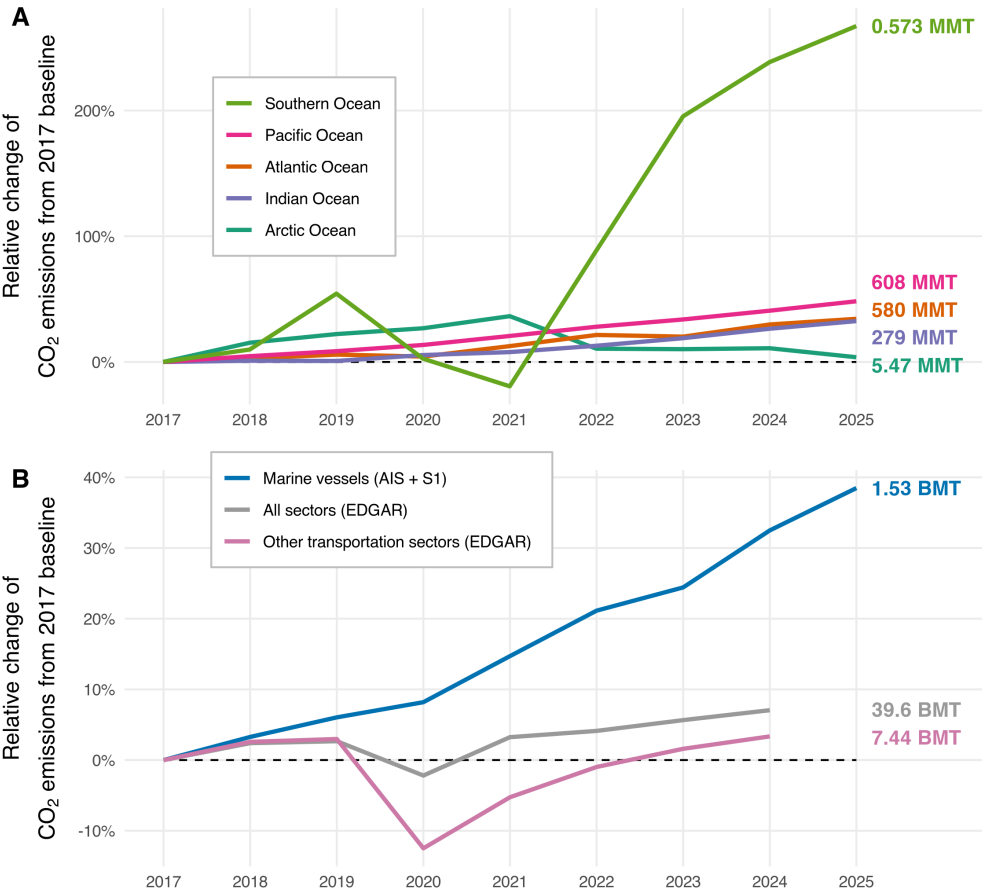


Fig. 4 (A) Trend in CO₂ emissions by ocean (normalized to 2017 baseline), use our fused AIS+S1 marine vessel data. (B) Comparison in trend of CO₂ emissions data (normalized to 2017 baseline) between marine transportation emissions from GFW, other transportation emissions from EDGAR, and all sector emissions from EDGAR. On the right-hand side of the figure, absolute values for the latest year of data (in millions of metric tonnes) are shown as text and color-coded for each line.

507 4 Discussion

508
509 High-resolution emissions data for marine vessels are essential for designing, imple-
510 menting, and evaluating policies aimed at reducing them. However, incomplete vessel
511 tracking data and technical data processing challenges have previously hindered
512 progress [9]. Our new data and methodology overcome these challenges by providing
513 the spatiotemporal data necessary for analyzing interventions that could mitigate the
514 GHG emissions that drive climate change, as well as other air pollutants that impact
515 ambient air quality and health. Our data could also be used to quantify the spatial
516 health impacts of air pollution related to shipping and fishing activity [23], along with
517 the associated environmental justice implications.

518 There are three main implications of the results: First, the spatial and temporal
519 detail of our dataset makes it directly relevant for climate change and air quality policy.
520 Carbon pricing, fuel standards, emissions control areas, port queuing reforms, and ves-
521 sel speed reduction programs all require credible baselines and measurable outcomes.
522 Our estimates provide such a baseline for both broadcasting and non-broadcasting
523 activity, including vessels that would be missed by AIS-only inventories. This is espe-
524 cially relevant for the IMO’s effort to move international shipping toward net-zero
525 greenhouse gas emissions by 2050 by implementing a carbon market for international
526 shipping emissions by vessels over 5,000 gross tonnage [24] [25]. Further, the European
527 Union Emissions Trading System has been expanded to include marine transportation
528 under the EU emissions cap, and to subject vessel activity to and from European
529 Economic Area ports to a carbon price [26]. Implementing a carbon market based on
530 limiting emissions intensity (the amount of emissions per unit of activity or produc-
531 tion) could also be an effective strategy for mitigating marine vessel emissions, even
532 with a growing number of vessels. Carbon markets based on emissions-intensity stan-
533 dards rather than absolute emissions caps, such as India’s emerging carbon market
534 [27], allow economic growth while continuing to incentivize improvements in emissions
535 efficiency. This policy design is consistent with a review paper on current marine ves-
536 sel emissions datasets, which concluded that measures aimed at increasing technical
537 efficiency of ships could be an effective way to decouple emissions growth from global
538 trade demand [28].

539 Second, our data can be used to calculate the climate externality social cost (SC)
540 associated with marine vessel activity, which can be compared against the economic
541 value of the activities; we conduct a simple version of that calculation here. We first
542 focus on non-fishing vessels (e.g., cargo ships and tankers) and compare their climate
543 externalities with the value of goods transported by sea. Multiplying our estimated
544 emissions of carbon dioxide, methane, and nitrous oxide from non-fishing vessels by the
545 corresponding social costs of carbon (283 USD/MT [29]), methane (2,900 USD/MT
546 [30]), and nitrous oxide (49,600 USD/MT [30]) yields an estimated annual global
547 climate externality of 344 billion USD in 2021 (Table S5). This amounts to about 5%
548 of the total value of goods transported by maritime trade, based on an estimated 6.63
549 trillion USD in global seaborne trade in 2021 [31]. Repeating this calculation for fishing
550 vessels yields an estimated annual global climate externality of 24 billion USD in 2022
551 (Table S5). This amounts to about 15% of the global first-sale value of capture fisheries,
552 which totaled 159 billion USD in 2022 [32]. Our voyage-level emissions dataset can

be used to estimate the climate externalities associated with country-to-country trade flows, enabling the construction of metrics such as the social cost per dollar of traded goods or the social cost per metric tonne of cargo transported. For vessels tracked with AIS, these externality costs can also be assigned to individual port-to-port voyages. Beyond their contribution to climate change, marine vessels emit air pollutants that impose substantial human health costs that have yet to be fully understood. [23].

Third, our methodology and data can help addressing a broader challenge in environmental monitoring: temporal trends in environmental variables are often confounded by changes in monitoring effort. For marine vessel emissions, this challenge arises because the adoption of AIS and the number of vessels being tracked have increased over time, causing existing methods to overestimate the growth of at-sea emissions by 15.9%. Similar biases may arise for other environmental variables when monitoring expands over time, such as the discovery of invasive species [33]. By combining two independent data sources, our approach is more robust to changes in monitoring effort than methods that rely on a single source and therefore produces more accurate estimates of environmental change. The same framework could be applied in other settings where changes in monitoring complicate the measurement of long-run environmental trends.

Several methodological limitations remain. One key assumption is that emissions from non-broadcasting vessels can be inferred from the relationship between Sentinel-1 detections and AIS-based emissions for matched vessels. We believe this is a reasonable assumption for several reasons. First, many unmatched S1 detections likely correspond to AIS-broadcasting vessels that are not observed in the AIS data because of transmission gaps or poor reception quality. In these cases, the relationship between S1 vessel detections and emissions would hold perfectly since these unmatched vessels are in fact just AIS-broadcasting vessels that happen to not be observed by AIS at a given point in time. Second, our model accounts for vessel length, a proxy for main engine power that is one of the primary determinants of emissions in AIS-based models and is applicable to virtually all vessels powered by internal combustion engines. Third, since fishing and non-fishing vessels are known to have different emissions factors [19], we account for this by differentiating detections between fishing and non-fishing vessels. Finally, our model incorporates spatiotemporal features that allow the relationship between detections and emissions to vary across space and time. Consequently, if broadcasting and non-broadcasting vessels operating in a given place and time have similar characteristics, the relationship between detections and emissions is expected to be similar as well.

Another limitation is that the relationship between detections and emissions may not hold for non-broadcasting vessels if their emissions factors differ substantially from the IMO values we use to convert energy use into emissions [4]. This could arise, for example, if non-broadcasting vessels disproportionately use newer engine or fuel technologies that were not widely represented when the IMO emissions factors were developed in 2020. This highlights one of the primary limitations of the AIS model our methodology builds upon. Since information on the fuel type used by individual vessels is generally not publicly available, we rely on weighted average emissions factors from the IMO [4]. Vessel-level emissions estimates based on average emission factors

599 may be biased if a particular vessel’s engine or fuel type is not well represented in
600 the global fleet. This could be an important consideration when using the model to
601 explore policies aimed at technological engine changes and fuel switching. However,
602 when this type of vessel-level information becomes publicly available, our modeling
603 framework could easily be adapted to incorporate it. Despite the fact that we do not
604 currently explicitly account for vessel-level fuel type, our vessel-level emissions model
605 still achieves high performance when validated against two fully independent vessel-
606 level datasets (Figs. 6 - 7). Finally, neither AIS nor Sentinel-1 have good coverage
607 for very small vessels (i.e., <15m), limiting our approach to only represent emissions
608 from large marine vessels. An exciting area of future work would be to incorporate
609 new satellite sensors into our data fusion framework to quantify emissions from small
610 vessels (e.g., optical imagery from Sentinel-2 or Planet).

611 Accurately measuring emissions is a necessary step towards reducing them. To date,
612 quantifying emissions from vessels in the ocean has suffered from incomplete monitor-
613 ing at-sea and the opaque nature of vessel characteristics. We fuse two satellite-based
614 data sources and by doing so, are able to track emissions from both broadcast-
615 ing and non-broadcasting activity. Furthermore, by estimating key emissions modeling
616 vessel characteristics for hundreds of thousands of vessels, we now have a compre-
617 hensive and high resolution picture of emissions from large marine vessels. This can
618 serve as the foundation for interventions aimed at decarbonizing the marine sector
619 and reducing the health impacts of marine air pollution.

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Conceptualization: GM and DK; Methodology: GM, PCM, OD, EFC, AH, DK,MP, ZW, and CC; Software: GM, PCM, EFC, AH, and ZW Validation: GM, PCM, DK,MP; Formal analysis: GM, PCM, OD, EFC, AH, DK,MP, ZW, and CC; Investigation: GM, PCM, OD, EFC, AH, DK,MP, ZW, and CC; Data curation: GM, PCM, EFC,

AH, DK,MP and ZW ; Writing - Original Draft: GM, PCM, OD, and CC; Writing
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There are no competing interests to declare.

Data and materials availability:

All code and data necessary for reproducing this figures and tables in this analysis
will be made available in a public GitHub repository with an accompanying Zenodo
repository and DOI.

Methods

This work relies on multiple sources of spatial and tabular data to assess marine vessel
emissions globally. We use AIS data, vessel registry information and Sentinel-1 derived
data to construct our AIS-based emission model and the non-broadcasting emissions
model (Fig. 5).

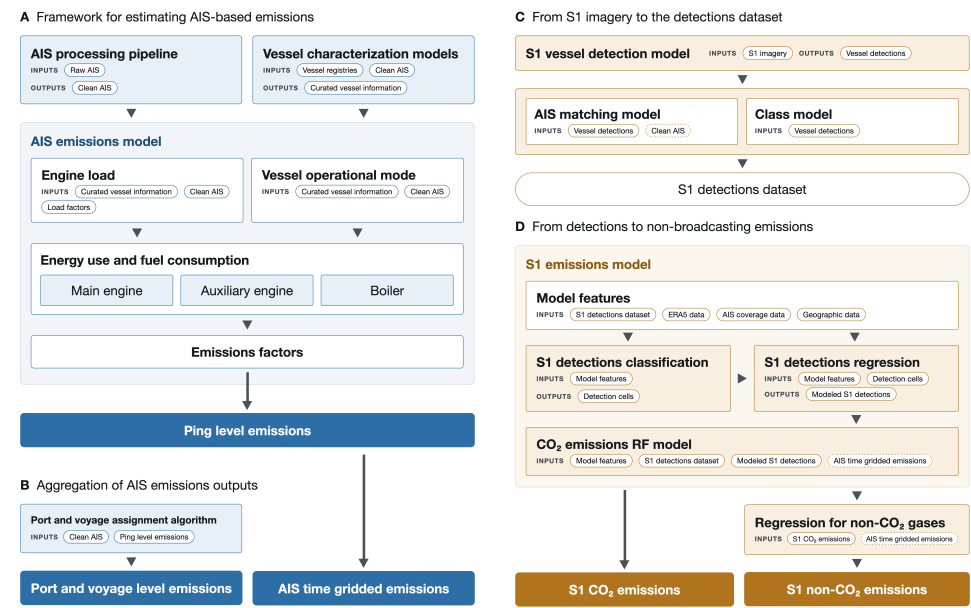


Fig. 5 Conceptual flowchart for estimating emissions using the AIS-based emissions model and the non-broadcasting emissions model.

967 The following sections describe our approaches for AIS-broadcasting vessels and
968 non-broadcasting vessels.

969

970 4.1 AIS-broadcasting vessels

971

972 4.1.1 AIS data description

973 *AIS messages, port visits, and voyages*

974 We use individual AIS messages (pings) from the latest version of GFW’s AIS pipeline
975 (‘Version 4’), which parses, cleans, and augments raw AIS data [12]. Since pings are
976 grouped into segments using a segmenter [12], the time for a single ping represents the
977 time since the previous ping in that segment; this value is typically small (the mean
978 value is 0.088 hours and the median value is 0.045 hours, or about 5 minutes and
979 3 minutes respectively), with virtually all pings representing < 24 hours of activity.
980 The activity associated with individual pings form the basis of our ping-level AIS-
981 broadcasting emissions estimates, described below. These AIS messages come from a
982 combination of satellite, terrestrial, and dynamic receivers (described in more detail
983 under the section below ‘Trends in AIS receiver types’ and Fig. S6, Fig. S7, Fig. S8)

984 This dataset also classifies whether a fishing vessel is actively engaged in fishing
985 activities at the ping level [12]. Finally, we leverage GFW’s algorithm for detecting
986 port visits [34], allowing us to aggregate ping-level emissions to the vessel-specific port
987 stays, as well as to vessel-specific port-to-port trips (both international and domestic).

988 In all, we analyze AIS data for over 775,000 AIS-broadcasting vessels, across 119
989 billion individual timestamped AIS messages which are assigned to 14.4 million inter-
990 national trips, 112 million domestic trips, and 96.4 million port visits that occurred
991 between 2017 and 2025.

992

993 *Vessel characteristics data*

994 Translating AIS vessel activity data into emissions estimates requires information
995 on numerous vessel-specific characteristics including vessel class, main engine power,
996 engine type, gross tonnage, maximum design speed, vessel construction year, and
997 engine operating speed. A key contribution of this work was therefore to refine and
998 expand GFW’s vessel characteristics database to include accurate estimates of these
999 attributes for all 775,000 AIS-broadcasting vessels included in our analysis. For each
1000 vessel, this information comes from: (1) official registry information when available
1001 [13], or (2) when registry data are unavailable, machine learning vessel characteristics
1002 models. We briefly define these models here, and provide full details on training data,
1003 features, model performance, and known limitations in the Supplementary Materials
1004 Sections 1.6 and 1.6.4.

1005 Leveraging work pioneered by [12], GFW uses a primary vessel classification model
1006 to assign each AIS-broadcasting vessel into one of 32 vessel classes (Section 1.6.1).
1007 For vessels classified as ‘cargo’ or ‘tankers’ by GFW’s primary vessel classification
1008 model, we developed new gradient boosting sub-models that refine the general vessel
1009 classification into sub-categories aligned with the Fourth IMO Greenhouse Gas Study
1010 vessel class types: five cargo categories (bulk carrier, container, general, refrigerated,
1011 and ro-ro) and four tanker categories (oil, chemical, liquefied gas, and other liquids)

(Section 1.6.2 and 1.6.3).. The sub-models are trained on port visit history and vessel activity features, using labels derived from over 26,000 vessels with known IMO vessel class types (S&P Maritime StatCode5 categories), and achieve weighted F1 scores of 0.91 (cargo) and 0.87 (tanker). These refined cargo and tanker vessel class categories allows us to more accurately assign auxiliary engine power and boiler power to each vessel, which are based on lookup tables according to vessel class and size [4].

For the main engine power, gross tonnage, maximum design speed estimates, vessel construction year, and engine operating speed, we develop separate random forest regression models (Section 1.6.4). All models are trained using the same features: inferred vessel classification scores, AIS self-reported shiptype messages, activity and location features derived from AIS usage, and values inferred from prior regression models. All models were trained on datasets of around 360,000 labeled registry samples, with a holdout test set of 90,000 for an 80/20 train-test split. The main engine power, gross tonnage, maximum design speed, and construction year regression models achieved traditional R^2 values exceeding 0.92, 0.94, and 0.86, and 0.45 respectively.

The resulting vessel characteristics dataset contains the necessary metadata for estimating emissions using an AIS-based methodology for over 775,000 active AIS-broadcasting vessels.

It should be noted that even when registries exist, vessel characteristics can be unrepresentative, particularly for fishing vessels. For instance, evidence from the European fishing sector suggests that vessels often use engines larger than those officially registered [35], while fraud also extends to reported vessel length and other characteristics [36]. Underreported main engine power would lead to an underestimation of AIS-based emissions from these vessels; and if underreported main engine power data are used for training models to infer main engine power, this may lead to a systematic underestimation of AIS-based emissions for vessels using inferred main engine power.

4.1.2 Emissions model for AIS-broadcasting vessels

We adapted our AIS-based model from the methodologies described in the 2017 ICCT study on global shipping emissions [3] and the 2020 IMO Fourth Greenhouse Gas Study [4], making modifications to better align with the structure and characteristics of our data sets. From a high-level, we: 1) determine the main engine energy use, auxiliary engine energy use, and boiler energy use for each individual AIS message based on dynamic vessel movement and static vessel characteristic data; 2) multiply the engine and boiler energy use numbers by pollutant- and engine-type-specific emissions factors in order to determine the amount of emissions for six pollutants associated with the engines and boiler for each ping; 3) add up the emissions for these six pollutants across the engines and boiler, to obtain the total emissions for each ping and each of these six pollutants; 4) estimate emissions for three additional pollutants that do not detailed information on emissions factors by simply multiplying the total CO_2 emissions for each ping by a conversion factor [37]. We should note that we focus our analysis on estimating emissions associated with fuel combustion in the main engine, auxiliary engine, and boiler; we do not include methane slip on vessels that use liquefied natural gas, or other non-combustion fugitive emissions associated with refrigeration, cooling, or venting of VOCs from vessels that transport liquid cargo such as oil and chemicals.

1059 Main engine energy use is estimated based on the time between each AIS ping
 1060 (hours), the main engine power from the vessel characteristics dataset, and a load fac-
 1061 tor used to adjust the estimate. The load factor is the product of four coefficients:
 1062 (1) a weather correction factor (represented by distance to shore, a proxy that can be
 1063 observed from AIS data - vessel operating beyond 5 nautical miles from shore receive
 1064 a value of 1.15; vessels operating within 5 nautical miles receive a value of 1.1 [4]);
 1065 (2) a hull fouling correction factor, which causes resistance in the water (set at 1.07
 1066 according to [3]); and (3) a vessel draft correction factor (set at 0.85, the emissions-
 1067 weighted average draft correction factor from [4]; and (4) a cubic term of vessel speed
 1068 divided by the vessel’s maximum design speed. To determine the vessel speed asso-
 1069 ciated with each AIS message, we use either: 1) the transmitted instantaneous speed
 1070 associated with the AIS message if less than an hour has passed since the last message
 1071 was transmitted; or 2) the implied speed for messages for which more than an hour has
 1072 passed since the previous message (calculated as the distance traveled divided by the
 1073 elapsed time between the two messages). We apply this hierarchy since implied speed
 1074 can be imprecise for short time intervals between pings due to signal latency; mean-
 1075 while, instantaneous speed can be unrepresentative of longer time intervals between
 1076 pings, in which case implied speed is more representative. Following [19, 38], for fish-
 1077 ing vessels we adjust the main engine load factor so that it falls between 0.2 and 0.9,
 1078 and assign a main engine load factor of 0.75 to trawlers and dredge fishing vessels that
 1079 are actively fishing due to the energy intensive nature of their fishing operations. We
 1080 also apply main engine low-load correction factors since engines operating at loads
 1081 below 20% operate inefficiently and emit more of certain pollutants based on how low
 1082 the engine is operating (CH_4 , CO , N_2O , NO_x , VOCs , $\text{PM}_{2.5}$, and PM_{10}). We apply
 1083 these correction factor on the main engine emissions for all AIS message with a main
 1084 engine load below 20%.

1085 In accordance with [4], our model also accounts for differences in auxiliary
 1086 engine power energy use and boiler energy use depending on the vessel’s operational
 1087 phase—whether it is at berth, at anchor, maneuvering, or at sea. These phases are
 1088 identified using AIS speed, distance to port, and distance to the coast. Power outputs
 1089 for auxiliary engines and boilers, based on vessel class and size, are extracted from [4]
 1090 and are applied accordingly for each phase.

1091 After calculating energy use, we apply pollutant-specific emission factors to the
 1092 main engine, auxiliary engine, and boiler for six GHGs and pollutants (CO_2 , CH_4 ,
 1093 CO , NO_x , SO_x , N_2O), following [3]. These factors are each averaged from across the
 1094 relevant pollutant factors associated with SSD, MSD, and HSD engine types, which
 1095 together represent approximately 98% of vessels engines [4]. To get total emissions for
 1096 each AIS message (i.e., emissions from the previous ping in a segment to the current
 1097 one), we sum emissions from both engines and boilers. For VOCs , $\text{PM}_{2.5}$, and PM_{10} ,
 1098 which do not have as detailed information on emissions factors, we instead multiply
 1099 the CO_2 emissions by pollutant-specific conversion factors [37].

1100 Starting January 1, 2020, the IMO mandated a low sulfur fuel switch, reducing the
 1101 allowable sulfur content from 3.5% to 0.5% [39]. To account for this, for data starting
 1102 on January 1, 2020 we apply a correction factor to SO_x , as well as $\text{PM}_{2.5}$, and PM_{10}
 1103 which are partially produced by sulfur. The SO_x emissions factor scales linearly with
 1104

fuel sulfur percentage content [4]. For pre-2020 data, we use an emissions factor based on a global average sulfur content of 2.5% [3]. Assuming the sulfur content drops from 2.5% to 0.5% starting in 2020, this would correspond to an 80% drop in sulfur content and an 80% drop in the sulfur emissions factor. Although this correction factor is assumed and does not explicitly account for non-compliance, scrubbers, or low sulfur areas, this 80% drop is consistent with several studies that looked at the impact this new requirement had on global sulfur emissions [40, 41]. PM emissions factors are based on an equation that is non-linear to sulfur fuel content [4]. Using this equation and again assuming an 80% drop in sulfur content we estimate a 79.6% drop in the PM_{2.5}, and PM₁₀ emissions factors. We apply these adjusted emissions factors for all SO_x, PM_{2.5}, and PM₁₀ emissions starting in January 1, 2020.

4.1.3 Validating AIS-broadcasting emissions model

We use two distinct emissions datasets for validation, one to assess the impact of updated vessel metadata from our random forest models on performance, and another to evaluate the model’s overall performance using emissions associated with actual vessel activity.

Registered data validation

We have access to a confidential registered dataset of vessels’ main engine fuel consumption measured during sea trials over a 24-hour period at a given speed, along with each vessel’s main engine power and design speed. Because these consumption values are not associated with specific trips that can be matched to our trip-level emissions estimates, direct comparison of derived emissions is not possible. However, this dataset allows us to validate the emissions generated by the main engine model module and assess the impact of updated metadata from our random forest models. Specifically, we compare the performance of the main engine model emissions estimates using main engine power and design speed values from the original IMO lookup table [4] compared with those obtained when using our vessel-specific random forest model predictions of main engine power and design speed. Furthermore, we define the performance ceiling achievable through improved vessel metadata by testing the model with the actual engine power and design speed recorded in the registered dataset.

For a validation sample of 15,679 vessels, we ran the main engine model to calculate main engine energy use, incorporating the weather correction factor for offshore navigation along with the other default hull-fouling and draft correction factors from our main model, over a 24-hour period at the consumption speed provided in the registered dataset. We test three different sets of vessel metadata. For each set, main engine power and design speed were taken from either: (1) Table 81 lookup values from [4]; (2) our vessel-specific random forest model predictions; or (3) the registered dataset. Main engine energy use was then converted to emissions using the same CO₂ emission factor as in our main model. Since the registered dataset provides vessel-specific main fuel consumption values and not actual emissions values, we converted the registered fuel consumption to emissions by applying a fuel-to-CO₂ conversion factor of 3.12 (kg CO₂/tonne fuel), derived from the heavy fuel oil (HFO) and marine diesel oil (MDO) factors weighted by their contribution to global fuel consumption, as reported in [4].

This finally allows us to assess the performance of our model-estimated CO₂ emissions to those from the registered dataset.

This comparison shows that the updated vessel metadata using random forest models substantially improves model performance compared to that achieved using the original [4] vessel metadata, bringing the results closer to the model performance ceiling.

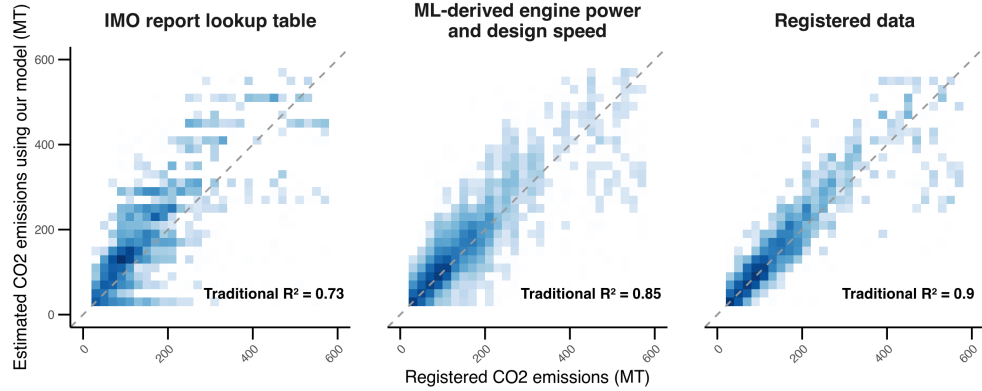


Fig. 6 Comparison of daily vessel-level CO₂ emissions estimated using our model with registered emissions from a proprietary vessel database (based on main engine fuel consumption over 24 hours). Estimates are shown using main engine power and design speed from IMO (left), our machine learning random forest models (center), and the same registered dataset (right). The dashed line represents the 1:1 line. Performance values for traditional R^2 are shown for each set of vessel characteristics. Over 15,000 vessels are represented in the figure, and blue shading indicates point density. All data are rule-of-three compliant, meaning each cell (20×20 mt) displayed contains at least three vessels.

(1) filter our trip-level emissions data to include only trips that begin and/or end in EEA countries; (2) aggregate these trip-level emissions to annual emissions for each vessel; (3) apply varying matching tolerance margins to the MRV-reported annual hours at sea and select vessels-year from our dataset within these limits, in order to ensure that the trips we use for comparison have a total hours at sea roughly equal to that reported in the MRV dataset; and (4) compare annual MRV emissions with the annual sum of our trip-level emissions for the same vessel and year to evaluate model performance.

Table 1 Performance summary across different thresholds for the hours at sea tolerance between datasets. Results are shown using the R^2 (measuring linear association between observed and predicted values) and traditional R^2 (proportion of variance explained compared to the mean-only baseline).

Tolerance	Number of observations	R^2	Traditional R^2
0.05	11,621	0.84	0.83
0.10	20,432	0.85	0.84
0.15	27,657	0.85	0.84
0.20	33,753	0.85	0.83
0.25	38,980	0.84	0.82
0.30	43,958	0.83	0.81
0.35	48,399	0.83	0.80
0.40	52,490	0.82	0.79

In Table 1, performance peaks at a traditional R^2 of 0.84 for vessels whose times at sea in our dataset are within 15% of the MRV-reported values (validation sample of 27,657 vessel-year observations, shown in Fig. 7). Beyond this threshold, performance declines as discrepancies grow between our annual aggregated trip activity and that recorded in the MRV dataset.

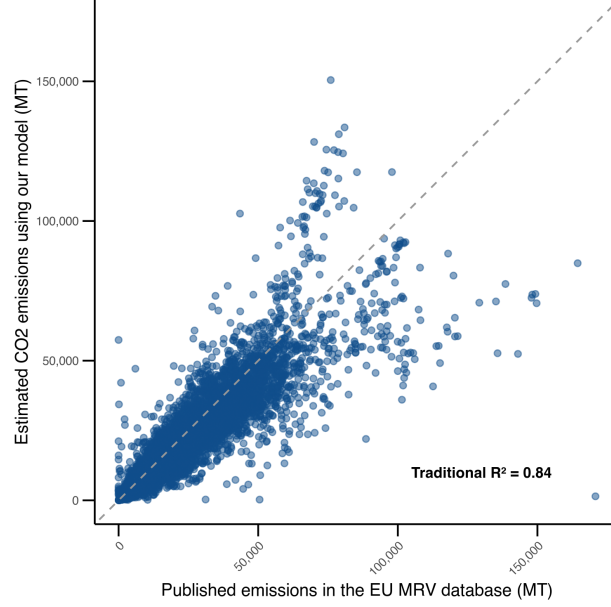


Fig. 7 Comparison of annual vessel-level CO₂ emissions estimated using our model with published emissions from the EU MRV database. The sample includes 27,657 observations from yearly aggregated vessel trips where hours at sea differ by no more than 15% between our data and the MRV dataset. The dashed line represents the 1:1 line.

4.2 Non-broadcasting vessels

4.2.1 Sentinel-1 SAR data description

For our non-broadcasting vessel emission estimates, we leverage GFW vessel detections from Sentinel-1 (S1) Synthetic Aperture Radar (SAR) imagery [6]. For each vessel detection, the dataset provides its inferred vessel length and likelihood of being classified as either a fishing vessel or non-fishing vessel. By spatiotemporally matching AIS-broadcasting vessels to S1 detections, the dataset also categorizes each vessel detection as either an AIS-broadcasting vessel ('matched') or a non-broadcasting vessel ('unmatched') [6, 42]. The detection-AIS matching utilizes a probabilistic framework that produces matching scores representing the likelihood of detection-AIS pairs being correct, and global threshold of the matching score [6, 42]. The matching considers AIS positions before and after the detection timestamp up to 12 hours away, and larger AIS position gaps typically result in lower matching scores with candidate detections. Therefore, a non-broadcasting vessel in this study detection may correspond to either a vessel not equipped with AIS or a vessel equipped with AIS but experiencing a transmission gap. The transmission gap can be intentional or a result of poor AIS reception. The data filtering involved in the GFW vessel detections include removing detections within a 1-km buffer from the shoreline to reduce false positives from land-based objects due to inaccurately defined and dynamic coastlines, repeated detections

of potentially fixed objects such as offshore infrastructure, detections from icy areas, moving vehicles close to shore, and other radar ambiguities [6]. In addition, we also remove detections from known vessel anchorages [34]. Vessels detected at anchorage may or may not be active and actually emitting, so we limit our non-broadcasting vessel emissions estimation to focus only on those vessels that are away from anchorage and actively operating in the open water. As a result, we used a dataset containing 31.9 million vessel detections from 2017 to 2025, of which 14.9 million detections were not matched to a broadcasting vessel ('unmatched'). These detections come from across more than 924,000 individual S1 scenes ((Fig. 8, Fig. 11).

4.2.2 Overview of non-broadcasting vessel emissions modeling framework

A common limitation in SAR-derived inventories is the sparse temporal coverage of Low Earth Orbit satellites, where dynamic vessel movements are inadequately captured by discrete, fixed-time overpasses. To directly addresses this spatial-temporal mismatch, we avoid assuming a steady-state temporal vessel activities from snapshots of imagery. Instead, we estimate emissions from non-broadcasting vessels using a multi-model framework that uses a combination of AIS-derived emissions, S1 vessel detections, and additional spatiotemporal model features. Let's first consider the case of modeling non-broadcasting CO₂ emissions. For the first model, we build a regression random forest to predict AIS-broadcasting CO₂ emissions using matched S1 detection density and additional spatiotemporal model features (subsequently referred to as the 'emissions regression' model). We can then use this model to predict non-broadcasting CO₂ emissions using unmatched S1 detection density (i.e., detections from non-broadcasting vessels). Because this model requires data for both AIS-broadcasting emissions and S1 detections, it is trained using data from pixel-month observations that are imaged by S1 (Fig. 8). For pixel-month observations that are imaged by S1, we can use the directly observed unmatched S1 detection density for this prediction.

Not all pixel-month observations are imaged by S1, either because there were simply no S1 overpasses in a given area and month, or because a given area is completely outside the S1 footprint (Fig. 8). For these observations without S1 imaging, we interpolate unmatched detection density using a two-stage hurdle framework which is composed of two models: (1) a classification random forest that predicts the presence or absence of unmatched detections (subsequently referred to as the 'detection classification' model); and (2) a regression random forest regression to estimate the density of unmatched detections, conditional on there being any detections present (subsequently referred to as the 'detection regression' model). Therefore, for pixel-months not imaged by S1, we can use the predicted density of unmatched detections from the 'detection regression' model as an input to the 'emissions regression' model in order to predict the amount of non-broadcasting CO₂ emissions in these areas.

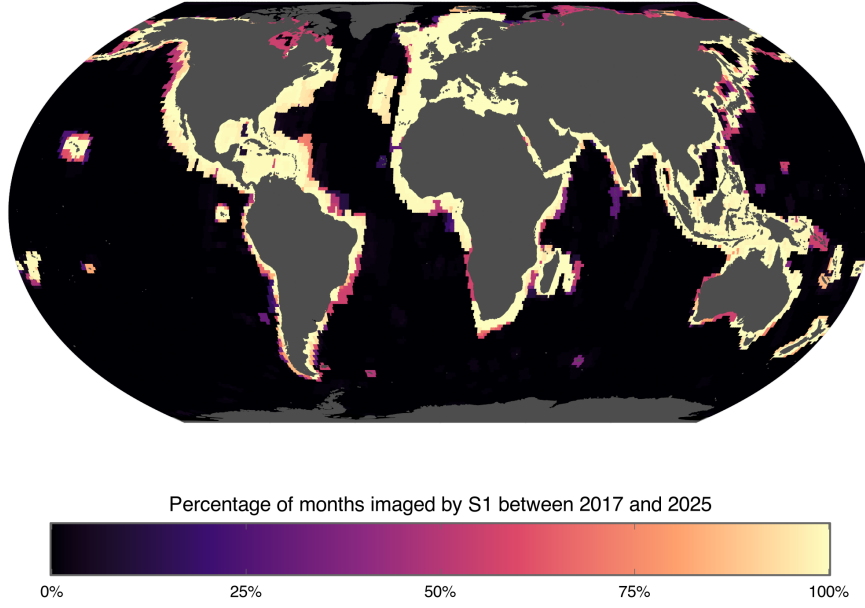


Fig. 8 Map showing the percentage of months from 2017 through 2025 that were imaged by S1 for each 1x1 degree pixel.

To reduce overall model complexity, we use a more simplified approach for estimating non-broadcasting emissions for the other eight gases (the other two GHGs and six pollutants). With our AIS-based emissions model, emissions of all gases are calculated using gas-specific emissions factors that linearly convert engine energy use (kilotowatt-hours) to emissions. Emissions of all non-CO₂ gases are therefore highly correlated with CO₂ emissions (although there are some non-linearities: different vessel types and sizes have different emissions factors for auxiliary engine and boiler energy use; we apply correction factors for some pollutants when engines are operating at very low loads; the 2020 IMO low sulphur policy affects emissions factors of some pollutants but not others). Therefore, rather than training nine separate random forest models to predict emissions for each of the nine gases, we simplify the overall modeling framework by building a single random forest to predict CO₂ emissions and then eight linear regression models to predict emissions of each gas from the emissions of CO₂. To account for the temporal discontinuity that occurs when the 2020 IMO low sulphur policy was implemented, we include an interaction term with a post-2020 dummy variable and CO₂ emissions such that regression coefficients are estimated separately for pre- and post-2020 periods.

$$emissions_{gas,i} = \beta_1 * emissions_{CO_2,i} * post2020_{FALSE} + \beta_2 * emissions_{CO_2,i} * post2020_{TRUE} + \epsilon_i$$

Therefore, for all non-CO₂ gases, we first predict non-broadcasting CO₂ emissions using the framework above, and then convert these emissions to non-CO₂ non-broadcasting emissions using these eight linear models.

In the sections below, we detail: 1) the unit of observation and outcome variable for each of the three models; 2) the features used in each of the three models; 3) the data pre-processing steps used for each model; 4) the performance testing procedures and results for each model; 5) the feature importance results for each model; and 6) the final prediction procedure for obtaining our global spatiotemporal non-broadcasting vessel emissions estimates.

4.2.3 Units of observation and outcome variables

Emissions regression

The observation unit for predicting emissions is the total emissions (of a given GHG or pollutant) in a given 1x1 degree ocean pixel, in a given month, for a given type of vessel (either non-fishing or fishing). For model training, we used the emissions estimated from the AIS-broadcasting model, extracting ping-level emissions from broadcasting vessels that occurred in ocean water beyond a 1 km buffer from shore, which corresponds to the spatial coverage of S1 detections processed by GFW and therefore excludes inland waterways, lakes, rivers, and ports. These ping-level emissions were aggregated to our unit of observation (by 1x1 degree pixel, month, and fishing/non-fishing vessel types), and observations with zero AIS-broadcasting emissions were excluded. Emissions for each observation are aggregated across all vessel sizes.

For the random forest to predict CO₂ emissions, in order to capture the vessel length distribution of S1 detections across different vessel sizes, for each observation we include several model features that represent the number of S1 detections in each of several vessel length bins. For non-fishing vessels, we use 10 length bins: 0-25m, 25-50m, ..., 200-225m, and 225+ m; for fishing vessels, which are typically smaller, we use five length bins: 0-25m, 25-50m, 50-75m, 75-100m, and 100+ m (Fig. 9). We use length bins because length is representative of main engine power (Fig. 10), which is directly related to emissions in our AIS-based model. We stratify by vessel type (fishing or non-fishing) because they have significantly different length size distributions (Fig. 9), and also because some fishing vessel types have fundamentally different emissions profiles than non-fishing vessels (i.e., trawlers and dredgers emit more at low speeds due to the nature of their energy-intensive gear deployment, which is reflected in their gear-specific main engine load factor adjustment.) Additionally, because emissions are highly right-skewed, we log-normalize the CO₂ emissions as the dependent variable.

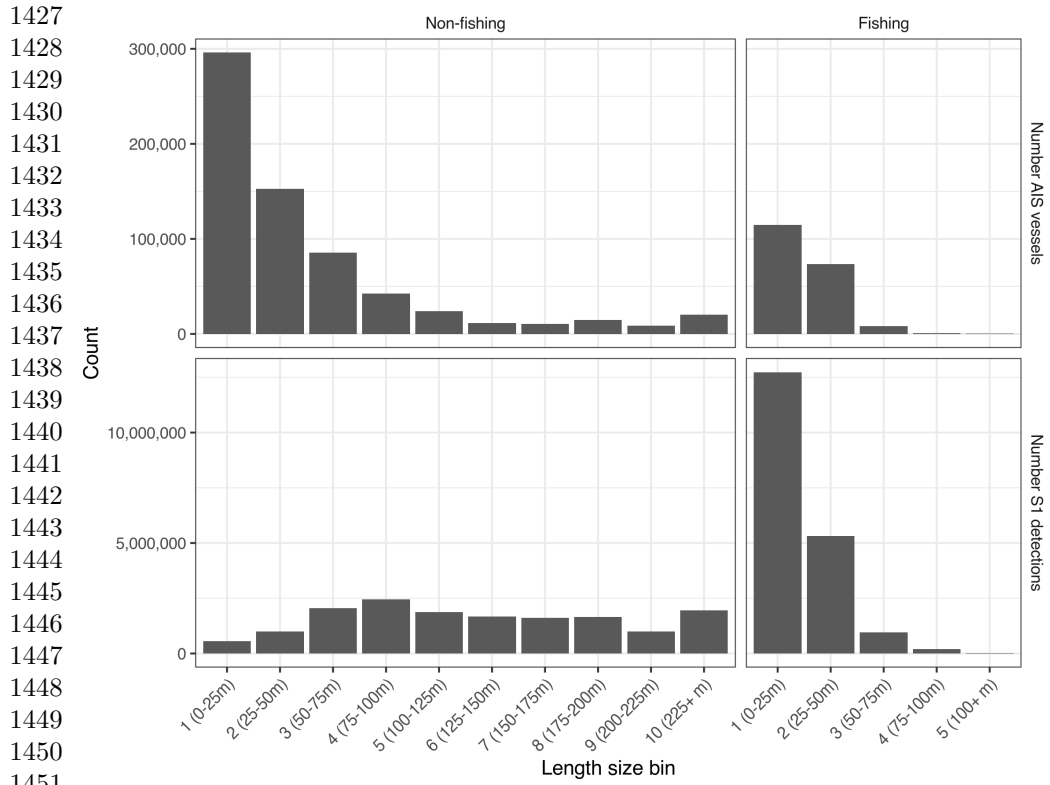


Fig. 9 Distributions showing the number of unique AIS vessels and number of S1 detections for each length size bin, disaggregated by non-fishing and fishing vessels.

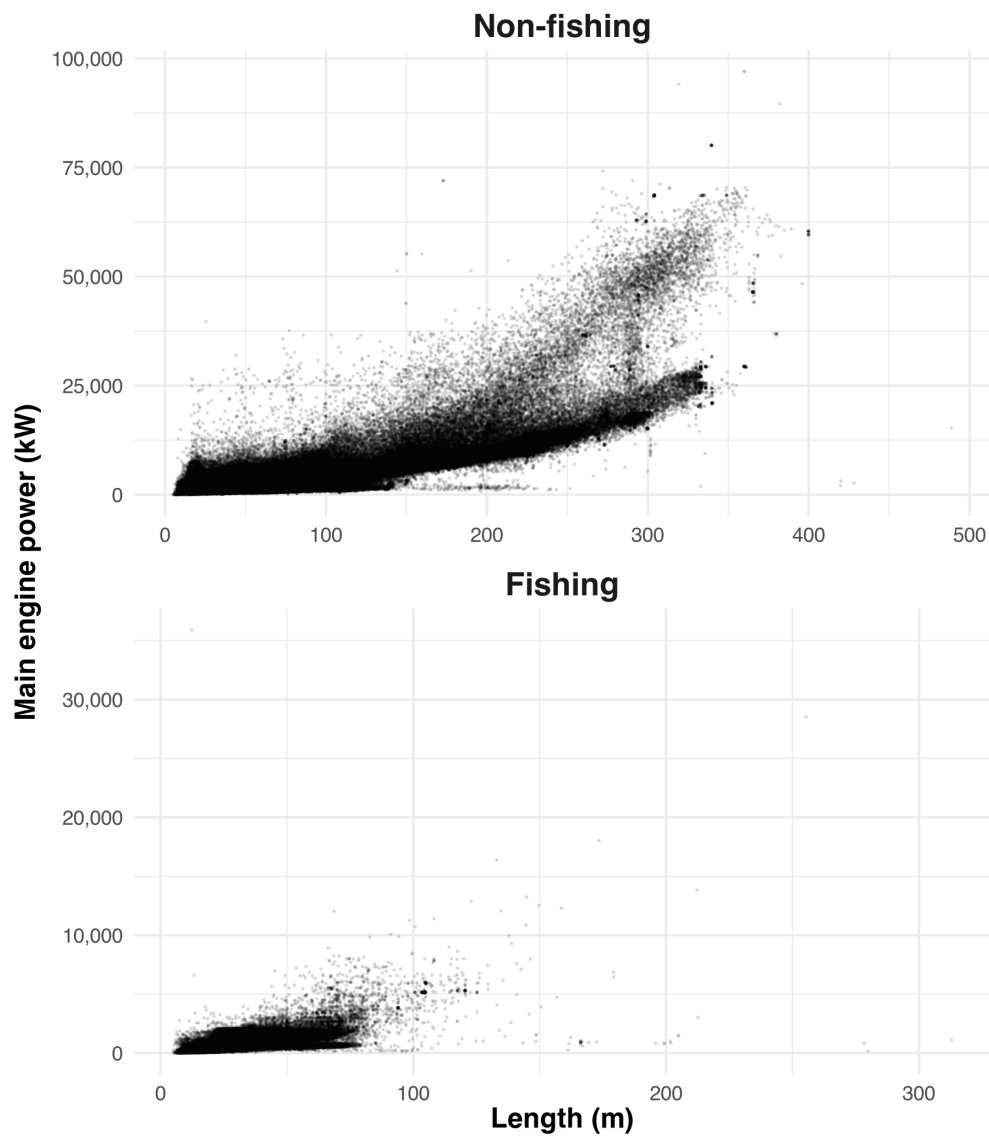


Fig. 10 Relationship between main engine power (kilotowatts) and length (meters) for all AIS-broadcasting vessels, disaggregated by non-fishing and fishing vessels.

Detection classification

Each observation unit is the boolean presence/absence of unmatched S1 detections in a given 1x1 degree ocean pixel, in a given month, for a given type of vessel (either non-fishing or fishing), for a given length bin.

1519 *Detection regression*

1520 Each observation unit is the detection density of unmatched S1 detections in a given
1521 1x1 degree ocean pixel, in a given month, for a given type of vessel (either non-fishing
1522 or fishing), for a given length bin. To calculate the detection density, we normalize
1523 detections by the total km² imaged across all S1 scenes within each pixel and month
1524 to control for variation in Sentinel-1 sampling effort.
1525

1526 **4.2.4 Model features**

1527
1528 We define a global spatial resolution of 1x1-degree which is used for all spatial model
1529 features, and use a monthly temporal units for all temporal model features. All models
1530 features used across the three models are summarized in Table 2 and described below.
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Table 2 Data sources for all included model features, including which random forest model uses the features (M1: detections classification; M2: detections regression; and/or M3: emissions regression), the units, feature type, whether it varies spatially, whether it varies temporally, and the source name and reference citation.

Model feature	Category	M1	M2	M3	Units	Type	Spatial	Temporal	Source	Ref.
Presence/absence of unmatched S1 detections	Dependent variable	Yes	No	No	-	Boolean	Yes	Yes	GFVW	[6]
Unmatched S1 detections per km ² imaged	Dependent variable	No	Yes	No	detects/km ²	Numeric	Yes	Yes	GFVW	[6]
Log(AIS emissions)	Dependent variable	No	No	Yes	log(MT)	Numeric	Yes	Yes	GFVW	-
Vessel type (fishing or non-fishing)	Dependent variable	Yes	Yes	Yes	-	Boolean	Yes	Yes	GFVW	[6]
Lower threshold of vessel length bin	S1 detection features	Yes	Yes	No	m	Numeric	No	No	GFVW	[6]
Vessel length bin category	S1 detection features	Yes	Yes	No	-	Factor	Yes	Yes	GFVW	[6]
S1 detections per km ² of area for each length bin	S1 detection features	No	No	Yes	detects/km ²	Numeric	Yes	Yes	GFVW	[6]
Total pixel area imaged by S1 across all time	S1 detection features	Yes	Yes	No	km ²	Numeric	Yes	No	GFVW	[6]
Unique AIS-broadcasting vessels in pixel-month	AIS reception quality	Yes	Yes	No	-	Numeric	Yes	Yes	GFVW	[12]
Unique AIS-broadcasting vessels globally	AIS reception quality	Yes	Yes	No	-	Numeric	No	Yes	GFVW	[12]
Positions reported by AIS type A	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	No	GFVW	[12]
Positions reported by AIS type B	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	No	GFVW	[12]
Proportion of messages, dynamic AIS receivers	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	Yes	GFVW	[12]
Proportion of messages, terrestrial AIS receivers	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	Yes	GFVW	[12]
Proportion of messages, satellite AIS receivers	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	Yes	GFVW	[12]
Month of year	Temporal features	Yes	Yes	Yes	-	Factor	No	Yes	GFVW	-
Post-2020 IMO Sulphur policy	Temporal features	No	No	Yes	-	Boolean	No	Yes	GFVW	[39]
Ocean basin	Geographic features	Yes	Yes	Yes	-	Factor	Yes	No	GFVW	[12]
Pixel water area	Geographic features	Yes	Yes	Yes	km ²	Numeric	Yes	No	GFVW	[6]
Pixel water fraction	Geographic features	No	No	Yes	-	Numeric	Yes	No	GFVW	[6]
Mean ocean depth	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	GFVW	[12]
Mean distance from shore	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	GFVW	[12]
Mean distance from nearest port	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	GFVW	[12]
Minimum distance to top 20 port	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	GFVW	[43]
Mean wind speed	Weather features	No	Yes	No	m/s	Numeric	Yes	Yes	ERA5	[44]
Mean significant wave and swell height	Weather features	No	Yes	No	m	Numeric	Yes	Yes	ERA5	[44]
Mean wave period	Weather features	No	Yes	No	s	Numeric	Yes	Yes	ERA5	[44]
Mean sea surface temperature	Weather features	No	Yes	No	K	Numeric	Yes	Yes	ERA5	[44]
Mean sea level pressure	Weather features	No	Yes	No	Pa	Numeric	Yes	Yes	ERA5	[44]
Total precipitation	Weather features	No	Yes	No	m	Numeric	Yes	Yes	ERA5	[44]

1611 *S1 vessel detection features*

1612 For the emissions regression model, we include several model features that capture the
1613 vessel length frequency distribution of S1 detections that occurred in any given pixel-
1614 month for each vessel type (fishing or non-fishing). This prevents assigning emissions
1615 to vessels for which no emission estimates are available. S1 detections smaller than
1616 the minimum vessel size represented in the AIS-broadcasting dataset are excluded to
1617 avoid assigning emissions to vessel sizes not characterized by the AIS emissions model.
1618 We normalize the detections by the total km² imaged across all S1 scenes within
1619 each pixel-month to account for spatiotemporal variation in Sentinel-1 sampling effort
1620 and therefore create a measure of detection density (Fig. 8 shows spatial variation of
1621 sampling effort; Fig. 11 shows temporal variation). The emissions regression model is
1622 trained with these features corresponding to matched detection density in order to
1623 predict AIS-broadcasting emissions; when it comes time for using the model to predict
1624 non-broadcasting emissions, the unmatched detection density is used for these features
1625 instead.

1626 For the detection classification and detection regression models, each unit of obser-
1627 vation represents one of the vessel length bins for each pixel-month and vessel type.
1628 We also include two model features describing the length bin for each observation: the
1629 numeric lower threshold for the vessel length bin, and a categorical factor represent-
1630 ing which of the length bins the observation belongs to. We finally include for each
1631 pixel the total area imaged by S1 across the entire 2017-2025 time period in order to
1632 account for the general amount each pixel is imaged.

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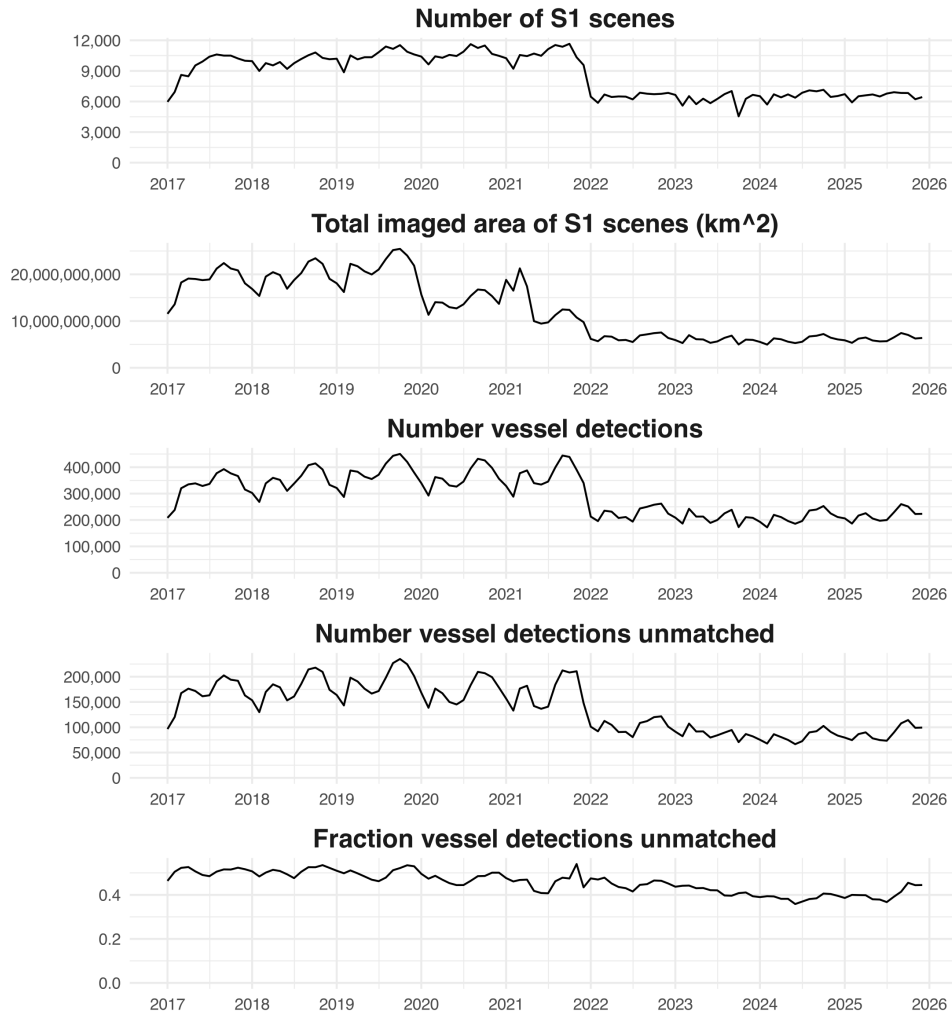


Fig. 11 Monthly statistics of global S1 coverage including total number of S1 scenes, total imaged area across all S1 scenes, total number of vessel detections, total number of vessel detections that could not be matched to an AIS-broadcasting vessels, and fraction of the total number of vessel detections that could not be matched to an AIS-broadcasting vessels.

Spatial features

For each pixel we extract spatial information, including depth, distance to shore, and distance to the nearest port, from a high-resolution GFW subproduct (0.01x0.01-degree) and aggregate it to our 1x1-degree grid resolution by computing the minimum value in each pixel. In addition to the distance from the nearest port (which can be of any size port), we also include the distance from each pixel centroid to the nearest

of the top 20 container ports according to the Lloyd’s list of 2024 [43]. We include this since traffic density, loitering practices, and navigation behavior in major ports differ from those in regular ports. Additional geographic features include the total area covered by water within each pixel, the fraction of area within each pixel covered by water, and a categorical variable indicating which of the five major ocean basins each pixel falls within.

Weather features

Weather influences both vessel activity as well as the ability for SAR to detect vessels. Storms or rough sea conditions may cause vessels, particularly smaller vessels, to stay in port during storms or avoid certain areas [45]. Sea surface temperature (SST) impacts the distribution of fish species, therefore impacting the spatiotemporal distribution of fishing vessels [46]. Weather and sea surface state also influences S1’s ability to detect vessels. Sea surface waves generated by wind modify SAR background signal, directly affecting the ability of the S1 detection model to accurately identify vessels [47]. Heavy rain can also affect sea surface scatter and background cluttering [48], as can swell and wave height [49], affecting the ability of SAR to detect vessels. To account for these effects, we extracted weather data from ECMWF Reanalysis v5 (ERA5), the fifth-generation atmospheric reanalysis of the global climate, produced by the Copernicus Climate Change Service [44]. We used Google Earth Engine to extract hourly ERA5 data and aggregate it to monthly average or total values for the ocean area in each of our 1x1 degree pixels. As our final model features that were used in both the detection classification and detection regression models, we used mean wind speed (which may influence vessel activity and S1 detection), total precipitation (which may influence vessel activity), mean sea surface temperature (which may influence vessel activity, particularly for fishing vessels), mean sea level surface pressure (which may influence vessel activity), mean wave and swell height which may influence vessel activity), and mean wave period (which may influence vessel activity).

AIS reception quality features

We included several model features related to AIS reception quality and signal type. To account for reception quality, we used two features: messages per day for AIS Type A and Type B transponders, capturing regional variation in AIS reception. Additionally, because AIS signals are received via different systems (terrestrial receivers, satellite receivers, and more recently dynamic receivers) we identified the associated receiver type for each ping and calculated the proportion of each type per pixel and month, summing to one. These features describing AIS coverage and the predominance of receiver types influence the fraction of broadcasting versus non-broadcasting vessels, and thus the associated emissions and detections.

Temporal features

To account for temporal variability in each of the three models, we include a factor feature representing the month of the year. We also include a boolean feature for whether or not the observation occurs on or after January 2020, to coincide with the start of the IMO low sulphur fuel policy. To account for temporal trends of general vessel activity

in the detection classification and regression models, we include features characterizing the total number of unique AIS-broadcasting vessels globally each month, as well as the number of unique AIS-broadcasting vessels in each pixel-month. Accounting for temporal variation is particularly important given both the variation in S1 coverage over time, as well as the variation in global fleet composition over time (Fig. 11).

4.2.5 Data pre-processing

We perform the following data pre-processing steps, in this order, each time a model is trained or predictions are inferred. In order to avoid data leakage during performance testing, data pre-processing that performs calculations that directly use the data (i.e., imputation of missing values) is always performed on the training data. To do so, we leverage the recipes package in R [50].

1. We treat several features as categorical (fishing or non-fishing; the post-2020 temporal boolean; ocean; and for the detection classification and regression models that are disaggregated by vessel length bin, we treat the length bin as categorical)
2. We remove any numeric model features that have near-zero variance
3. For any missing values of numeric model features, we impute their value using the median of that feature’s non-missing values.
4. We remove any numeric model features that are highly correlated with other features (i.e., that have a Pearson’s correlation coefficient of > 0.9).

4.2.6 Model training

We defined three independent Random Forest models using the tidymodels framework in R [51]. Model fitting was done using the ranger engine [52] in parsnip [53]. We use 500 trees for each forest, and otherwise use default hyperparameter settings for ranger. We perform several out-of-sample performance tests for each model using independent training and testing data splits to assess how well it does for predictions inside the S1 footprint, as well as a simulation of how well it would do for predictions outside the footprint. We train each of the three final models using the entire dataset.

For the detection presence/absence classification model, we use an additional validation dataset to optimize the classification cutoff threshold to maximize the F1 score (the harmonic mean of precision and recall). So for performance tests, we train the model on a training dataset, calibrate the classification threshold on the validation dataset, and then test performance on the testing dataset using the optimized threshold. For the final model, we train the model using a training dataset, calibrate the classification threshold on a validation dataset, and then train a final model using the entire dataset and the optimized threshold.

For the eight linear regression models to convert CO₂ emissions to emissions of other GHGs and pollutants, we use the lm function from the base R stats package [54].

4.2.7 Model testing

We assessed model performance for each of the three models using two different evaluation approaches designed to test: 1) predictive power for areas within the S1 footprint but that don’t have overpasses in a given month (subsequently referred to as our ‘inside

footprint' test); and 2) the models' ability to generalize to locations completely outside the S1 footprint (subsequently referred to as our 'simulated outside footprint' test).

Inside S1 footprint testing

Because both emissions and detection predictions inside the S1 footprint are made for locations where emissions and detections data is available for training, (i.e., inside the footprint we are not generalizing to regions with unknown emissions and detections), we first evaluated performance inside the footprint using a stratified random 80%/20% training/testing split. For each of the three models, we stratify the random split by the dependent variable (presence/absence of unmatched detections for the detections classification model; density of unmatched detections for the detections regression model; and log-transformed emissions for the emissions regression model).

For the detection presence/absence classification model, we use an additional validation dataset to optimize the classification cutoff threshold. In this case, we use a 60%/20%/20% training/ validation/ testing split.

Simulated outside S1 footprint testing

Then, to assess spatial transferability when predicting emissions in areas completely outside the S1 footprint (i.e., areas far from shore), we implement a spatially structured split, training on nearshore pixels ($\leq 100\text{km}$ from shore) and testing on offshore pixels ($> 100\text{km}$ from shore) (Fig. 12).

For the detection presence/absence classification model, we again use an additional validation dataset to optimize the classification cutoff threshold. We therefore use data from nearshore pixels to create a stratified random 80%/20% training/validation split, and then test using the offshore pixels.

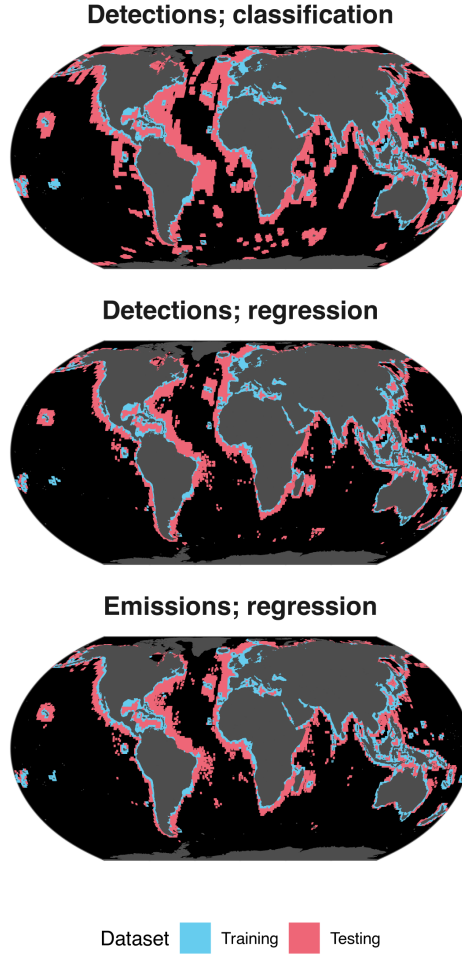


Fig. 12 Maps of pixels that were used in the training and testing datasets for the simulated outside S1 footprint performance tests for each model. Pixels $\leq 100\text{km}$ from the shore were used for training; pixels $> 100\text{km}$ from shore were used for testing.

Classifier performance is reported using precision-recall curves (Fig. 13), receiver operating characteristic (ROC) curves (Fig. 14), and confusion matrices (Fig. 15). Table 3 also summarizes regression performance for detections, both inside and outside the footprint, alongside the emissions model performance. To assess the performance of the emissions model, predicted log emissions were back-transformed using Duan's smearing coefficient, calculated from the residuals of the training data [55]. The same transformation was applied to the final model prediction.

For the CO_2 emissions regression random forest, we simply train the model on the training data and test the model on the testing data (using the two sets of splits

described above for inside and outside the S1 footprint tests). For the other eight gases, we: 1) train the linear regression using the training data (to convert CO₂ emissions to emissions of each gas); 2) use the CO₂ emissions regression random forest to predict emissions for the testing data; and then 3) use the predicted CO₂ emissions for the testing data and then convert this to emissions for each of the other eight gases using the eight linear regressions built in step 2; and then 4) compare the predicted emissions of each of the eight gases against the observed emissions of each of the eight gases in the training data. In this way, we are able to measure fully out-of-sample performance for predicting not only CO₂, but also all of the other two GHGs and six pollutants.

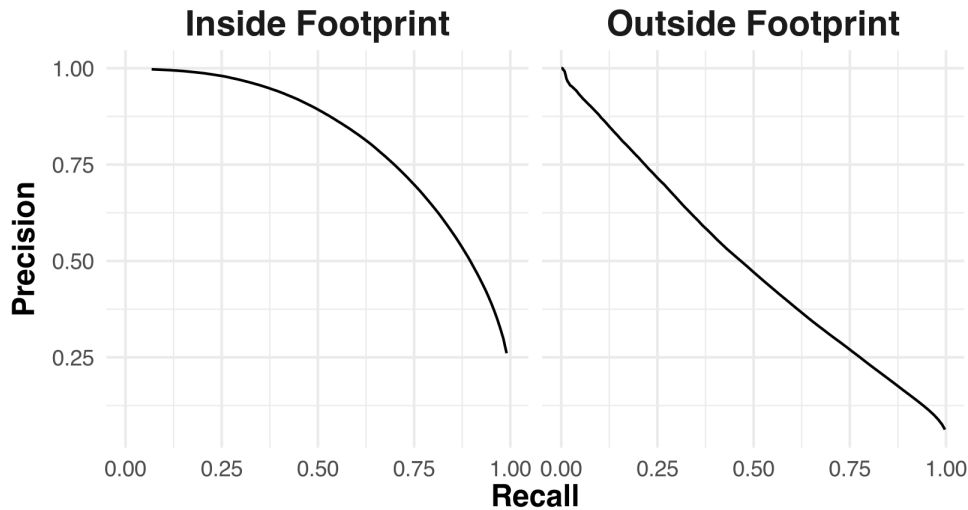


Fig. 13 Precision-recall curves of the detection presence/absence classification model, shown for both the inside S1 footprint test and simulated outside S1 footprint test.

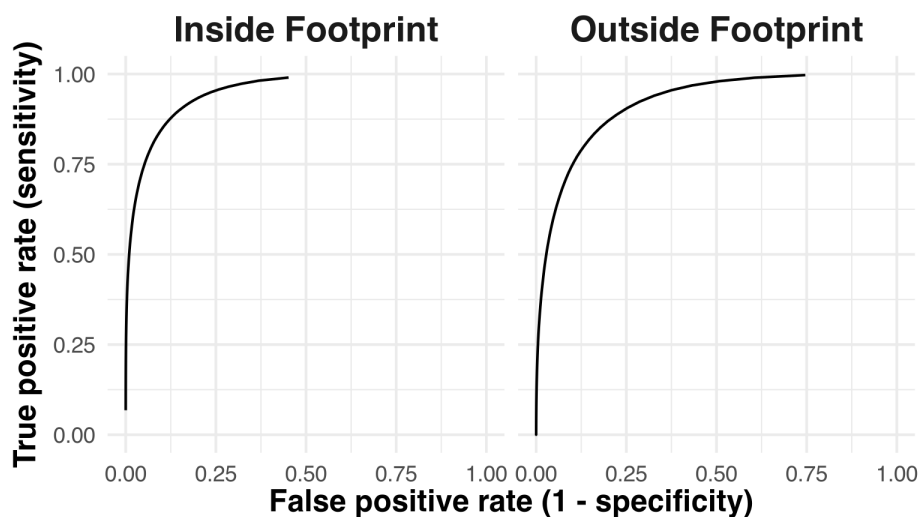


Fig. 14 Receiver Operating Characteristic (ROC) curves of the detection classification model, shown for both the inside S1 footprint test and simulated outside S1 footprint test.

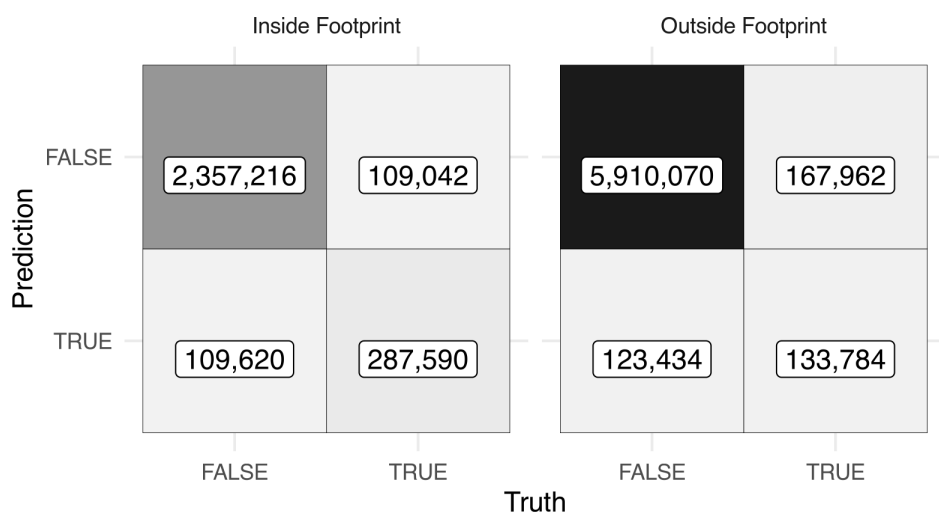


Fig. 15 Confusion matrices of the detection classification model, shown for both the inside S1 footprint test and simulated outside S1 footprint test.

Table 3 Summary of dark fleet model performance metrics for the: 1) emissions regression model; 2) detections classification mode; and 3) detections regression model. Performance estimates for each metric are shown for both the inside S1 footprint and outside S1 footprint tests. All metrics are calculated at the unit of observation except for 'Total percent error', which is the percentage error in the total sum of all predictions relative to the total sum of all truth values.

Model	Pollutant	Metric	Inside footprint	Outside footprint
Emissions regression	CO2	Traditional R ²	0.94	0.73
Emissions regression	CH4	Traditional R ²	0.90	0.63
Emissions regression	N2O	Traditional R ²	0.94	0.71
Emissions regression	CO	Traditional R ²	0.92	0.63
Emissions regression	NOX	Traditional R ²	0.87	0.57
Emissions regression	SOX	Traditional R ²	0.95	0.73
Emissions regression	PM2.5	Traditional R ²	0.94	0.65
Emissions regression	PM10	Traditional R ²	0.94	0.65
Emissions regression	VOCs	Traditional R ²	0.88	0.60
Emissions regression	CO2	Total percent error	-8.61	-10.08
Emissions regression	CH4	Total percent error	-20.26	-30.01
Emissions regression	N2O	Total percent error	-10.70	-15.51
Emissions regression	CO	Total percent error	-19.10	-29.29
Emissions regression	NOX	Total percent error	-23.68	-36.44
Emissions regression	SOX	Total percent error	-8.23	-9.01
Emissions regression	PM2.5	Total percent error	-16.21	-25.52
Emissions regression	PM10	Total percent error	-16.21	-25.52
Emissions regression	VOCs	Total percent error	-22.23	-32.45
Detections classification	-	Accuracy	0.92	0.95
Detections classification	-	F1 score	0.72	0.48
Detections classification	-	Precision	0.72	0.52
Detections classification	-	Precision-recall AUC	0.81	0.49
Detections classification	-	Recall	0.73	0.44
Detections classification	-	ROC AUC	0.95	0.92
Detections classification	-	Sensitivity	0.73	0.44
Detections classification	-	Specificity	0.96	0.98
Detections regression	-	Traditional R ²	0.73	0.46
Detections regression	-	Total percent error	1.18	28.46

4.2.8 Feature importance

After assessing model performance, we perform a final fit using the entire dataset to assess feature importance for the three non-broadcasting emissions random forest models (Fig 16). Feature importance was quantified using the bias-corrected Gini index [56], as implemented for random forests in the ranger R package [52]. This metric quantifies how much each model feature reduces node impurity when it is used to make splits in the random forest's decision trees.

4.2.9 Linear regression fit statistics for non-gas models

Here we look at the linear regression fit statistics for the final models to convert CO₂ emissions to emissions of non-CO₂ gases (other GHGs and non-GHG pollutants), trained using the full dataset (Table 4).



Fig. 16 Feature importance plots for the: 1) detections (presence/absence) classification model; 2) detections regression model; and 3) CO₂ emissions regression model.

Table 4 Linear regression fit statistics for the final models trained to estimate emissions of non-CO₂ gases from CO₂ emissions. The model for each gas is a zero-intercept linear regression with 5 terms: CO₂ emissions interacted with each combination of period (pre/post-2020 IMO sulphur policy) and vessel type (fishing/non-fishing), plus CO₂ emissions interacted with the log of mean distance from shore.

Pollutant	Term	Estimate	SE	T-stat.	p-value
CH4	CO2 : log(distance from shore + 1)	1.25e-06	3.29e-09	379.9	<0.001
CH4	CO2 : pre-2020 : non-fishing	-1.21e-06	3.17e-08	-38.0	<0.001
CH4	CO2 : post-2020 : non-fishing	-1.92e-06	3.14e-08	-61.3	<0.001
CH4	CO2 : pre-2020 : fishing	3.69e-06	8.42e-08	43.9	<0.001
CH4	CO2 : post-2020 : fishing	3.73e-06	4.52e-08	82.6	<0.001
N2O	CO2 : log(distance from shore + 1)	7.13e-07	2.32e-09	308.0	<0.001
N2O	CO2 : pre-2020 : non-fishing	4.00e-05	2.23e-08	1869.4	<0.001
N2O	CO2 : post-2020 : non-fishing	4.00e-05	2.21e-08	1865.7	<0.001
N2O	CO2 : pre-2020 : fishing	4.00e-05	5.92e-08	727.0	<0.001
N2O	CO2 : post-2020 : fishing	4.00e-05	3.18e-08	1379.9	<0.001
CO	CO2 : log(distance from shore + 1)	6.00e-05	1.57e-07	382.3	<0.001
CO	CO2 : pre-2020 : non-fishing	3.00e-05	1.51e-06	19.2	<0.001
CO	CO2 : post-2020 : non-fishing	-5.49e-06	1.50e-06	-3.7	<0.001
CO	CO2 : pre-2020 : fishing	2.70e-04	4.02e-06	67.9	<0.001
CO	CO2 : post-2020 : fishing	2.80e-04	2.16e-06	128.1	<0.001
NOX	CO2 : log(distance from shore + 1)	2.02e-03	4.33e-06	466.7	<0.001
NOX	CO2 : pre-2020 : non-fishing	-7.34e-03	4.00e-05	-176.3	<0.001
NOX	CO2 : post-2020 : non-fishing	-8.45e-03	4.00e-05	-205.2	<0.001
NOX	CO2 : pre-2020 : fishing	-9.70e-04	1.10e-04	-8.8	<0.001
NOX	CO2 : post-2020 : fishing	-2.14e-03	6.00e-05	-36.1	<0.001
SOX	CO2 : log(distance from shore + 1)	-1.00e-05	7.15e-08	-173.1	<0.001
SOX	CO2 : pre-2020 : non-fishing	6.30e-03	6.88e-07	9164.2	<0.001
SOX	CO2 : post-2020 : non-fishing	1.66e-03	6.81e-07	2441.1	<0.001
SOX	CO2 : pre-2020 : fishing	6.37e-03	1.83e-06	3486.1	<0.001
SOX	CO2 : post-2020 : fishing	1.67e-03	9.81e-07	1705.6	<0.001
PM2.5	CO2 : log(distance from shore + 1)	4.00e-05	9.11e-08	401.0	<0.001
PM2.5	CO2 : pre-2020 : non-fishing	4.10e-04	8.76e-07	472.5	<0.001
PM2.5	CO2 : post-2020 : non-fishing	5.00e-05	8.67e-07	58.1	<0.001
PM2.5	CO2 : pre-2020 : fishing	5.30e-04	2.33e-06	228.9	<0.001
PM2.5	CO2 : post-2020 : fishing	2.10e-04	1.25e-06	168.4	<0.001
PM10	CO2 : log(distance from shore + 1)	4.00e-05	9.90e-08	401.0	<0.001
PM10	CO2 : pre-2020 : non-fishing	4.50e-04	9.52e-07	472.5	<0.001
PM10	CO2 : post-2020 : non-fishing	5.00e-05	9.43e-07	58.1	<0.001
PM10	CO2 : pre-2020 : fishing	5.80e-04	2.53e-06	228.9	<0.001
PM10	CO2 : post-2020 : fishing	2.30e-04	1.36e-06	168.4	<0.001
VOCs	CO2 : log(distance from shore + 1)	1.00e-04	2.02e-07	495.7	<0.001
VOCs	CO2 : pre-2020 : non-fishing	-3.90e-04	1.94e-06	-203.0	<0.001
VOCs	CO2 : post-2020 : non-fishing	-4.40e-04	1.92e-06	-228.0	<0.001
VOCs	CO2 : pre-2020 : fishing	-2.00e-04	5.15e-06	-39.0	<0.001
VOCs	CO2 : post-2020 : fishing	-2.10e-04	2.77e-06	-75.7	<0.001

4.2.10 Predicting non-broadcasting emissions

To generate the final global estimates of non-broadcasting vessel emissions, we: 1) train all models (the three random forests and eight linear regressions) using the entire dataset of observations that have both S1 detections and AIS emissions (Fig 16, Table 4; noting that again for the presence/absence detection classification model, we use a

random stratified training/validation split to first calibrate the classification threshold before fitting the final model on all the data); 2) then applied the final trained models across all ocean pixels on a monthly basis for the entire 2017-2025 study period. The prediction workflow followed a hierarchical logic based on the availability of S1 detections:

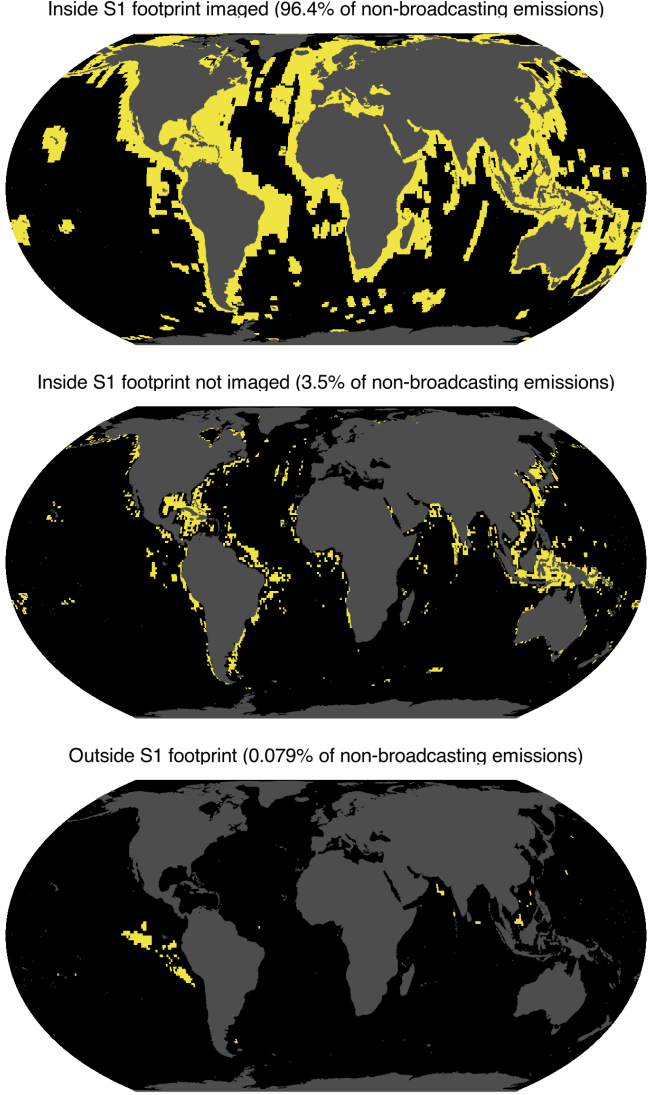
1. Direct estimation: If unmatched S1 detections are available for a given pixel-month, we use the unmatched detection density to directly predict non-broadcasting CO₂ emissions using the emissions regression model. We then use the other eight gas-specific linear regression models to convert CO₂ emissions to emissions of each of the other eight gases.
2. Indirect estimation: If unmatched S1 detections are not available for a given pixel-month, we: a) predict whether or not unmatched detections are present using the classification first stage of the hurdle model; then b) if unmatched detections are predicted to be present, we predict the density of unmatched detections using the regression second stage of the hurdle model; then c) we use the predicted density of unmatched detections in the emissions regression model to predict CO₂ emissions; and finally d) we then use the other eight gas-specific linear regression models to convert CO₂ emissions to emissions of each of the other eight gases.

The final output provides a continuous global dataset of monthly non-broadcasting emissions, for all three GHGs and six pollutants, from 2017 to 2025 at 1×1-degree resolution, disaggregated by fishing and non-fishing vessel types.

Direct estimation

For this prediction, we replaced the matched S1 detection density features for each length bin used during model training with the observed unmatched S1 detection density for each of the length bin features and separately for non-fishing/fishing vessel types. We were able to use this direct estimation approach for 96.4% of the total estimated non-broadcasting CO₂ emissions (Fig. 17). This approach predicts non-broadcasting emissions fully within the S1 footprint, and thus also represents the models with the highest predictive performance (Table 3).

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2198 **Fig. 17** Spatial coverage of the non-broadcasting emissions model. Areas are categorized according
2199 to whether they fall inside or outside the S1 footprint and whether they have been imaged by S1 at
2200 any point in time. A given pixel may be imaged in some months but not in others. Pixels outside
2201 the footprint have never been imaged. The percentage of total CO₂ emissions from non-broadcasting
2202 vessels attributed to each category is shown.

2203 *Indirect estimation*

2204 For pixel-months where S1 detections were unavailable, either because they were not
2205 imaged due to spatiotemporal satellite heterogeneity or because they fall fully outside
2206 the S1 footprint, we first used the detections classification model to predict whether
2207 detections were present or absent for each of the length bins and fishing/non-fishing
2208

vessel types. If detections were predicted, the subsequent detection regression model was used to predict the number of unmatched S1 detections per km² for each length bin and fishing/non-fishing vessel type. These predicted unmatched detection densities were then fed into the emissions regression model as the separate length-bin-related model features (for each fishing/non-fishing vessel type) to predict the resulting emissions. We do this for all nine emissions regression models corresponding to the three GHGs and six pollutants. Of the total predicted non-broadcasting CO₂ emissions, 3.5% corresponded to indirect estimations within the S1 footprint, while only 0.08% occurred entirely outside the footprint (Fig. 17).

Supplementary materials

Supplementary Text and Figures
Figures S1 to S9
Tables S1 to S7

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Supplementary Materials for

Gavin McDonald*, Pol Carbó-Mestre, Olivier Deschenes, Jennifer Bone,
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Supplementary Figures, Tables, and Text

Figures S1 to S9

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Supplementary Figures, Tables, and Text

1.3 Supplementary Figures

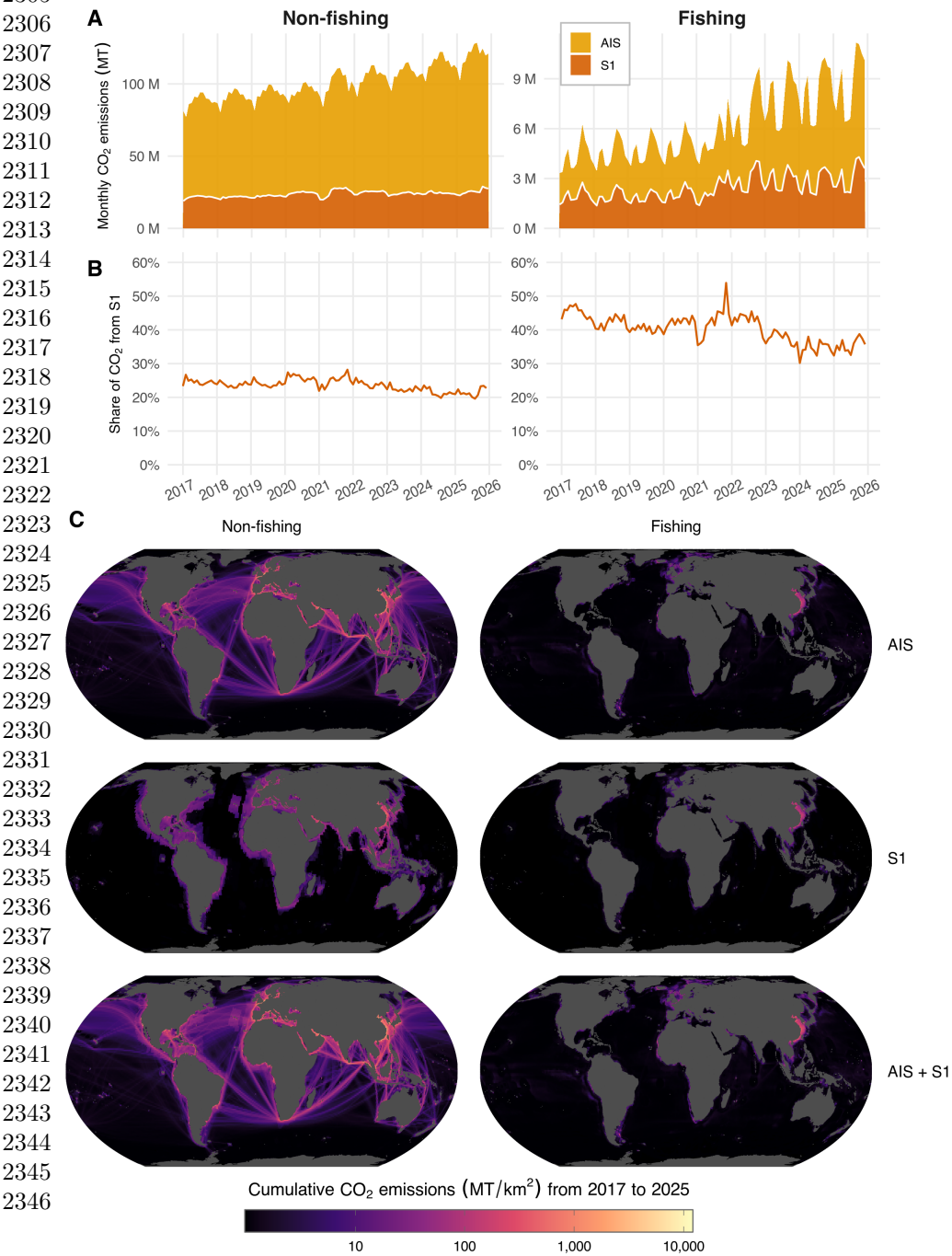


Fig. S1 (A) Time series of monthly global CO₂ emissions, shown for both AIS-broadcasting and S1-detected non-broadcasting vessels, and for both fishing and non-fishing vessels. (B) Time series of the monthly share of global CO₂ emissions attributable to S1-detected non-broadcasting vessels, for both fishing and non-fishing vessels. (C) Spatial distribution of cumulative 2017-2025 CO₂ emissions from AIS-broadcasting vessels, S1-detected non-broadcasting vessels, and AIS-broadcasting and S1-detected vessels combined (rows), for both fishing and non-fishing vessels (columns), and shown at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale).

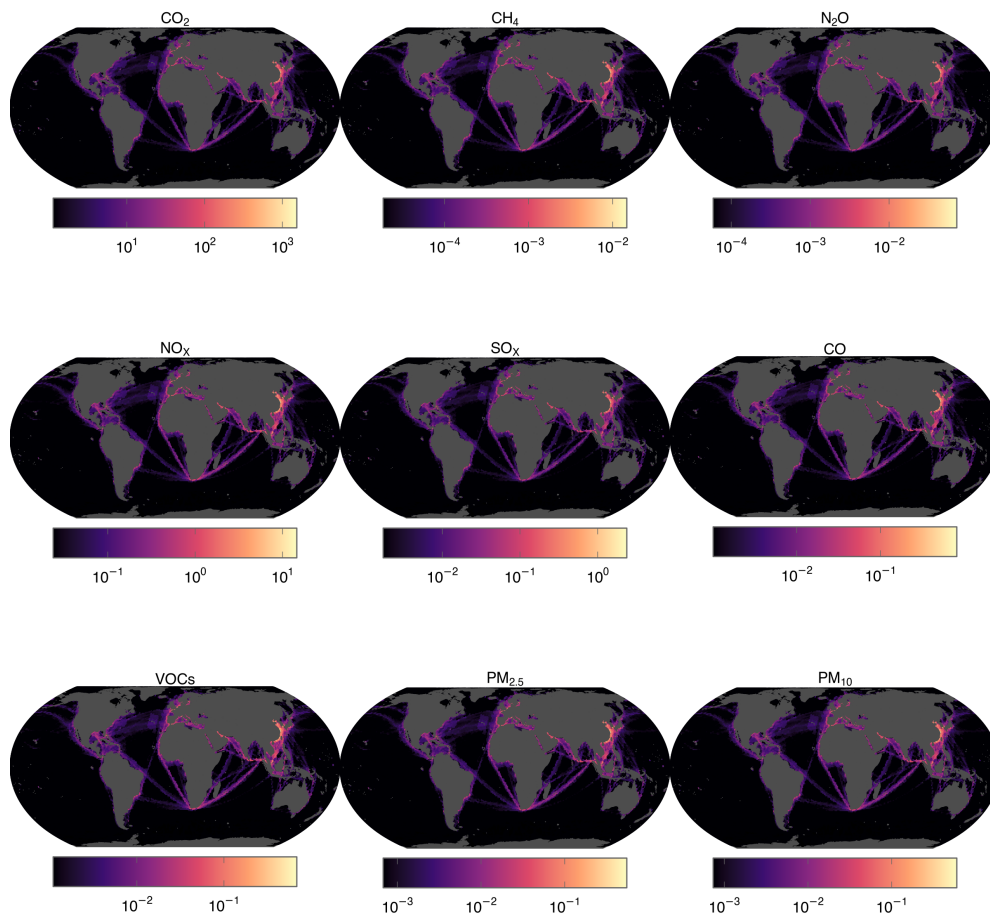


Fig. S2 Spatial maps of 2025 emissions for each GHG and pollutant, aggregated across AIS-broadcasting vessels and non-broadcasting vessels, and shown at a 1x1 degree spatial resolution. Emissions are area-normalized for each pixel (metric tonnes per square kilometer) and shown using a log-10 scale. Each panel is scaled independently, spanning the 75th percentile to the maximum of that pollutant, with lower values drawn in the darkest colour; colours are therefore not comparable between panels.

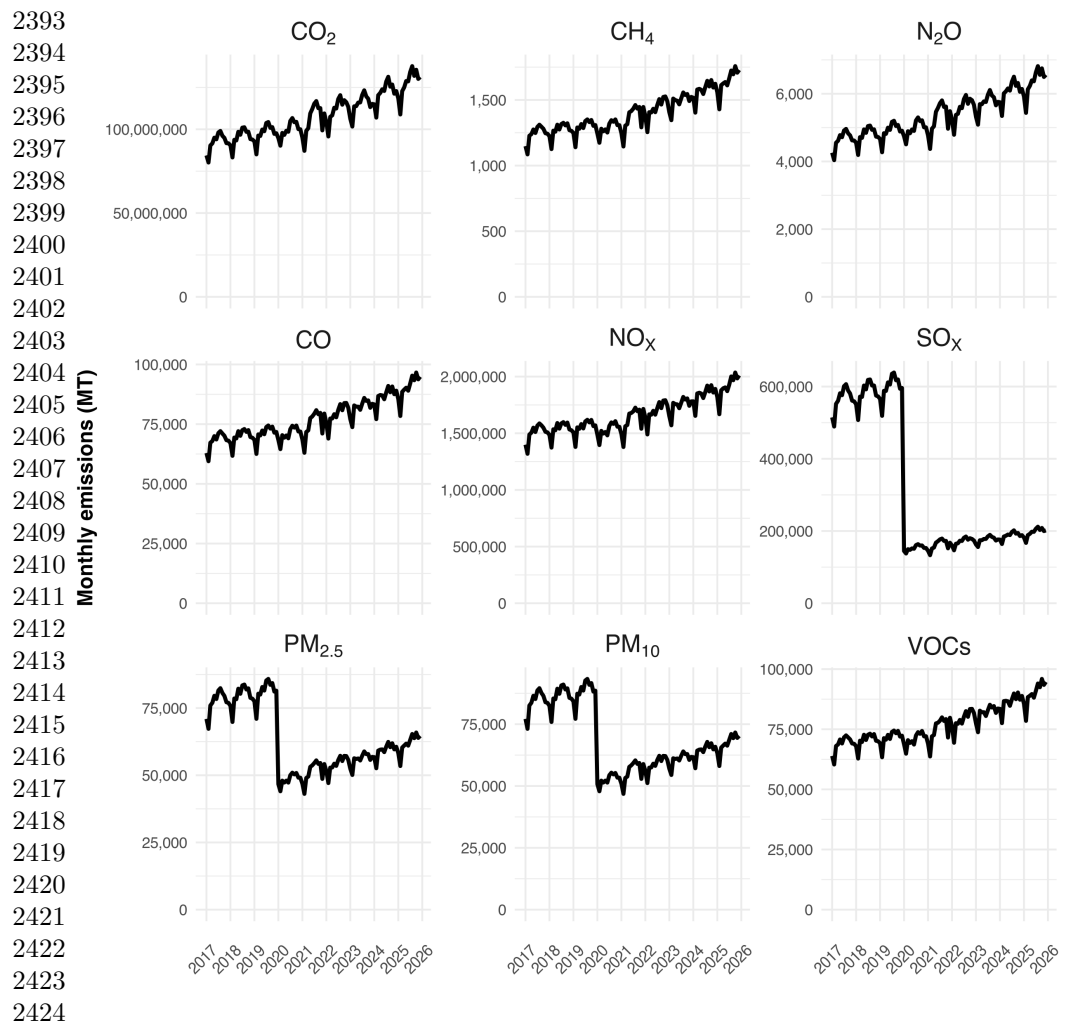


Fig. S3 Time series of monthly global emissions for each GHG and pollutant, aggregated across AIS-broadcasting vessels and non-broadcasting vessels for which we only have Sentinel-1 data.

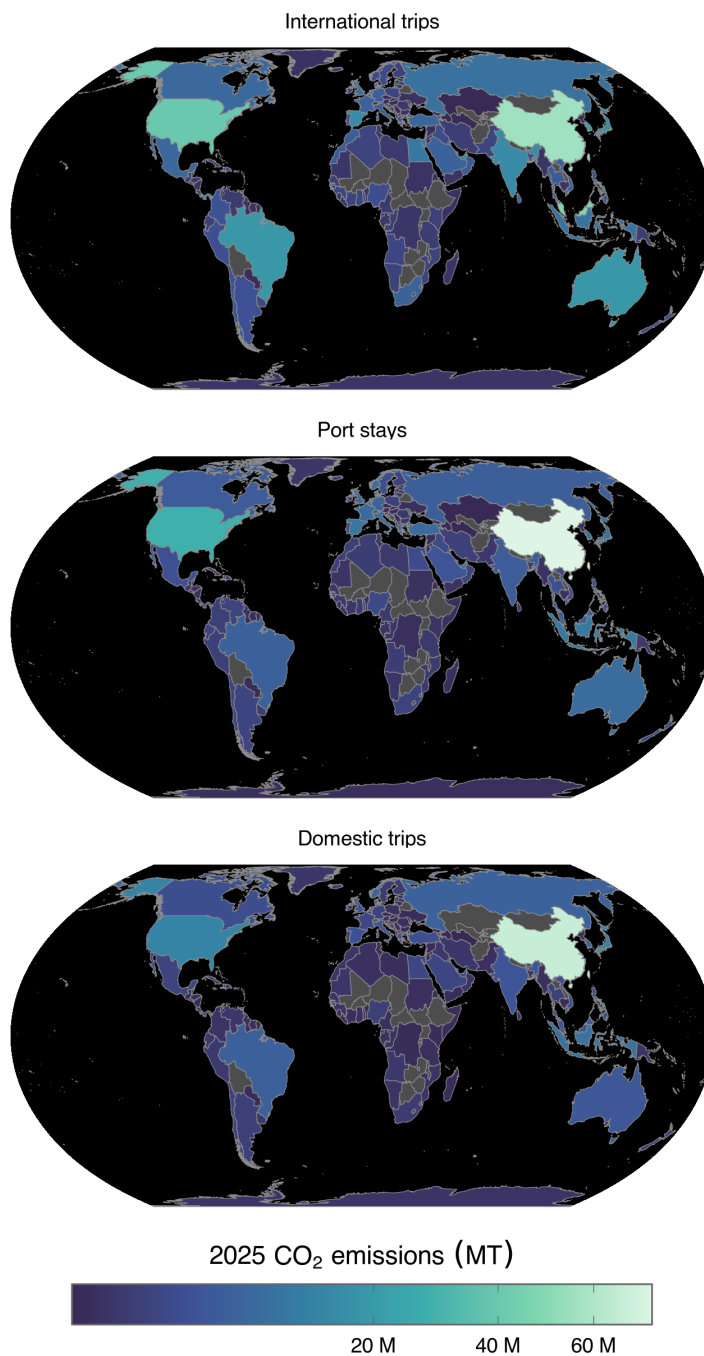


Fig. S4 CO₂ emissions by country for AIS-broadcasting vessels during different activity types (international trips, port stays, and domestic trips), aggregated for activities ending in 2025. Emissions for each international trip are partitioned equally to its departure and arrival country. The color scale is square-root transformed, so that lower-emitting countries remain distinguishable alongside the largest emitters. Countries with no values are shown in gray.

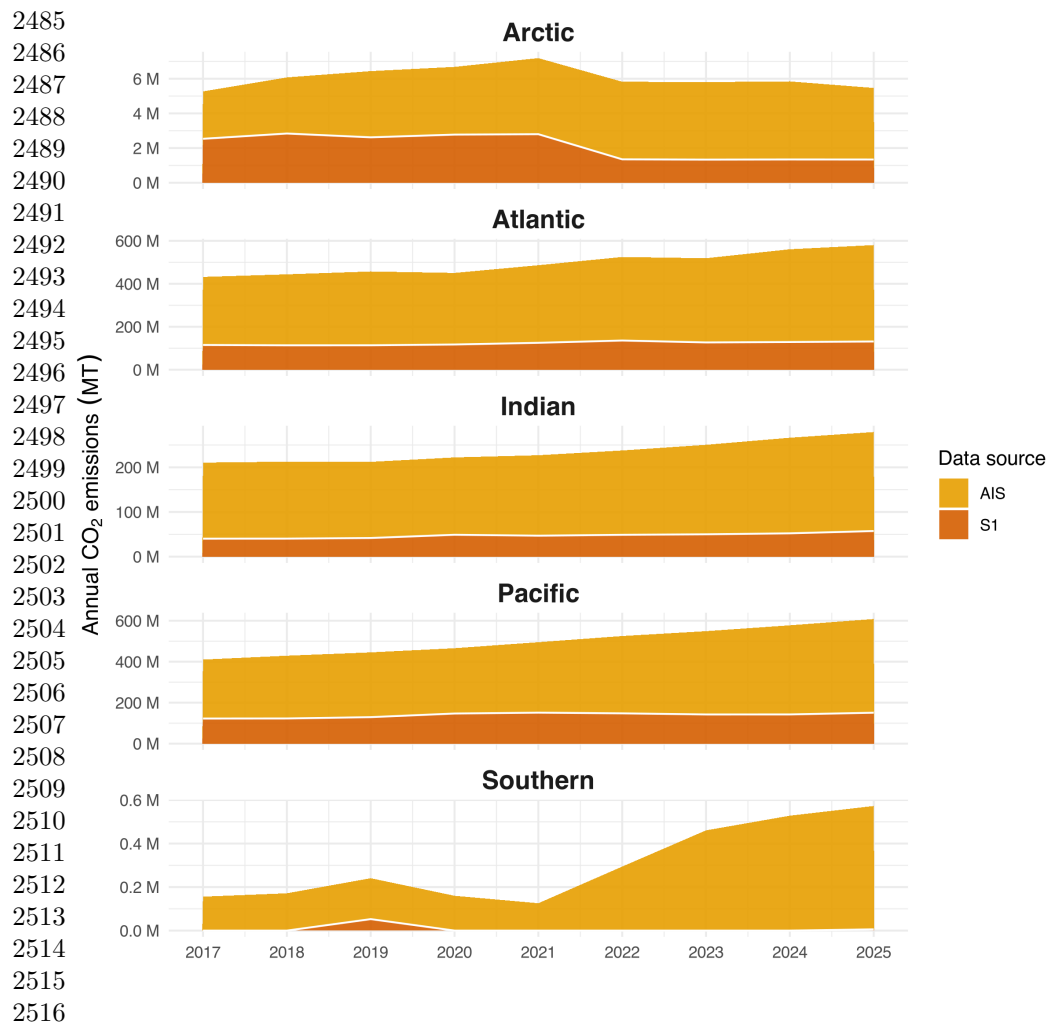


Fig. S5 Annual total CO₂ emissions by ocean, disaggregated by data source (AIS-broadcasting vessels, and non-broadcasting vessels detected by S1).

1.4 Supplementary Tables

Table S1 Summary of 2017 and 2025 emissions, data source and pollutant.

Source	Pollutant	2017 emissions, MT	2025 emissions, MT	Percent 2025 total	2017 to 2025 change
AIS	CO2	8.24e+08	1.19e+09	78%	44%
S1	CO2	2.83e+08	3.43e+08	22%	21%
AIS + S1	CO2	1.11e+09	1.53e+09	100%	38%
AIS	CH4	11,500	15,800	80%	37%
S1	CH4	3,420	3,990	20%	17%
AIS + S1	CH4	14,900	19,700	100%	33%
AIS	N2O	41,800	59,500	78%	42%
S1	N2O	13,900	16,800	22%	21%
AIS + S1	N2O	55,700	76,300	100%	37%
AIS	CO	628,000	865,000	80%	38%
S1	CO	189,000	222,000	20%	17%
AIS + S1	CO	817,000	1,090,000	100%	33%
AIS	NOX	14,100,000	18,500,000	80%	31%
S1	NOX	3,960,000	4,490,000	20%	14%
AIS + S1	NOX	18,100,000	2.3e+07	100%	27%
AIS	SOX	5,030,000	1,830,000	78%	-64%
S1	SOX	1,750,000	527,000	22%	-70%
AIS + S1	SOX	6,770,000	2,360,000	100%	-65%
AIS	PM2.5	701,000	590,000	79%	-16%
S1	PM2.5	226,000	153,000	21%	-32%
AIS + S1	PM2.5	927,000	743,000	100%	-20%
AIS	PM10	762,000	641,000	79%	-16%
S1	PM10	246,000	166,000	21%	-32%
AIS + S1	PM10	1,010,000	807,000	100%	-20%
AIS	VOCs	641,000	868,000	80%	35%
S1	VOCs	185,000	213,000	20%	15%
AIS + S1	VOCs	825,000	1,080,000	100%	31%

Table S2 Underestimation of 2025 emissions using only AIS data, and overestimation of 2017 to 2025 emissions trend using only AIS data, by pollutant.

Pollutant	Underestimation of 2025 emissions	Overestimation of 2017-2025 emissions trend
CO2	22%	15%
CH4	20%	15%
N2O	22%	15%
CO	20%	14%
NOX	20%	14%
SOX	22%	-2%
PM2.5	21%	-20%
PM10	21%	-20%
VOCs	20%	15%

2577
2578 **Table S3** For each ocean, summary of: 2025 total (AIS+S1) CO₂ emissions (metric tonnes); percentage of
2579 global total (AIS+S1) 2025 CO₂ emissions from across all oceans; percent of 2025 total (AIS+S1) CO₂
2580 emissions by non-broadcasting vessels; percent change in total (AIS+S1) CO₂ emissions from 2017 to 2025; and
2581 the percentage of all observations in the dataset (pixel-months) from 2017-2025 that were imaged by S1.

2582	Summary statistic	Atlantic	Southern	Pacific	Indian	Arctic
2583	2025 total CO2 emissions (million MT)	449	0.567	457	222	4.13
2584	Percent of total CO2 2025 emissions	39.37%	0.04%	41.28%	18.93%	0.37%
2585	Percent of total CO2 emissions by non-broadcasting	22.64%	1.02%	24.81%	20.58%	24.50%
2586	Percent change in total CO2 emissions (2017 to 2025)	34.31%	267.19%	48.27%	32.49%	3.74%
2587	Percent observations imaged by S1 (across 2017-2025)	35.94%	0.06%	17.74%	16.99%	13.55%

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2590 **Table S4** For each data source, summary of: total 2024 CO₂ emissions (metric tonnes); and percent change
2591 in CO₂ emissions from 2017 to 2024.

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2593	Data source	2024 emissions, billion MT	Change in emissions, 2017 to 2024
2594	Other transportation sectors (EDGAR)	7.44	3.35%
2595	All sectors (EDGAR)	39.6	7.06%
2596	Marine vessels (AIS + S1)	1.47	32.49%

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2600 **Table S5** For both non-fishing vessels and fishing vessels, we summarize: the total
2601 emissions (metric tonnes) of each GHG gas; the social cost associated with each GHG
2602 (USD/metric tonne); the social cost of each (USD); the total social cost summed across
2603 all three gases (USD). We use 2021 for non-fishing vessels since it is the year for which we
2604 have the total value of all traded goods at sea. We use 2022 for fishing vessels since it is
2605 the year for which we have the total value of wild capture fisheries.

2606	Year	Vessel type	GHG	Emissions (MT)	SC (USD/MT)	SC (billion USD)
2607	2021	Non-fishing	CH4	15,296	2,900	0.044
2608	2021	Non-fishing	CO2	1,205,568,569	283	341.176
2609	2021	Non-fishing	N2O	60,152	49,600	2.984
2610	Total	Non-fishing	-	-	-	344.204
2611	2022	Fishing	CH4	1,373	2,900	0.004
2612	2022	Fishing	CO2	82,662,425	283	23.393
2613	2022	Fishing	N2O	4,211	49,600	0.209
2614	Total	Fishing	-	-	-	23.606

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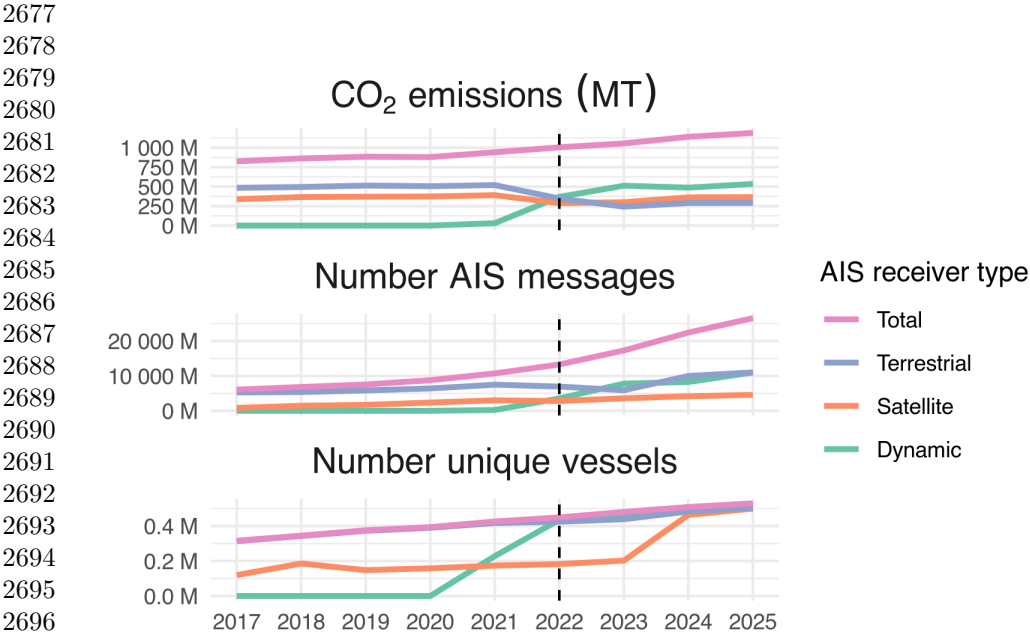
Table S6 Top 30 countries by 2025 CO₂ emissions (millions of metric tonnes) from AIS-broadcasting vessels, shown separately for each activity type, with all remaining countries pooled into a single Other row. Emissions for each international trip are partitioned equally to its departure and arrival country. These are the totals mapped in Figure S4.

International trips		Port stays		Domestic trips	
Country	CO ₂	Country	CO ₂	Country	CO ₂
China	55.5	China	74.0	China	66.8
Malaysia	48.1	United States	27.8	Japan	13.1
United States	39.9	Indonesia	10.6	United States	13.1
South Korea	20.2	Spain	9.8	Indonesia	8.8
Australia	19.1	South Korea	9.0	South Korea	5.7
Brazil	18.9	Netherlands	9.0	Russia	5.6
Spain	15.7	Japan	8.9	Norway	5.4
Japan	15.5	United Arab Emirates	8.4	Brazil	5.2
India	15.2	Norway	8.4	Australia	3.8
United Arab Emirates	14.2	Italy	8.1	Malaysia	3.4
Panama	13.8	Australia	7.6	India	3.4
Singapore	12.7	Turkey	7.4	Italy	3.2
Egypt	10.4	France	7.2	Spain	3.1
Indonesia	9.0	United Kingdom	6.7	Turkey	2.7
Russia	8.5	Germany	6.0	United Kingdom	2.6
United Kingdom	7.7	Greece	5.8	Canada	2.5
Netherlands	7.4	Brazil	5.3	Vietnam	2.3
Canada	6.9	Russia	5.0	United Arab Emirates	2.2
Mexico	6.4	India	4.8	Philippines	2.2
Turkey	6.3	Malaysia	4.7	Greece	1.8
Taiwan	6.1	Canada	4.5	Taiwan	1.8
Belgium	6.0	Denmark	4.4	France	1.7
France	5.6	Belgium	3.9	Egypt	1.7
Saudi Arabia	5.5	Saudi Arabia	3.3	Saudi Arabia	1.6
South Africa	5.5	Mexico	2.8	Mexico	1.3
Vietnam	5.4	Thailand	2.8	Germany	1.1
Italy	5.0	Sweden	2.5	Chile	1.1
Germany	5.0	Philippines	2.3	Thailand	1.1
Sri Lanka	4.9	Taiwan	2.2	New Zealand	1.0
Morocco	4.7	Egypt	2.2	Argentina	1.0
Other	107.2	Other	59.3	Other	19.5

1.5 Trends in AIS receiver types

AIS messages are broadcast from vessels and then received through one of three technologies: satellite-based receivers, terrestrial receivers, and dynamic receivers. Satellite and terrestrial receivers are the oldest receiver types, while 2022 saw the widespread adoption of 'Dynamic-AIS' in several regions throughout the world. Dynamic AIS is a technology that was first implemented by Spire, one of the major AIS data providers, in which satellite-enabled AIS receivers are installed on vessels that travel through some of the world's busiest shipping lanes, such as the South China Sea [8]. Through GFW, we can determine the receiver type for every single AIS message in the dataset

2669 over time. In Fig. S6, we summarize the amount of estimated emissions and the num-
 2670 ber of distinct vessels that were detected by the three different types of AIS receivers.
 2671 In 2022, we can clearly see a sharp rise in the amount of emissions that were esti-
 2672 mated from Dynamic AIS, which intuitively coincides with a decrease in the emissions
 2673 that are estimated from satellite and terrestrial receivers. On aggregate, this led to
 2674 an increase in total estimated emissions. The number of distinct vessels detected by
 2675 dynamic AIS also sharply rose in 2022, almost equaling the number of vessels detected
 2676 by all receiver types.



2699 **Fig. S6** Annual total CO₂ emissions (metric tonnes) by AIS-broadcasting vessels, AIS pings (i.e.,
 2700 individual vessel timestamped location messages), and number of unique AIS-broadcasting vessels.
 2701 Trends are disaggregated by AIS receiver type (satellite, terrestrial, and dynamic), and also shown
 2702 as a total aggregated across receiver types. A dashed vertical line is shown for 2022, the first year in
 2703 which dynamic AIS was widely adopted.

2705 In Fig. S7, we see the amount of quantified emissions for the top 10 vessel flags
 2706 broken apart by AIS receiver type. The trends are similar to those seen on aggregate (Fig. S6),
 2707 although there is variation in terms of the increase in Dynamic AIS detections across different flags.
 2708 This is likely driven by whether or not each flag travels through regions where Dynamic AIS was
 2709 widely adopted. It is important to note that towards the end of 2021, China experience a blackout
 2710 of terrestrial AIS signals due to new privacy laws that were being implemented [57].

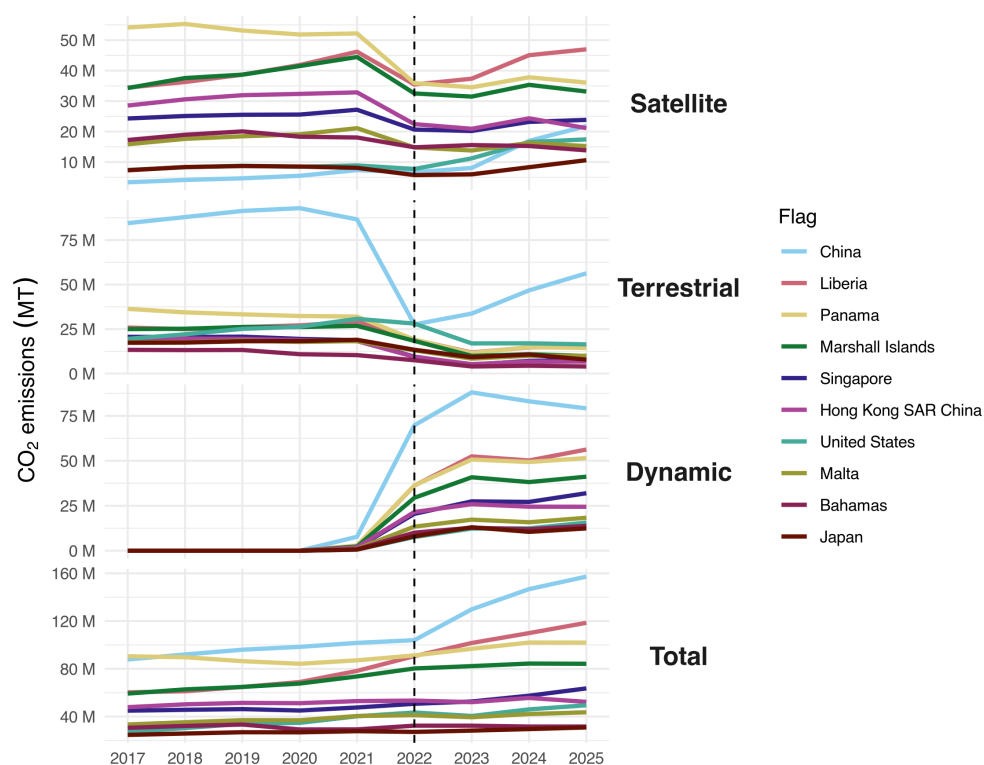


Fig. S7 Annual total CO₂ emissions (metric tonnes) by AIS-broadcasting vessels, disaggregated by AIS receiver type (satellite, terrestrial, and dynamic) and total aggregated across receiver types, for the top 10 flags with the most emissions in 2025. A dashed vertical line is shown for 2022, the first year in which dynamic AIS was widely adopted.

Finally, in Fig. S8, we see how the use of the three AIS receiver types has changed spatially over time between 2017 and 2025. Dynamic AIS was mostly implemented in busy shipping lanes, with clear concentration in areas including the South China Sea, Europe, and the Gulf of Mexico. These regions also generally correspond to decreasing use of satellite and terrestrial receivers. Overall, the largest total change in emissions is generally concentrated in the same regions that saw the largest uptake of Dynamic AIS.

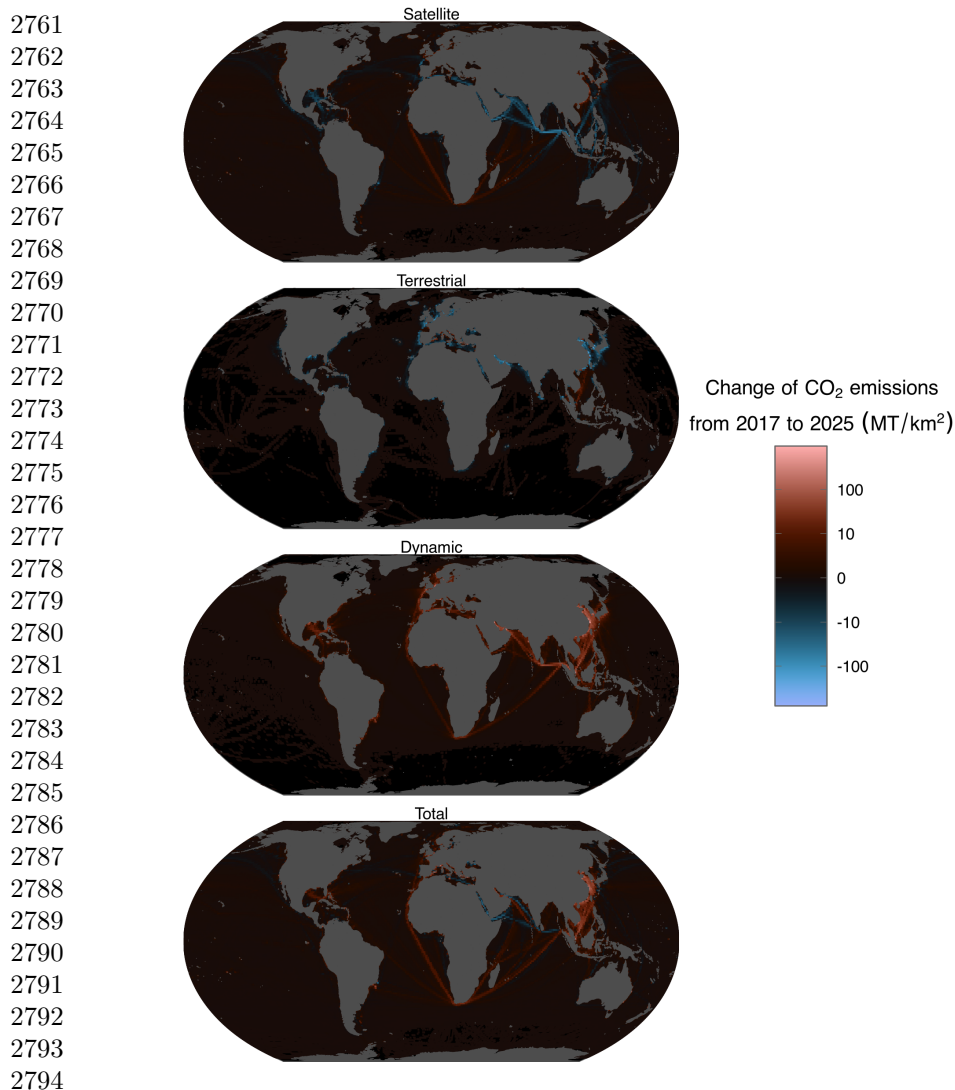


Fig. S8 Spatial change in CO₂ emissions by AIS-broadcasting vessels between 2017 and 2025 (metric tonnes) by receiver type (satellite, terrestrial, and dynamic) and total across receiver types, at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale).

1.6 Vessel class and characteristics models

Two families of machine learning models are used to estimate vessel classes and properties. Vessel class is determined by a combination of a broad, primary vessel class model and focused cargo and tanker sub-type models. Vessel properties are determined by six separate models used to estimate the following vessel characteristics: gross tonnage (GT), main engine power (kW), design speed (kn), year-of-build (year),

and engine type (class). Random forest classifiers are used for the primary vessel class and the engine subtype, random forest regression models are used to estimate gross tonnage, engine power, design speed, and year of build. Finally, gradient boosting classifiers are used for the cargo and tanker sub-type models. Details about each model can be found in the relevant subsections below.

All models are trained on three categories of features derived from AIS information: inferred features (37), identity features (14), and activity features (13). The cargo and tanker sub-type models are trained on an additional 50 features derived from port visits. These features were built with Word2Vec (50 dimensions, window 10, minimum count 5) over each vessel’s sequence of anchorage visits, from high-confidence port visits in the 365 days to 2025-06-01; a vessel’s vector is the visit-weighted mean of the anchorages it used. Inferred features consist of vessel class neural network scores and neural network estimates for vessel length, tonnage, engine power, and crew size [12]. Identity features include identification data broadcast through a vessel’s AIS, such as the first digit and the broadcast vessel class. Activity features describe patterns in a vessel’s AIS activity over time, such as fraction of active hours, average speed, distance from shore, etc.

Training/test data for all models has an 80/20 split, and is stratified by vessel class and the first digit of each MMSI to mitigate bias by vessel type or region. The number of training samples varied for each model as labeled data was available, with the primary GFW vessel class model having the maximum 500,000 total samples. Test sample sizes available for all models are provided in the metrics Table S7. For each vessel characteristic model, Table S8 summarizes the percentage of vessels that had information already available from public sources, private sources, or for which inferred values from our models were used.

Table S7 Performance for every model. Held-out values use the metric named in the *Metric* column, so rows are comparable within a family but not across families.

Model	Target	<i>n</i> train	<i>n</i> test	Metric	Held-out
<i>Classification models</i>					
Vessel class	32 classes	510 486	127 645	macro-F1	0.878
Engine type	4 classes	46 108	11 527	macro-F1	0.550
Cargo sub-type	5 classes	11 744	2 939	macro-F1	0.855
Tanker sub-type	4 classes	5 903	1 477	macro-F1	0.712
<i>Regression models</i>					
Main engine power	kW	13 842	3 542	R^2	0.929
Gross tonnage	GT	62 636	15 687	R^2	0.944
Design speed	knots	8 808	2 241	R^2	0.861
Engine RPM	rev/min	45 868	11 467	R^2	0.533
Construction year	year	45 868	11 467	R^2	0.447

Table S8 Availability of vessel characteristics used in emissions modeling, by source. “Inferred” is the percentage of vessels for which the characteristic were not matched to any registry and was therefore inferred by our machine-learning models. “Public Sources” is the percentage sourced from any publicly available registry, including that maintained by IMO’s GISIS (Global Integrated Shipping Information System). “Private Sources” is the percentage sourced from proprietary vessel databases, chiefly S&P Global Market Intelligence Ships Dataset and to a smaller extent non publicly available lists maintained by Global Fishing Watch.

Characteristic	Inferred (%)	Public Sources (%)	Private Sources (%)
Vessel Class	82.5	17.5	0.0
Gross Tonnage	85.4	14.3	0.2
Length	89.8	6.9	3.3
Main Engine Power	92.5	3.1	4.4
Engine Type	93.9	0.0	6.1
Year of Build	94.0	0.0	6.0
Engine RPM	95.1	0.0	4.9
Design Speed	98.2	0.0	1.8

The primary GFW vessel class model, used to predict broad classes such as cargo, tanker, trawler, and gear, identifies the cargo and tanker vessels used in this analysis. Based on those classifications, the cargo and tanker sub-type models further refine those predictions into the more granular classes. Note that nearly half of AIS-broadcasting devices are likely associated with non-vessel objects (such as buoys, offshore structures, FADs, etc). These objects are identified as their own vessel class ‘gear’ by the primary GFW vessel class model, and are excluded from the emissions estimates in this paper. Cargo and tanker sub-type labels are drawn from S&P Global Maritime records, with StatCode5 levels 2–5 mapped to align with the fourth IMO Greenhouse Gas Study vessel class types, keeping only records that report a single MMSI so each label attaches to one vessel identity.

1.6.1 Primary GFW vessel class model

Target

The primary GFW vessel class model made vessel class predictions at the MMSI level for 1.8M active, AIS-broadcasting vessels. Vessels are assigned a unique class, separated into fishing, non-fishing, and gear, where gear refers to non-vessel AIS-enabled objects. The final assigned vessel class follows a hierarchy determined by the model’s score for that class, where low-scoring classes get a final assignment at a coarser level. For example, a low-scoring “tanker” might be assigned a final vessel class “non-fishing”.

Training data

The primary GFW vessel class model makes use of all inferred, identity, and activity features described above. Labels are drawn from clean, aggregated registry data, ensuring that re-used MMSIs and MMSIs with conflicting classes by registry are dropped

from training. 500,000 vessels with aggregated registry information were used to train the model, with a holdout set of 130,000 for testing.

Performance

The model achieves an accuracy of 98.57%, with weighted precision, recall, and F1 all at 98.7%. Macro-averaged precision, recall, and F1 are 87.8%, 64.8%, and 71.6%, respectively. The gap between weighted and macro reflects a class imbalance, with rare classes performing worst. Cargo and tanker vessels are well-represented in the training data and perform well individually. Cargo achieves an F1 score of 97.0%, and tanker reaches an F1 score of 95.6%.

Key predictive features

The top features are determined using the Gini index, an imperfect but useful measure. Transmitted shiptype unavailable: 0.10, median speed: 0.09, most common AIS class: 0.08, transmitted shiptype cargo: 0.07, neural net passenger vessel score: 0.06, neural net gear score: 0.06, neural net cargo score: 0.05.

Interpretability

The feature importances are relatively spread out across different training features, with the most indicative feature whether or not the vessel broadcasts a shiptype. This makes sense, as vessel class is gear-dominated, and gear typically does not broadcast a shiptype. Other features, such as median speed, AIS class, and cargo shiptype, indicate a combination of vessel behavior and AIS information itself, while the neural net scores for specific classes show the propagation of one model's predictions into the next.

Overfitting controls and known weaknesses

The model uses regularized hyperparameters ($n_{estimators} = 300$, $max_{depth} = 35$, $min_{samples_{leaf}} = 10$, $min_{samples_{split}} = 20$) rather than an unconstrained forest to balance accuracy against model size. In order to mitigate overfitting or regional bias, the 80/20 split in the train/test data is stratified by MMSI and MMSI region, and the label-exclusion rules reduce noisy or duplicated MMSI from training. The tight cross-validation variance (0.05% accuracy across folds) suggests the model isn't overfit to a particular split.

Despite the training data stratification, high-volume classes such as gear and cargo perform very well, while rarer classes where limited data is available struggle significantly. Larger classes, particularly gear, may be over-classified. Since the model score threshold to mitigate low-scoring, incorrect classifications, some genuinely correctly classified vessels may end up assigned too broad of a label.

As the vessel sub-types models rely on the output of the primary vessel classification model predictions, biases and errors are propagated from this model to the next. This effect similarly happens between the neural net scores and the primary GFW vessel class model.

2945 **1.6.2 Cargo sub-type model**

2946 *Target*

2947 One of five cargo sub-types (bulk carrier, container, general cargo, refrigerated cargo,
2948 ro-ro cargo), for 186 689 vessels the primary vessel class model classified as cargo.

2950 *Training data*

2951 11 744 vessels with S&P registry sub-types: bulk carrier 5 201; general cargo 3 200;
2952 container 2 166; ro-ro 1 100; refrigerated 77.

2954 *Performance*

2955 Accuracy 90.5%; macro-F1 85.5%; weighted F1 90.5%. Per-class F1: bulk carrier 94.3,
2956 container 93.9, general cargo 84.7, ro-ro 84.6, refrigerated 70.3.

2958 *Key predictive features*

2959 Importance is the loss in held-out macro-F1 when a feature is permuted, used in place
2960 of the Gini index, which is undefined for the gradient boosting models used here.
2961 Anchorage embeddings, 50 dimensions permuted together: 0.220. Vessel class neural
2962 network scores, 33 permuted together: 0.157. Top speed (95th percentile): 0.029.

2965 *Interpretability*

2966 Vessel class neural network scores form a substantial part of this model's classification
2967 skill, indicating it is potentially inherited from the primary model rather than learned
2968 from vessel behaviour.

2970 *Overfitting controls and known weaknesses*

2971 Gradient boosting classifiers with early stopping on a held-out internal validation
2972 split, rather than a fixed number of boosting iterations. Unlike the primary model, the
2973 train/test split here is stratified by sub-type only rather than by MMSI region, with
2974 the split fixed by an MMSI hash so it is stable as data is added.

2975 Refrigerated cargo has 77 training vessels; its F1 of 70.3 is indicative only. Ro-ro
2976 and general cargo are the main confusion pair among the well-populated classes: 17%
2977 of ro-ro vessels are predicted as general cargo. General cargo is 27% of the training set
2978 but 66% of predictions, so the model is likely extrapolating beyond its training mix.

2979 As with the primary model, errors are inherited from it: a vessel it misclassifies is
2980 refined against the wrong sub-type vocabulary. Labels exist only for S&P-registered,
2981 single-MMSI vessels, and 4.5% of training vessels have no port visit features at all.

2983 **1.6.3 Tanker sub-type model**

2985 *Target*

2986 One of four tanker sub-types (oil, chemical, liquefied gas, other liquids), for 43 530
2987 vessels the primary vessel class model classified as tanker.

2988
2989
2990

<i>Training data</i>	2991
5 903 vessels with S&P registry sub-types: oil 2 664; chemical 2 015; liquefied gas 1 188;	2992
other liquids 36.	2993
	2994
<i>Performance</i>	2995
Accuracy 87.5%; macro-F1 71.2%; weighted F1 87.4%. Per-class F1: liquefied gas 91.8,	2996
oil 88.6, chemical 84.3, other liquids 20.0.	2997
	2998
<i>Key predictive features</i>	2999
Top speed (95th percentile): 0.169. Anchorage embeddings, 50 dimensions permuted	3000
together: 0.145. Vessel class neural network scores, 33 permuted together: 0.061.	3001
Average distance traveled per trip: 0.030.	3002
	3003
<i>Interpretability</i>	3004
One feature, top speed, carries more importance than any other in either sub-type	3005
model. That makes the model easy to explain, but it also means it leans on a single	3006
behavioural signal that degrades wherever AIS coverage is patchy.	3007
	3008
<i>Overfitting controls and known weaknesses</i>	3009
Gradient boosting classifiers with early stopping on a held-out internal validation split.	3010
As with the cargo model, the train/test split is stratified by sub-type only, and the	3011
same label-coverage limits and inheritance of errors from the primary model apply.	3012
Performance for <i>other liquids</i> is poor: recall 11.1%, with 89% of those vessels pre-	3013
dicted as oil. With 45 labeled vessels in total the class does not have enough signal	3014
to be separated. Oil and chemical tankers are often confused with each other: 13% of	3015
chemical vessels are called oil and 10% the reverse. 5.6% of training vessels have no	3016
port visit features at all.	3017
	3018
	3019
1.6.4 Other vessel characteristics models	3020
	3021
<i>Targets</i>	3022
Continuous numeric vessel characteristics - main engine power, gross tonnage, design	3023
speed, and construction year - are estimated using random forest regression models,	3024
while the categorical engine type value is estimated using a random forest classifier. All	3025
models make predictions at the MMSI level for 1.8M active, AIS-broadcasting vessels.	3026
	3027
<i>Training data</i>	3028
All models make use of the inferred, identity, and activity features described above.	3029
Labels are drawn from clean, aggregated registry data, ensuring that re-used MMSIs	3030
and MMSIs with conflating classes by registry are dropped from training. The engine	3031
power model had a training set of 13,000 samples, the gross tonnage model had a	3032
training set of 63,000 samples, the design speed model had a training set of 8,808	3033
samples, the construction year and engine RPM models had a training set of 45,868	3034
samples, and the engine type model had a training set of 46,108 samples.	3035
	3036

3037 **Performance**

3038 *Engine power model*

3039 The engine power model achieved an R^2 of 0.929 with a mean absolute error of 548.71
3040 kW and a mean absolute percentage error of 43.8%.

3041

3042 *Tonnage model*

3043 The tonnage model achieved an R^2 of 0.944 with a mean absolute error of 2,830.15
3044 GT and a mean absolute percentage error of 209.6%.

3045

3046 *Design speed model*

3047 The design speed model achieved an R^2 of 0.861 with a mean absolute error of 0.82
3048 kn and a mean absolute percentage error of 4.9%.

3049

3050 *Year-of-build model*

3051 The year-of-build model achieved an R^2 of 0.447 with a mean absolute error of 6.68
3052 and a mean absolute percentage error of 0.3%.

3053

3054 *Engine RPM model*

3055 The engine RPM model achieved an R^2 of 0.533 with a mean absolute error of 360.56
3056 and a mean absolute percentage error of 33.2%.

3057

3058 *Engine type model*

3059 Accuracy of 99.08%; macro-F1 54.6%; weighted F1 0.991%. Per class F1: oil 99.5%,
3060 gas 61.7%, turbine 57.1%, reciprocating 0.00.

3061

3062 **Key predictive features**

3063 *Engine power model*

3064 Average engine power inferred by neural network: 0.535. Top speed (95th percentile):
3065 0.149. Average tonnage inferred by neural network: 0.104.

3066

3067 *Tonnage model*

3068 Average tonnage inferred by neural network: 0.626. Fraction of hours spent at port:
3069 0.0968. Average distance traveled per trip: 0.041.

3070

3071 *Design speed model*

3072 Top speed (95th percentile): 0.609. Median speed: 0.054. Neural network supply
3073 vessel score: 0.051.

3074

3075 *Year-of-build model*

3076 Average engine power inferred by the neural network: 0.162. First digit of the MMSI
3077 code: 0.101. Average tonnage inferred by the neural network: 0.063.

3078

3079 *Engine RPM model*

3080 Average gross tonnage inferred by the neural network: 0.310. Average length inferred
3081 by the neural network: 0.093. Average percentage of self-identified shiptype messages

3082

as fishing: 0.074.	3083
	3084
<i>Engine type model</i>	3085
Top speed (95th percentile): 0.096. Neural network patrol vessel score: 0.080. Average	3086
engine power inferred by the neural network: 0.062.	3087
	3088
<i>Interpretability</i>	3089
Both the engine power and the tonnage predictions are largely driven by the neural	3090
network's inferred values, indicating that these models are refining existing predic-	3091
tions.	3092
The design speed predictions are most heavily reliant on the vessel's top speed and	3093
median speed, which makes intuitive sense as the two act as a proxy for design speed.	3094
The year-of-build predictions are less singularly reliant on one strong signal, but are	3095
driven the most by the neural network's inferred engine power and the first digit of	3096
the MMSI code. The latter likely acts as a proxy for build/registration era for each	3097
flag state.	3098
The engine RPM predictions are largely weighted by physical characteristics of the ves-	3099
sel, but similarly less reliant on a single feature. The physical characteristics describing	3100
a vessel likely act as a proxy for RPM, while the self-identified shiptype likely distin-	3101
guishes fishing and non-fishing vessel properties.	3102
The top features for engine type predictions are much more diffuse, suggesting that	3103
no single feature strongly separates engine types and the model relies on combinations	3104
of weaker signals.	3105
	3106
<i>Overfitting controls and known weaknesses</i>	3107
To mitigate overfitting, all models are trained with a fixed random seed and a train/test	3108
split stratified by vessel class and first-digit MMSI. The engine type model additionally	3109
used class-weight balancing to counteract severe class imbalance in the training labels.	3110
However, because the engine type labels are so heavily skewed towards oil engines,	3111
the model only performs reliably for that class, and semi-reliably for gas. As higher	3112
volume data becomes available for a greater diversity of engine types, future models	3113
will be more sophisticated and better predictive of the less represented types.	3114
The year-of-build model has a 6-7 year mean error, suggesting that it is useful for	3115
predicting general build era up to around a decade, but not reliable for year-by-year	3116
predictions. The engine RPM model has mean error of 360 revolutions per minute,	3117
indicating that it can capture very broad patterns but with more relative error than	3118
the year-of-build model. The engine power, tonnage, and design speed models all show	3119
the same pattern: performance is strongest in the mid-range of each variable and	3120
degrades substantially at the extremes, where training data is sparse.	3121
	3122
	3123
	3124
	3125
	3126
	3127
	3128

3129 1.7 Systematic comparison of our methods

3130 1.7.1 Scope of the comparison and modeling approaches

3132 Global ship emission estimates fall into two methodological families, distinguished by
3133 whether emissions are derived from aggregate statistics or from the activity of indi-
3134 vidual vessels. Top-down, fuel-based inventories combine reported marine fuel sales
3135 with fuel emission factors, without reference to individual ships. Such estimates rest
3136 on fleet-average parameters, cannot resolve operational behaviour, and are known
3137 to underestimate emissions relative to activity-based inventories [58]. Bottom-up,
3138 activity-based approaches instead derive emissions mechanistically from the observed
3139 movement of individual vessels, calculating the pollutants emitted by each ship in a
3140 specific spatial context and aggregating over time and fleet [59]. Since the widespread
3141 adoption of AIS, which resolves ship activity and operational mode at high spatial and
3142 temporal resolution, bottom-up AIS-based models have become the reference method-
3143 ology for global inventories and the approach adopted by the IMO [4]. We therefore
3144 restrict our methods comparison to this family, which includes our AIS-based model,
3145 and retain top-down estimates only as an external check on global totals.

3146 Any AIS-based inventory is assembled from four components, and the dif-
3147 ferences between published inventories are attributable to how each is specified.
3148 First, the AIS record itself, which determines which vessels and which movements
3149 are observed at all. Second, a vessel-characteristics dataset supplying the param-
3150 eters the model requires. Third, an engine model converting observed speed into
3151 instantaneous main engine load, plus assumptions on auxiliary engine and boiler
3152 demand. Fourth, fuel consumption and pollutant specific emission factors, condi-
3153 tioned on engine type, fuel type and the regulation in force at the time and place of
3154 the emission. Supplementary Tables `bottom-up_inventory_comparison_methods.`
3155 `csv`, `bottom-up_inventory_comparison_inputs.csv` and `bottom-up_inventory_`
3156 `comparison_outputs.csv` compare our methods, input data and outputs against
3157 the most recent and representative global bottom-up inventories: the STEAM model
3158 [15, 45, 60–62], SEIM [17, 63], MariTEAM [16], the OECD’s recent update transi-
3159 tioning from a machine-learning to a bottom-up approach [64, 65], the SAVE model
3160 developed by the ICCT [3, 18, 66], and the Fourth IMO GHG Study [4].

3161

3162 1.7.2 Activity and vessel-characteristics data

3163 All seven inventories in the supplementary tables are global in spatial extent, so the
3164 differences lie in AIS coverage, fleet size, and temporal continuity rather than in geo-
3165 graphic scope (Supplementary Table `bottom-up_inventory_comparison_inputs.`
3166 `csv`). The inventories diverge substantially in the number of vessels represented.
3167 Restricting the comparison to vessels explicitly modelled from observed AIS activity,
3168 we model 864,738 unique vessels, an average of 396,221 vessels per year rising from
3169 249,732 in 2015 to 529,848 in 2025. This exceeds the AIS modelled fleet of the Fourth
3170 IMO GHG Study, the OECD update, SAVE, MariTEAM and SEIM. Our fleet size
3171 is only comparable to the STEAM inventory, which reported 376,219 vessels in 2015,
3172 exceeding our fleet size for the same year. However, STEAM does not report fleet sizes
3173 in subsequent updates, preventing a direct comparison for the later years in which our
3174

fleet is larger. The difference with STEAM is also reflected in the underlying AIS data, which used approximately 7.8 billion AIS positions in 2015, compared with approximately 2.3 billion in our 2015 dataset after data cleaning. Our own message volume increases to approximately 26.5 billion positions by 2025, totaling approximately 125 billion positions over the study period (an average of 11.4 billion per year), reflecting the expansion of terrestrial and satellite AIS. It also reflects differences in the AIS data types available to us. Terrestrial and satellite AIS constitute the core AIS data types. In addition, GFW obtains a data add-on for both class A and class B Dynamic-AIS, released by the data provider Spire in 2022, which further increases the density and coverage of our AIS record. To our knowledge, D-AIS is generally not included in base AIS data products from most providers. However, we note that other inventories covering from 2022 onward report conventional AIS data types without explicitly indicating whether Dynamic-AIS (also marketed as shipborne AIS) is included. Therefore, we cannot determine whether the differences arise from variations in input data or simply from a lack of reporting. Among the remaining inventories, SEIM and MariTEAM report lower AIS message volumes, whereas the remaining do not indicate message counts.

Temporal coverage differs in extent. STEAM and MariTEAM are single-year inventories (2015 and 2017 respectively), with STEAM subsequently updated through 2021 to account for ambient conditions [45]. SEIM covers 2013 and 2016-2021, the Fourth IMO GHG Study 2012-2018, SAVE 2016-2023, and the OECD from 2019 onward. Our use of consistent GFW AIS products, though input density evolves, enables a continuous AIS inventory from 2015—and AIS + S1 inventory from 2017—to the present without methodological discontinuities.

Every inventory uses official registries as the primary source of vessel characteristics and applies gap-filling where information is unavailable. The differences are in how far the registry is extended and how the remainder is inferred. The simplest filling approaches substitute central tendencies. The OECD update restricts to vessels matching between commercial and public registries and imputes from vessel class and size group means [65], while SAVE fills missing characteristics from averages values [66]. STEAM retains some of the within class variation by replicating the characteristics of a similar vessel already listed in its database [15]. Regression-based approaches such as that used for MariTEAM, fills gaps using regression models, though its modelled fleet is fully restricted to the registered fleet [16], while the Fourth IMO GHG Study applies similar regressions as its primary method and falls back to class and size medians, on a dataset expanded with registered and public datasets [4]. Of the published inventories, SEIM is the only one besides ours to describe a more involved machine-learning approach to fill missing characteristics, applied after extending commercial registries with open data [17]. In our vessel information dataset, characteristics are taken from data purchased from S&P Global, from publicly available official registries consolidated through extensive prior scraping work [13], and where neither is available, from inferred characteristics, building on [12] using a new random-forest model trained on commercial registry data, enabling inclusion of a substantially larger number of vessels than most previous studies and reach a fleet dominated by small domestic and fishing vessels that the more registry-anchored inventories cannot represent.

Finally, the inventories differ in how they treat vessels absent from the AIS feed. STEAM, SEIM and MariTEAM rely exclusively on AIS or port-call data and make no allowance for unobserved activity. The OECD, SAVE and the Fourth IMO GHG Study models impute activity and emissions for small registered vessels missing from AIS, scaling from explicitly modelled vessels of the same type and size. Both approaches are bounded by the registry, where imputation can only recover a vessel that some registry lists. We instead estimate emissions from non-broadcasting vessels detected in Sentinel-1 SAR imagery using machine learning and data fusion (Section 4.2), which recovers vessels that appear in neither AIS nor any registry. To our knowledge this is the first global shipping inventory to do so.

1.7.3 Engine and emissions model

In terms of propulsion formulation, the literature defines two main modeling approaches: load-based (i.e., SEIM, SAVE, the OECD update, the Fourth IMO GHG Study) and resistance-based (i.e., STEAM and MariTEAM). Every AIS-based inventory ultimately estimates the propulsion power required to overcome hull resistance at the observed speed, and the models differ in the resolution at which the components of that resistance are resolved. While load-based models are generally regarded as roughly representative of more complex methods, resistance-based models are often avoided because of the number of inputs needed. Resistance models require hull parameters, which restricts their applicability to vessels with such documented information in the registries. Imputing class-average hull parameters may be acceptable for fleet-level aggregates and well-defined classes with limited variation in hull design, but performance declines for underrepresented and heterogeneous classes and can substantially alter single-vessel estimates [67]. This limitation is particularly relevant because we model a broader fleet dominated by small domestic and fishing vessels, for which hull geometry is often absent or incomplete, and report emissions at a finer resolution, by trip and by vessel. The resistance model’s advantage in physical realism would therefore be offset by uncertainty in its inputs. The load-based formulation, by contrast, requires only installed power, design speed, observed speed and a small number of further characteristics that are readily available in registries and can be modeled with confidence where they are not.

In terms of other attributes, such as engine and fuel parameterization, we adapted the methods from the Fourth IMO GHG Study [4], which established standard assumptions now adopted by most of the shipping emissions models. Building an expanded activity dataset around a community reference also allows us to attribute differences from existing inventories to fleet coverage and activity representation, which is the claim this paper makes. Deviations from [4] methods were dependent to data availability or the quality of inferred values, but remains in line with other load-based models [17, 65, 66]. Correction factors for main engine activity are implemented, with draft corrections applied at a lower resolution than in [4], [66] and [65], as draft is not reliably reported in our AIS records. Weather and fouling corrections follow comparable approaches to other load-based models, and an additional new correction is introduced to account for fishing activity. Similarly, SFOC calculations and decomposition in emissions by main auxiliary and boiler engines come from a general consensus across

models [4, 17, 65, 66], even among non load-based ones [15, 16]. See (Supplementary Table `bottom-up_inventory_comparison_methods.csv`)

However, our approach is limited on its fuel types characterization which are not explicitly modeled due to lack of available vessel-level fuel type data. For a given engine type, SFOC values and emissions factors are therefore calculated as the average value from across the fuel types associated with each engine type. LNG was not included as a fuel type due to the lack of available vessel-level fuel type data, the very limited number of vessels that currently use LNG, and the vastly different emissions factors this fuel type produces. Our treatment of policy instruments is also narrower than that of comparable models, which apply fuel-sulphur and NOx limits geographically within Emission Control Areas [4, 15, 17, 66]. Additionally, STEAM and SAVE account for technologies such as scrubbers and dual-fuel operation. While our NOx limits are assigned by engine Tier from build year, these are not resolved by NOx ECA boundaries, and SOx and PM estimates are expected to be biased upward within ECAs, an effect concentrated before 2020, when the difference between ECA and global sulphur limits was largest.

1.7.4 Differences in validation

Validation of existing inventories has rested largely on aggregate benchmarking against independent statistics, top-down approaches and against other bottom-up inventories. Our global totals were cross-compared against all six inventories discussed here, and against the multisector EDGAR and CEDS inventories [68, 69]. As for the other models, STEAM benchmarked global fuel use against IEA statistics and the Third IMO GHG Study, modelled maritime transport work against UNCTAD seaborne-trade statistics and regional results against the EMEP inventory. SEIM, which rests entirely on this class of evidence, compares global totals and trends against EDGAR, CEDS and the Fourth IMO GHG Study, with an activity and fleet benchmark against UNCTAD. MariTEAM cross-compares against the Fourth IMO GHG Study for the same year and STEAM3 for a different year. The OECD update is the most extensive in this respect, benchmarking global totals against IEA and the Fourth IMO GHG Study, fleet activity against UNCTAD, average efficiencies by class and size against IMO (2018), and country-level results against national Air Emission Accounts for more than 35 countries. SAVE benchmarks aggregated fuel consumption against IMO DCS for ships of 5,000 GT and above, and the Fourth IMO GHG Study against the Third IMO GHG Study, their own top-down estimates, the ICCT inventory, and global fuel sales reported by the IEA and UNSD.

We also validate against two independent in situ measurement datasets: ship-specific measured fuel consumption for 15,679 unique vessels from a registered dataset, and EU MRV records covering 20,432 vessel-years. Together these constitute the largest ship-level validation sample reported for a global bottom-up shipping inventory. Other validations using in situ measurements include that of the Fourth IMO GHG Study, which compares yearly estimates against 9,739 vessel records in the 2018 EU MRV dataset and additionally draws on engine power and fuel consumption measurements from 94 vessels' Continuous Monitoring Dataset [4]. SAVE also validates per-vessel-year carbon intensity against EU MRV [66]. The OECD update likewise

draws on EU MRV, but compares totals aggregated by vessel class rather than by individual vessel [65]. MariTEAM validates modelled engine power against in situ measurements for a single vessel [16]. STEAM was validated in earlier versions against a limited sample of in situ measurements [62], and its recent ambient-conditions update validates modelled fuel consumption against 6,445 vessel-years records in the EU MRV data [45]. This last comparison bears directly on the choice of propulsion formulation. Applied to an identical fleet and AIS record, resistance-based methods have been reported to yield lower fleet-level CO₂ than their load-based equivalents [67], but that offset has so far been established only between models, or against limited samples of in situ measurements [16, 62]. Here the two families, represented by STEAM and our model, are each tested against the same independent benchmark dataset (i.e., EU MRV) and perform comparably, both achieving an R² of approximately 0.8 [45], supporting our methodological choices and indicating that the load-based formulation as implemented here does not exhibit the systematic upward bias reported by [67].

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3328 **1.8 Comparing our emission estimates to other inventories**

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While validation against the registered and EU MRV datasets tests the emissions model itself, establishing whether it reproduces observed fuel consumption and emissions for individual vessels, it says nothing about whether the modelled fleet is complete. Aggregate benchmarking against independent inventories tests whether totals are of plausible magnitude. Below we first compare our estimates with the activity-based inventories discussed in Section 1.7, and then with the fuel-statistics-derived inventories EDGAR and CEDS. Each comparison is conditioned by data availability, and is limited to the emissions, vessel counts, fleet composition, etc. that each inventory reports publicly.

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3340 **1.8.1 Comparing our estimates with other bottom-up inventories**

Our AIS-based emissions estimates agree closely with the other activity-based inventories (STEAM, SEIM, MariTEAM, OECD, SAVE, and the Fourth IMO GHG Study) over the first years of the study period (Fig. S9, panel A, tabulated in Table S9). This close agreement in emissions coexists with substantial disagreement in fleet scope and AIS coverage—not only between our inventory and the others, but among the other inventories themselves—with our own vessel count and message volume falling within the range of the other inventories in the early years of the series (Table bottom-up_inventory_comparison_inputs.csv). The convergence in totals therefore likely reflects where emissions are concentrated rather than agreement on fleet extent: although total fleets differ, all of these inventories rely on a registry-matched core of roughly 45,000-90,000 large vessels, which accounts for most emissions [4, 15–17, 65, 66]. In our inventory, registry-matched vessels likewise account for the majority of estimated emissions in the early years (Fig. S10, panel C). Differences in emissions instead reflect how much of the low-information tail of smaller vessels and other unregistered vessels each inventory includes. Residual differences in totals are not attributable to scope alone and may also reflect methodological choices (Table bottom-up_inventory_comparison_methods.csv).

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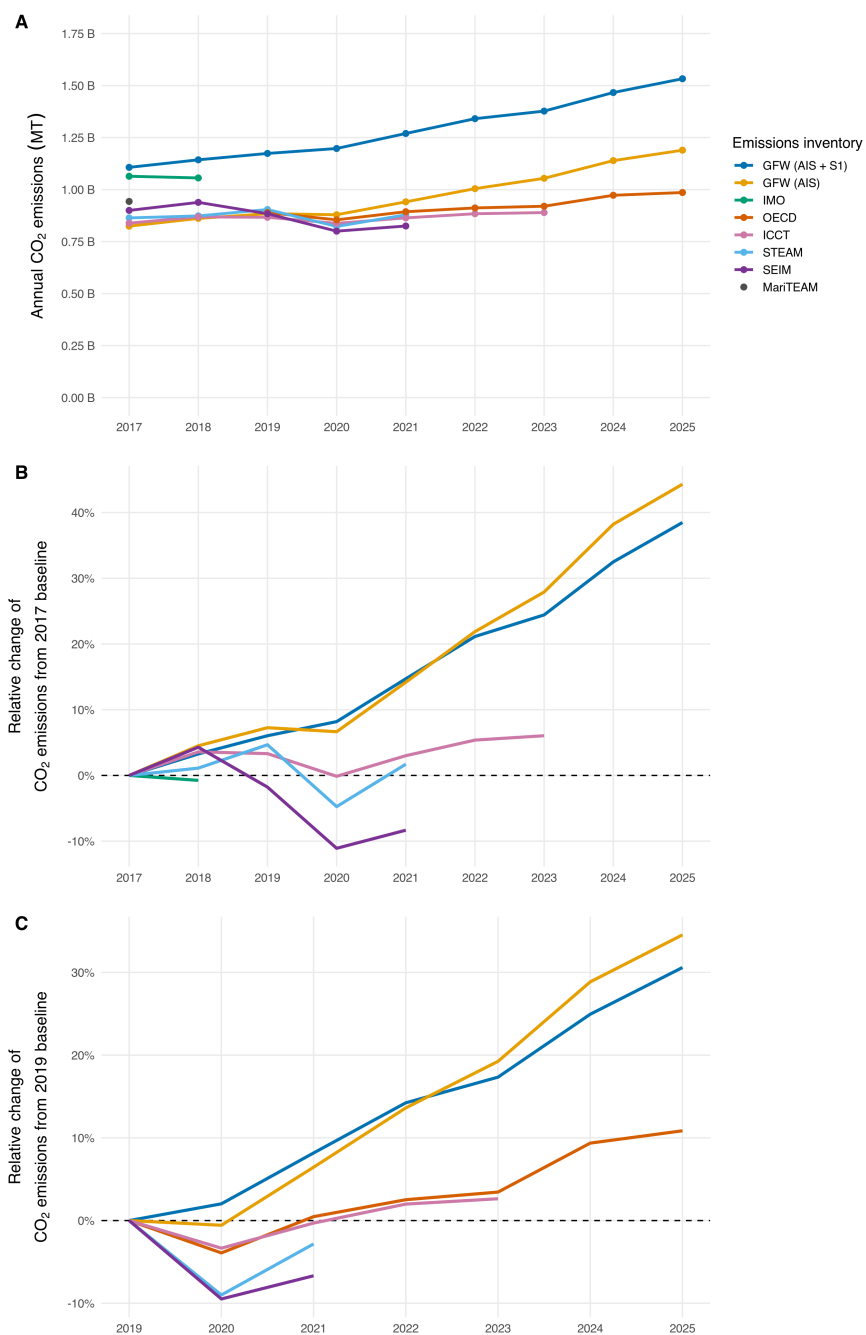


Fig. S9 Our AIS and fused estimates against the published bottom-up inventories. Panel A shows levels; panels B and C show change relative to 2017 and 2019.

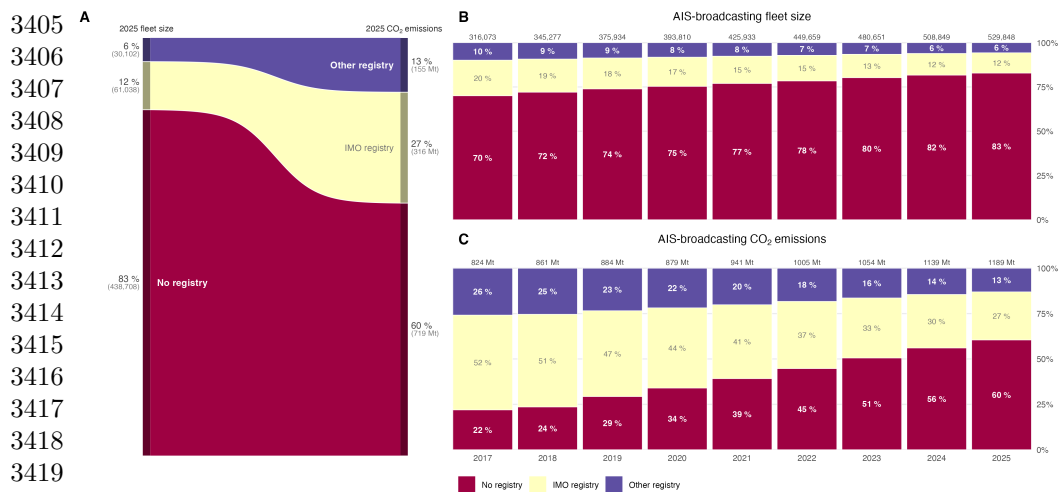


Fig. S10 Registry status in 2025: share of the fleet against share of CO₂ for registry-matched and unmatched vessels, with the supporting series.

Table S9 Annual marine vessel CO₂ emissions (billions of metric tonnes) for our AIS-based and fused (AIS + S1) estimates and for the published bottom-up, activity-based inventories. A dash marks a year for which an inventory publishes no estimate.

Year	GFW (AIS + S1)	GFW (AIS)	IMO	OECD	SAVE (ICCT)	STEAM	SEIM	MariTEAM
2017	1.107	0.824	1.064	-	0.839	0.864	0.900	0.943
2018	1.143	0.861	1.056	-	0.869	0.874	0.939	-
2019	1.174	0.884	-	0.889	0.867	0.904	0.884	-
2020	1.198	0.879	-	0.855	0.838	0.823	0.800	-
2021	1.270	0.941	-	0.894	0.864	0.879	0.825	-
2022	1.341	1.005	-	0.912	0.884	-	-	-
2023	1.377	1.054	-	0.920	0.890	-	-	-
2024	1.467	1.139	-	0.973	-	-	-	-
2025	1.533	1.189	-	0.986	-	-	-	-

Two subsequent changes drive the divergence in our emissions estimates that follows in the most recent years. First, our AIS-broadcasting fleet grows faster than the OECD and SAVE fleets, the only inventories with vessel-count series overlapping the recent period (Fig. S11). These contrasting trends likely reflect differences in how each fleet is constructed (Table `bottom-up_inventory_comparison_inputs.csv`). Inventories that are more constrained by commercial registries (even when they also consider to some extent non-registered vessels) change primarily at the pace of registry updates. Our fleet, by contrast, is more AIS observation-driven where both the number and the share of vessels that cannot be matched to a registry rose over time carrying all the

increase in emissions (Fig. S10 and Fig. S12). Our trends are therefore conditioned more by the expansion of AIS carriage than are those of other inventories, all of which rely more heavily than ours on registry-based fleet definitions holding their emissions more constant (Fig. S9, panel B and C). Sentinel-1 detections make this directly observable. Among detected large non-fishing vessels—the segment where AIS carriage is mandated and registry coverage is strongest, and thus the closest analogue to a registry-based fleet—the share that cannot be matched to an AIS broadcast is essentially unchanged since 2017, whereas that share falls steeply among small non-fishing and fishing vessels as AIS carriage and reception expand (Fig. S13).

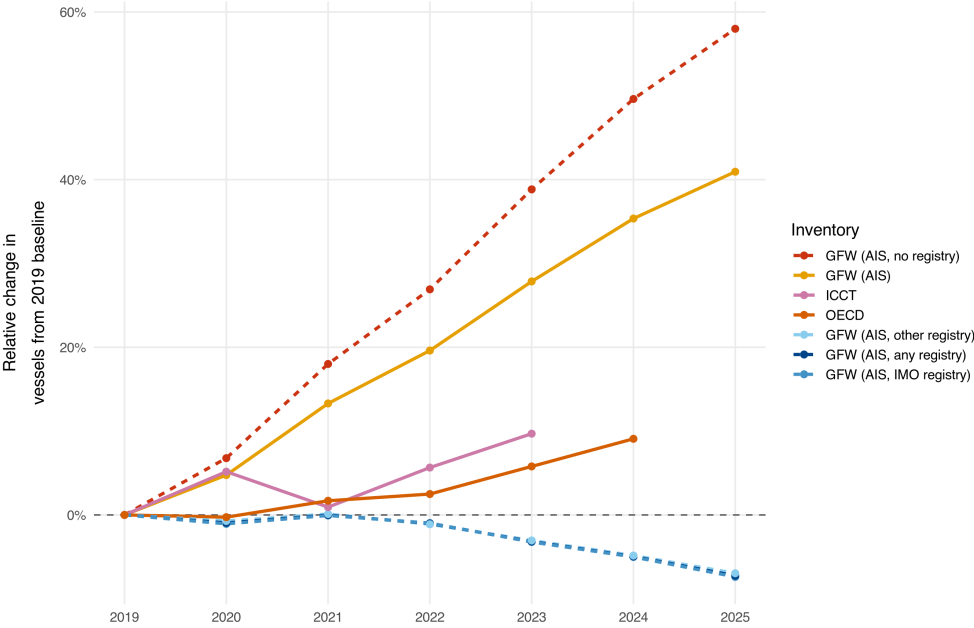


Fig. S11 Modelled fleet size over time for the inventories publishing a vessel count per year, split by registry status for ours.

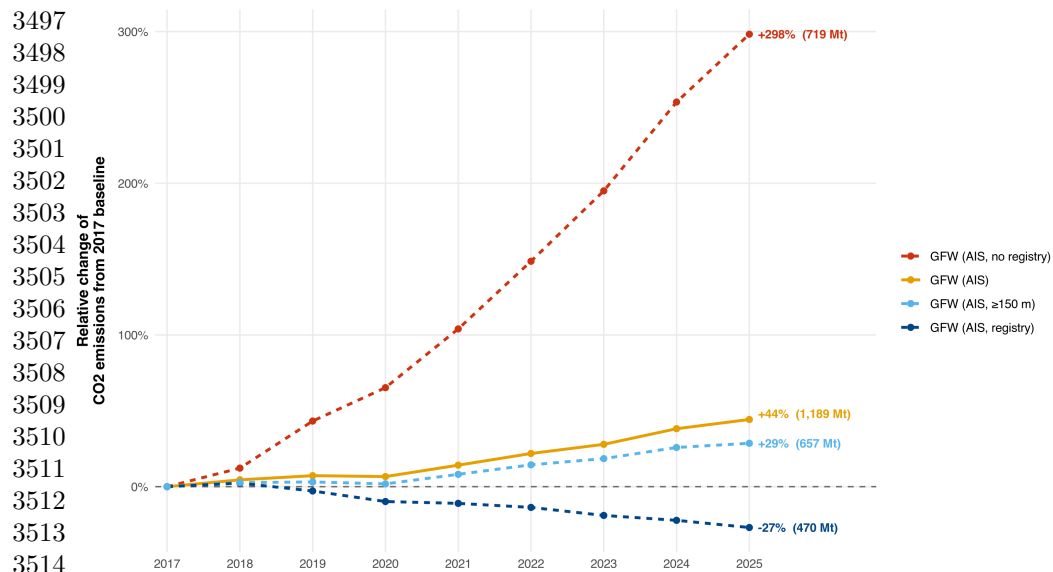


Fig. S12 Emissions from registry-matched and unmatched vessels, relative to 2017, with the size-restricted fragment for comparison.

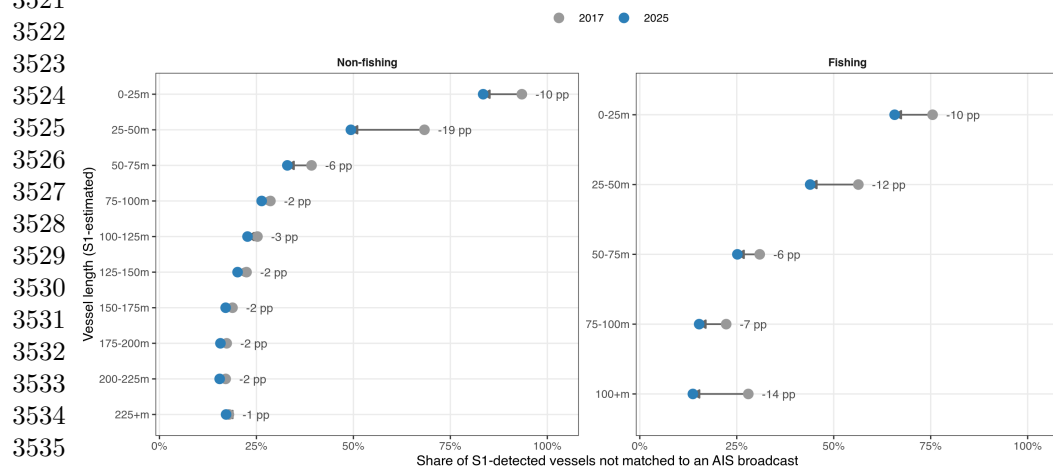


Fig. S13 AIS carriage saturation by vessel size: the unmatched share is flat for the largest vessels and falls steeply for the smallest.

Second, our observed activity is also shaped by the phase-in of dynamic AIS (D-AIS), after which the emissions gap to the other inventories widens most rapidly

(Fig. S9). D-AIS primarily increases AIS-estimated emissions where it fills gaps that were previously unobserved, particularly in high-traffic zones where satellite reception degrades. Activity that was formerly interpolated across, or missed altogether, becomes measured, adding newly observable emissions to the inventory. Alongside this, D-AIS also acts on resolution, densifying segments that were already observed so that trajectories are more tightly constrained and speed profiles better resolved, without additional activity entering the inventory. Both mechanisms operate, but the second predominates: AIS positions received more than double between 2021 and 2025, while observed distance travelled and AIS emissions rise together by only $\sim 27\%$, with emissions per message falling accordingly (Fig. S14). Densification also improves S1 detection-AIS matching, so where this occurs activity moves from the modelled dark component to the AIS one, and its emissions get computed directly from AIS rather than estimated by the dark model, a reallocation between the fused components rather than an addition.

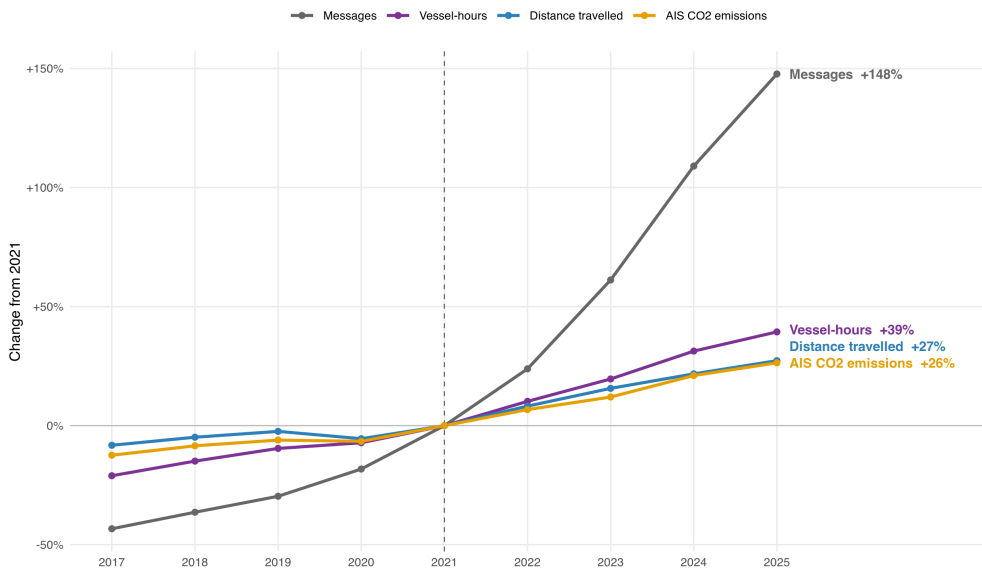


Fig. S14 AIS messages received against observed activity and emissions, indexed to 2021, separating densification from new activity.

The above suggests that the increase in emissions occurs mostly via AIS carriage and, to a lesser extent, AIS coverage. The emissions divergence against the other inventories therefore does not indicate that our totals are inflated—a conclusion also supported by our global per-vessel emissions and emission intensities relative to other models (Table S10)—but rather that the other inventories, and our own early years, are missing part of the fleet and activity at sea. Global fleet emissions are likely understated wherever limited AIS carriage and/or high reliance on registry-defined

3589 fleets leaves low-information vessels out of the scope. This is supported by the S1
3590 detection record where the share of detections that cannot be matched to an AIS
3591 broadcast has declined in all size classes since 2017, most steeply among fishing vessels
3592 and small non-fishing vessels, indicating an increase in AIS adoption (Fig. S15). Indeed,
3593 the largest emissions growth we observe comes from passenger vessels, driven by the
3594 smallest group (Fig. S16), and fishing vessels, a fleet segment comprising numerous
3595 small, largely coastal vessels that are poorly covered by registries and enter the model
3596 only as their AIS carriage expands (Fig. S17 and Fig. S18). This is precisely the
3597 segment that registry-based methods are least equipped to capture.

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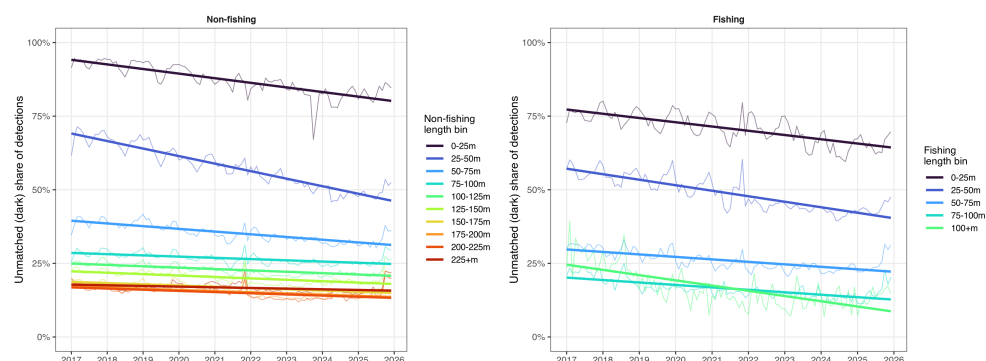


Fig. S15 Share of Sentinel-1 detections with no matching AIS broadcast, by length bin, over the full period.

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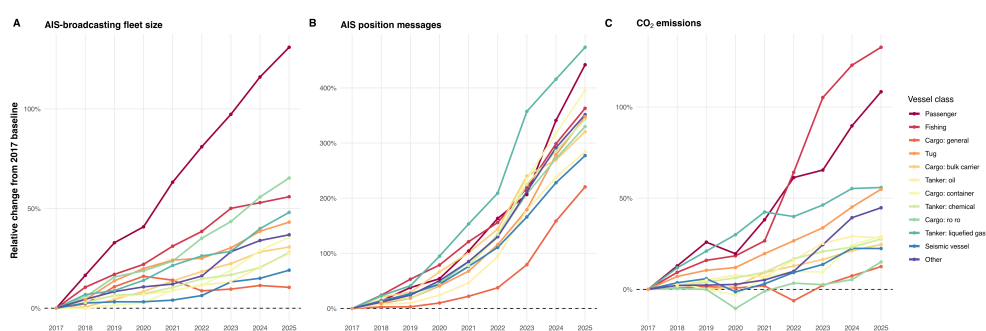


Fig. S17 Growth by vessel class since 2017 in emissions, vessel count and activity.

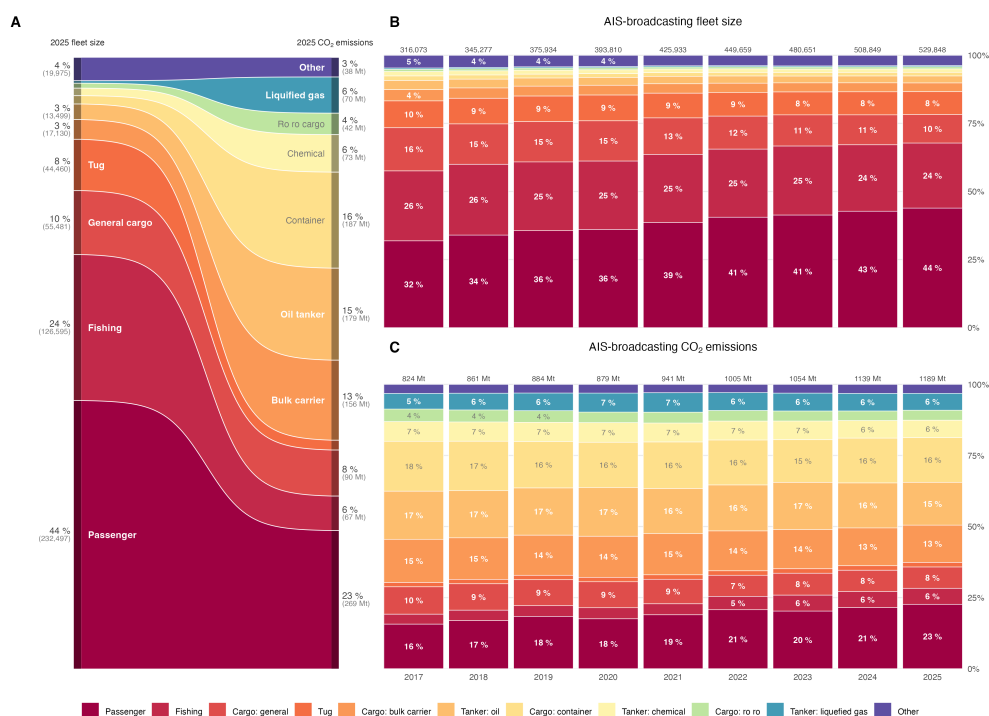


Fig. S18 Fleet composition in 2025: each vessel class's share of the fleet against its share of CO₂, with the supporting series.

Table S10 CO₂ intensity per vessel for each inventory in the comparison, ordered by intensity. Intensities are computed on the fleet count each model publishes: single-total models use their reported figure, per-year models the mean over years where both a count and emissions exist, and the GFW rows our own activity summary on the same basis. Per-distance intensity is shown only for the models reporting distance travelled: SAVE (ICCT) restricted to its distance-reporting classes, and our AIS series whole and registry-matched.

Inventory	CO ₂ per vessel (kt)	CO ₂ per distance (kg nm ⁻¹)
MariTEAM	20.55	—
SEIM	7.90	—
OECD	7.83	—
SAVE (ICCT distance-reporting classes)	6.78	289.5
GFW (AIS, any registry)	6.08	317.8
IMO	4.92	—
SAVE (ICCT)	3.39	—
GFW (AIS)	2.34	322.4
STEAM	2.33	—

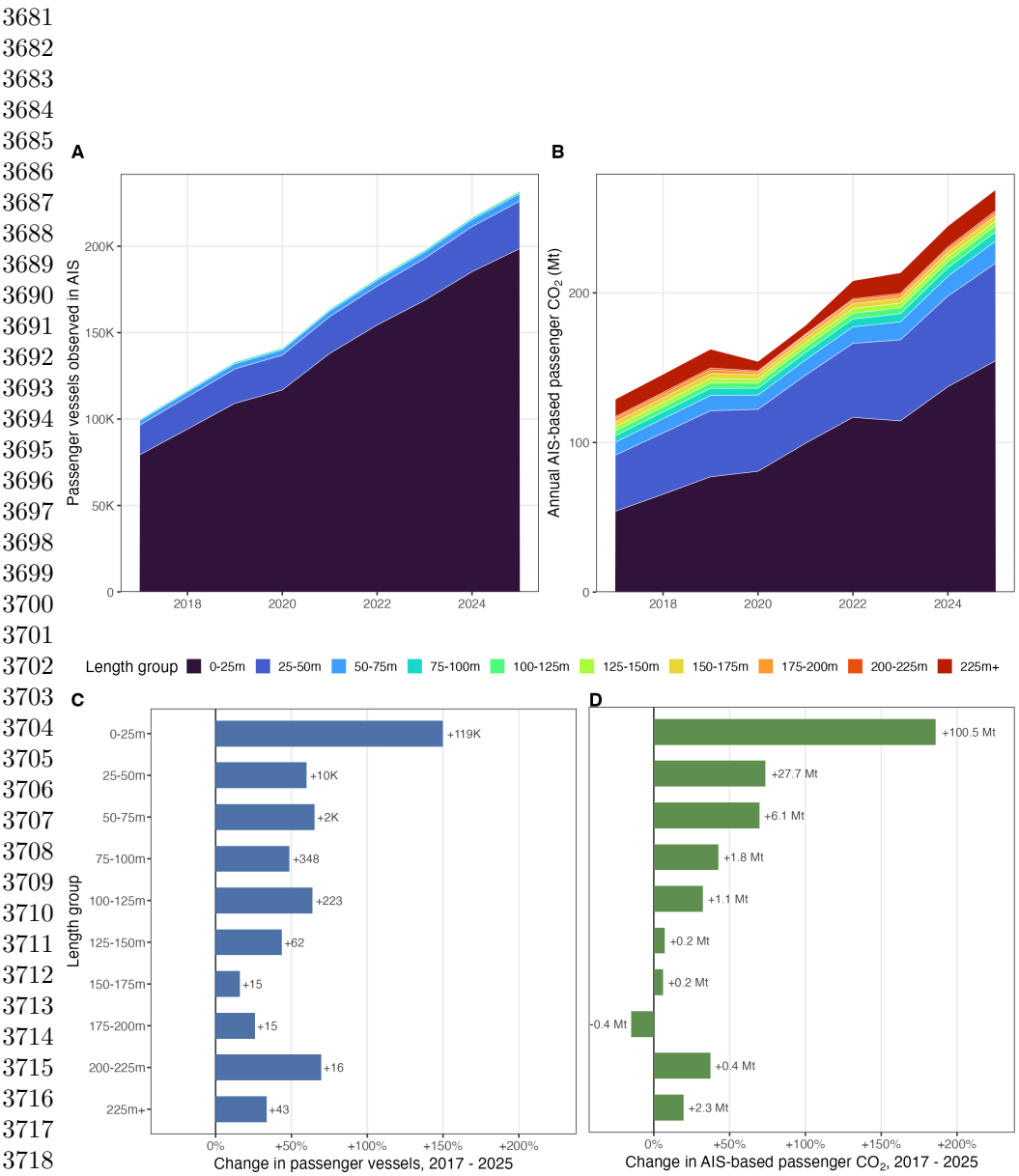


Fig. S16 Passenger vessels by length bin, 2017 against 2025, showing growth concentrated in the smallest sizes.

Still, full coverage of vessel activity remains limited by what the modeling is anchored to, whether AIS transmissions or registry data. Our total estimates (summed across broadcasting and non-broadcasting vessels) extend the inventory through satellite detections, capturing vessels that appear in neither registries nor AIS records. As a result, emissions diverge even further from every other inventory across the entire time range (Fig. S9). The Fourth IMO GHG Study, OECD, and SAVE also attempt to account for vessels with no AIS activity by assigning small registered vessels without AIS signals the activity profiles of explicitly modeled vessels. Yet only the IMO study approaches our total emissions. This again highlights the limits of anchoring an inventory to a registry, where its coverage can reach no further than the registry itself, blind both to the low-information segment of the fleet our AIS series captures and to the dark fleet beyond it.

1.8.2 Comparing our estimates with top-down inventories

We also compared our total estimates against two top-down inventories, the Emissions Database for Global Atmospheric Research (EDGAR) and the Community Emissions Data System (CEDS). EDGAR builds its shipping estimates from reported fuel sales, representing international shipping from international marine bunkers and domestic activity from domestic navigation fuel statistics, with recent years extrapolated from Energy Institute fuel oil trends [68]. EDGAR can therefore only count fuel that was sold and reported. CEDS also starts from reported fuel sales but treats them as incomplete, on the view that marine fuel is systematically underreported, and raises its global shipping total using bottom-up estimates, mainly from the IMO GHG Studies [69, 70]. Its limits are therefore not only what gets sold, but also the coverage of the bottom-up studies its global total rests on, whose limitations were described in the previous section.

Our total estimates (summed across broadcasting and non-broadcasting vessels) exceed both inventories, each of which is bounded by the fleet and activity they can resolve (Fig. S19, tabulated in Table S11). The differences are in magnitude and in trend, reflecting differences in fleet scope and composition, respectively.

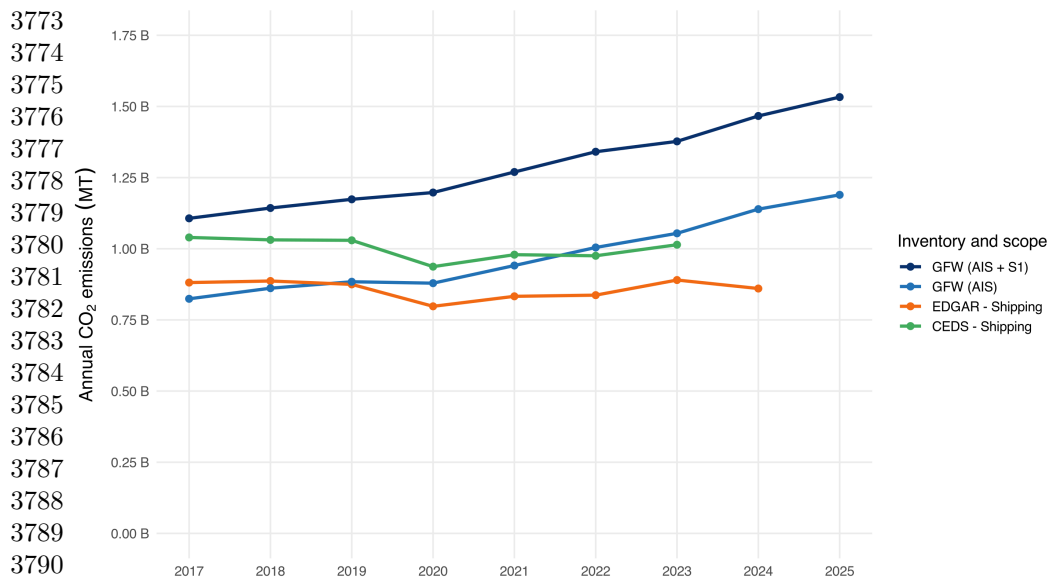


Fig. S19 Our shipping totals against the shipping sector of the top-down EDGAR and CEDS inventories.

Table S11 Annual marine vessel CO₂ emissions (billions of metric tonnes) for our AIS-based and fused (AIS + S1) estimates against the shipping sector of the top-down EDGAR and CEDS inventories, whose shipping scope combines international and domestic navigation. A dash marks a year for which an inventory publishes no estimate.

Year	GFW (AIS + S1)	GFW (AIS)	EDGAR - Shipping	CEDS - Shipping
2017	1.107	0.824	0.881	1.040
2018	1.143	0.861	0.887	1.031
2019	1.174	0.884	0.875	1.030
2020	1.198	0.879	0.798	0.937
2021	1.270	0.941	0.833	0.979
2022	1.341	1.005	0.837	0.975
2023	1.377	1.054	0.890	1.014
2024	1.467	1.139	0.860	-
2025	1.533	1.189	-	-

The magnitude gap arises from three main differences in the fleets these inventories can account for. First, fishing largely falls outside the shipping sector in both inventories. They report fishing fuel combined with agriculture and forestry, so EDGAR's shipping estimates contain no fishing vessels at all, while in CEDS only some unreported fishing fuel is absorbed into its global shipping sector through the bottom-up

anchor. Second, small coastal and inland vessels are only partly covered. Domestic navigation fuel is intended to capture this traffic, but the sector is known to be inconsistently defined across reporting countries, with emissions double-counted or omitted [71]. Third, neither inventory can capture fuel that never passes through reported channels, affecting vessels supplied informally or illegally. Further, even when sales are reported correctly, bunker statistics change only as fast as national energy reporting is revised.

The trend gap is different in kind. Growth in our AIS series can reflect expanding carriage and coverage rather than new activity, but the fused total should be largely insensitive to that: a vessel that starts broadcasting moves from the non-broadcasting component to the AIS one, reallocating emissions between them rather than adding to the sum. Growth in the fused total emissions should therefore reflect a real activity increase, at least in the segments carrying most of the emissions. Satellite detections offer independent confirmation that this could be the case, since S1 detections depend on neither broadcasting nor registration. Detected activity per unit of observing effort stayed roughly flat overall, but its composition shifted, with activity growing at both ends of the size range—among small non-fishing vessels and, most strongly, among the largest—while mid-size classes were flat and fishing declined (Fig. S20 and Fig. S21 panel A). In the smaller segments, part of this growth in new activity can move directly into AIS coverage along any existing activity from the non-broadcasting component becoming visible as AIS carriage expands (e.g., fishing emissions rise in our AIS series even as satellite detections show a decline in fishing activity). For the largest vessels that explanation is unavailable: AIS carriage is already near-universal (i.e., >75% coverage) there and the unmatched share has barely changed since 2017 (Fig. S13), so their growth must largely reflect new activity. Additionally, because a large ship emits on the order of a hundred times what a small one does, that compositional shift matters for emissions even if the total number of vessels did not change.

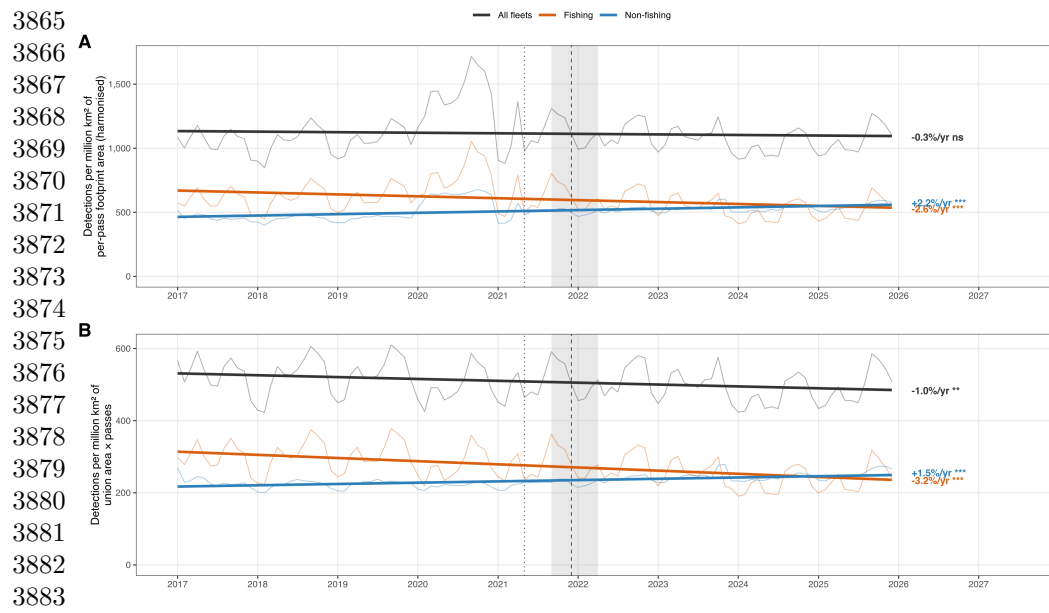


Fig. S20 Sentinel-1 detection density under two denominators, and by fixed length bin, showing detected activity per unit of observing effort.

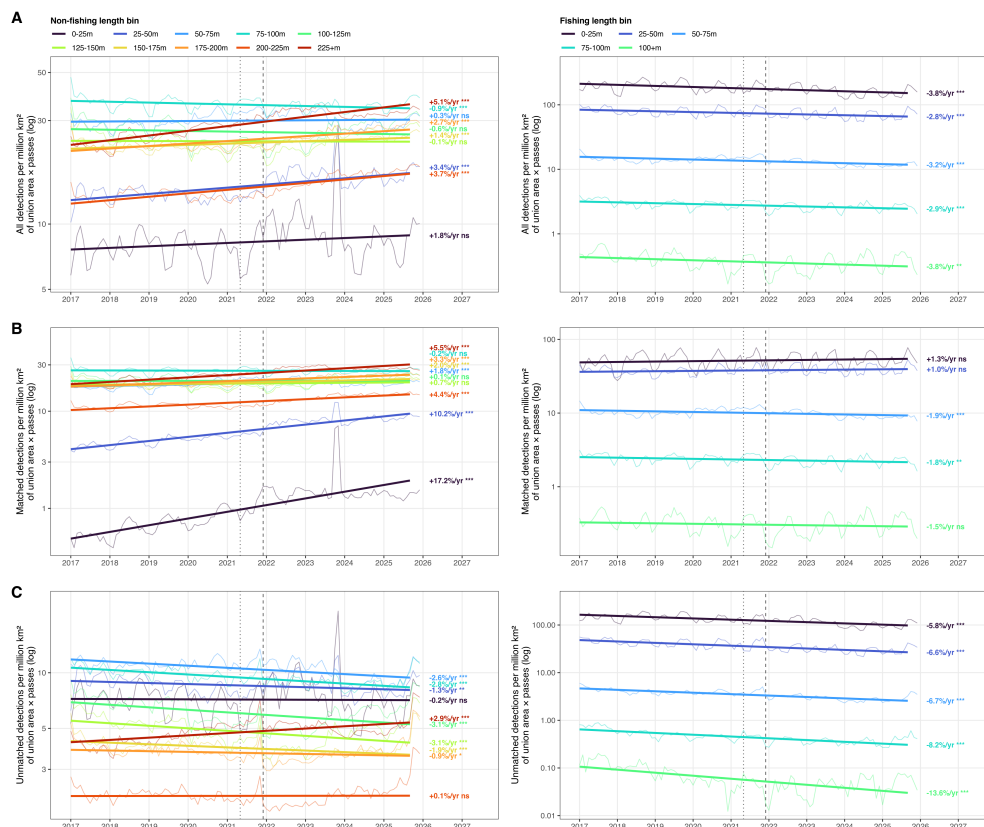


Fig. S21 Sentinel-1 detection density by length bin, split into total, AIS-matched and unmatched detections, for fishing and non-fishing fleets.

The magnitude gap rests on firm ground: each of these mechanisms adds emissions the top-down inventories cannot account for, and our own detection limits can only mean we understate the total further. However, for the trend, whether the increase in activity explains it depends on whether the increase reflects real activity rather than the limits of what our fused approach can observe.

Reallocation between our components requires that a vessel beginning to broadcast was already represented in the non-broadcasting estimate, and that holds only where Sentinel-1 detects it. Vessels too small or too close to shore to be reliably detected, or operating outside the imaged footprint—which accounts for a fifth of our AIS emissions—contribute nothing to that component and might enter the fused total for the first time when they begin broadcasting. Because AIS carriage and reception expanded over the study period, the unobserved fraction was larger at the start of the series than at the end, so part of the measured growth in the fused total is that gap closing rather than activity rising. Figure S21 (panel B and C) shows that this might be the case. Among non-fishing vessels under 25 m, matched detections (i.e., AIS-broadcasting)

3957 per unit of observing effort rise at 17% per year while unmatched detections stay
3958 flat, so the growth there is added to the fused total rather than transferred within
3959 it. That addition could, in principle, reflect new vessels entering service. However, a
3960 150% increase in passenger vessels under 25 m over eight years in our AIS record (Fig.
3961 [S16](#)) is difficult to reconcile with the idea that new construction of small craft could
3962 plausibly account for it. The more likely interpretation is that most of these vessels
3963 were already at sea but unobserved by S1, and became visible as AIS carriage and
3964 reception expanded, representing activity newly entering our record rather than newly
3965 occurring. A fused trend closely tracking the AIS-only trend is consistent with this.
3966 That the fused slope is slightly shallower indicates some genuine reallocation from the
3967 non-broadcasting component, but the similarity between the two implies that some of
3968 the measured growth enters directly through AIS rather than being transferred within
3969 the fused total.

3970 Our fused trend is therefore likely inflated by the expansion of our own observation
3971 capacity, and is better read as an upper estimate of the underlying growth than as
3972 a measurement of it. This does not, however, imply that the flat trends of the other
3973 inventories are correct either. The reconciliation comes from the part of the fleet our
3974 approach captures without ambiguity. Among non-fishing vessels of 150 m and above,
3975 where AIS carriage is the highest and already settled at the start of our series (Fig.
3976 [S13](#)), AIS-based emissions rose 29% between 2017 and 2025, compared with 44% for
3977 our full AIS total and 38% for our fused total, respectively (Fig. [S12](#)). S1 emissions
3978 are not disaggregated by vessel size, so we cannot provide the fused growth for this
3979 segment separately. The growth we report is therefore lower than our headline figure
3980 once observational expansion is set aside, but it is nowhere near the stability these
3981 inventories describe.

3982 If these larger magnitudes and steeper trends were confirmed, the implication is not
3983 only that marine emissions have been systematically underreported, but that marine
3984 transport is among the fastest-growing sources in the global emissions picture. Our
3985 fused estimate grows roughly 4-5 times faster than global emissions across all sectors,
3986 and 7-10 times faster than all other transportation modes combined across EDGAR
3987 and CEDS (Fig. [S22](#)). Even confined to the segment of the fleet where observation
3988 capacity did not expand (i.e, non-fishing vessels of 150 m and above) the increase
3989 in AIS-based emissions remains more than 3 times that of global emissions across all
3990 sectors and more than 5 times that of all other transportation modes combined.

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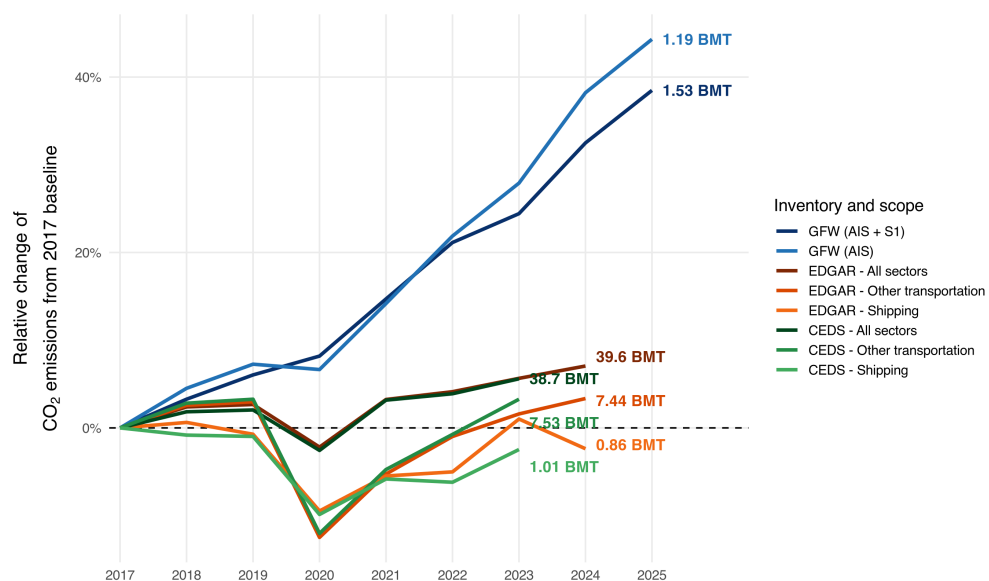


Fig. S22 Marine emissions growth against all sectors and against other transport modes, in EDGAR and CEDS, indexed to 2017.